



The Expanding Frontier of Agentic AI: A User's View

August 6, 2026

The most important change in AI today may not be the arrival of a single new frontier model, but the speed at which frontier-like capability is becoming cheaper. My own use of AI agents since March suggests the shift is already visible: models that once looked like expensive high-end choices are being matched by cheaper alternatives within months. This report looks at that expanding frontier and what it means for adoption.

Since I started using AI agents in March, trying new models has become something of a hobby—and partly a necessity. I could not rely on the most advanced models to power my OpenClaw Bots and other tools, not only because access is not always guaranteed, but also because they are extremely expensive for agentic usage. I therefore spent the past few months switching among more accessible and affordable alternatives, many of them models developed by Chinese labs.

The effect of each switch was often not obvious. A model might seem more reliable or require less instruction, but the improvement was difficult to separate from chance. Over a longer period, however, the change became unmistakable. My AI agents could complete the same tasks with fewer prompts, use tools more effectively and travel further from a simple instruction before requiring intervention.

That said, two recent model releases did feel different to me. Both Kimi-K3 and DeepSeek-V4-Flash-0731 seemed notably more powerful than the previous alternatives I had been using. Was this merely a subjective experience, or had the underlying models genuinely improved? This prompted me to do some databased analytical research.

Measuring agentic intelligence against cost

The research I did was a simple plot of popular models along two dimensions: capability and cost. The horizontal axis is API cost per million tokens on a log scale, using a simple blend of 90% input tokens and 10% output tokens. The vertical axis measures model capability. There are multiple possible measures, but I chose the [Agent Index from LLM Stats](#) for two reasons: first, I care more about agentic capability than other dimensions such as coding or writing because of my use case; second, LLM Stats offers a composite measure based on benchmarks involving tool use, multi-step execution, web search, terminal operation, software engineering and other agentic tasks.

The broad pattern is intuitive: more capable models tend to be more expensive. US models are shown in blue, and Chinese models in red in the chart. A clear upward correlation can be observed across all the dots. The difference is that US models occupy much of the upper-right part of the chart, while Chinese models are more densely represented at lower price points. The regression lines capture this difference: both groups exhibit a positive

Authors

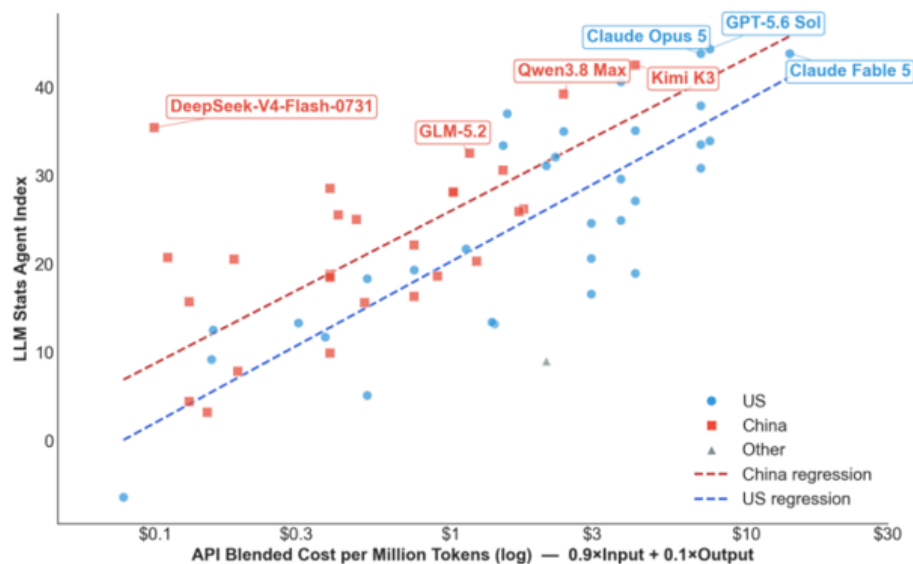
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relationship between cost and agentic performance, but the US cluster generally extends further into the highcost, high-capability region.

US and Chinese AI Models copared on cost and Agentic capability



Source: Deutsche Bank Research, LLM Stats

A frontier moving rapidly

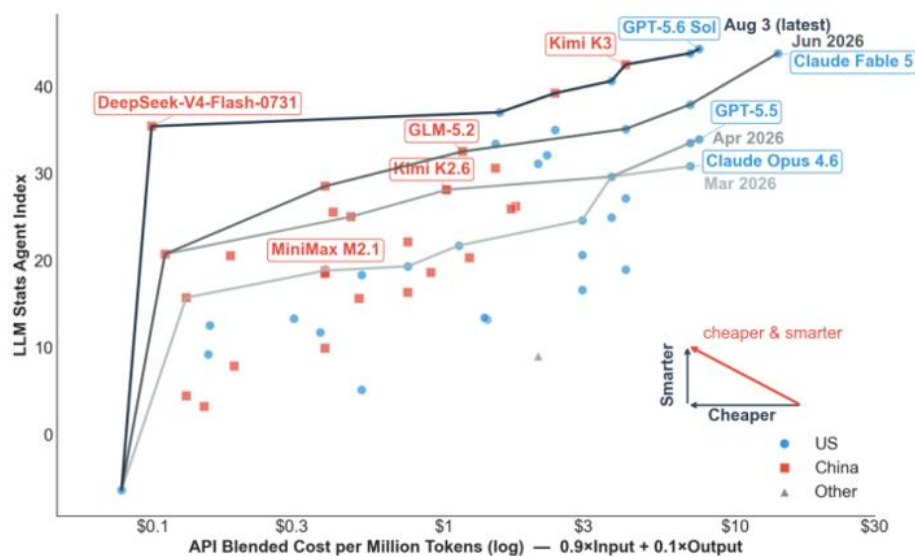
The more useful concept for model selection—at least for me as an economist—is the Pareto frontier. A model is Pareto optimal if no cheaper model has a higher score. Connecting all Pareto optimal models gives the Pareto frontier: the boundary of best available model performance at each cost level. A model below that frontier may still suit a particular task, but on these two dimensions it is dominated by another choice. The chart below shows the Pareto frontier for all models in my dataset.

For example, at the top of the current frontier, GPT-5.6 Sol has the highest Agent Index in the dataset, at 44.3, with a blended cost of \$7.50 per million tokens (the dot next to it is Opus 5). This is just too expensive for me. If I want to reduce cost, the next best choice according to the chart would be Kimi-K3: with only a modest loss in capability, it offers a meaningful cost saving and could be a good choice for slightly less complex tasks. But it is still too expensive for simple tasks, for which I would go even further down the line to maybe DeepSeek-V4-Flash.

But the analysis did not stop there. The chart also shows how the Pareto frontier has shifted as new models became available. By including only models released by each cutoff date — March 31, April 30, June 30 and August 3 — I was able to reconstruct earlier frontiers and compare them with the latest one.



Expanding Pareto Frontier: Agentic AI is becoming cheaper and smarter at speed



Source: Deutsche Bank Research, LLM Stats

The speed of Pareto improvement is striking. Every month, the Pareto frontier moves upward and left by a large margin. This means users can achieve better performance at the same cost, or cut spending significantly while maintaining similar performance. For example, GLM 5.2 on the June frontier has almost the same score (32.5) as Opus 4.7 (33.5) on the April frontier, while its blended cost of \$1.16 is far below Opus 4.7's \$7. In merely two months, a user could potentially reduce token cost by more than 80% without losing performance.

The latest frontier line makes the point even more clearly. As of today, almost all models on the latest Pareto frontier have exceeded GPT-5.5's score. In other words, **what sat on the high-cost frontier in April is now available across the cost spectrum.**

Chinese AI models are catching up

It would be easy to summarize the chart by saying that US models remain ahead... At the absolute frontier, that is still true: GPT-5.6 Sol holds the highest Agent Index in the dataset. But that reading misses the more consequential competitive dynamic.

...but Chinese models' advancement from the lower-left is also remarkable.

In March, when I first started using AI agents, the Pareto frontier was dominated by US models. Kimi K2.6 moved into the middle by April; GLM-5.2 approached the performance of recent leaders by June; and Kimi K3 narrowed the gap near the top in July. Kimi K3, being the largest (2.8T parameter) and most costly openweight model to date, shows that Chinese models are no longer competing only on price; they are increasingly present near the capability frontier as well

The DeepSeek Outlier

What strikes me most is how the latest DeepSeek release altered the shape of the chart. DeepSeek-V4-Flash-0731, released last week, was a large improvement over the earlier low-cost version included in the April frontier. The wearlier model was already inexpensive, with an Agent Index of 20.7 at \$0.11 per million tokens. The July release has effectively the same cost, but its Agent Index rises sharply to 35.4. It moves DeepSeek from the low-cost edge of the chart into the performance range of much more expensive frontier models.



DeepSeek demonstrates how much potential remains for AI models to improve. DeepSeek-V4-Flash is far smaller than models near the current frontier. At 284B parameters in total (13B active under a Mixture-of-Experts architecture), its agentic capabilities outperformed much larger models such as the 744B GLM-5.2 and the 1.6T DeepSeek-V4-Pro. The improvement was achieved not through increasing model size (making it more knowledgeable) but through post-training (teaching it to better handle tasks). The contrast illustrates that good agent performance does not require the largest model; architecture and post-training can move the effective frontier more than parameter count alone would suggest.

The Jevons Paradox is Working on Me

Another interesting fact: the steep decline in AI model costs did not bring me any savings. The opposite is happening—my AI bills have surged. My total spending on AI was effectively zero in February before I started using AI agents. Then I started using OpenClaw and made my first token subscription plan at \$20/month. A few months later, my recurring spending on AI is now reaching \$100-200/month. My costs increased because I now use AI agents to do many more things and must subscribe to new plans as tokens run out under existing ones. The increase also includes cloud storage and an \$800 Mac Mini to host my “little lobsters” (OpenClaw Bots).

This is a familiar economic mechanism: the Jevons paradox. Improvements in efficiency did not reduce consumption; rather, they encouraged wider use and led to an increase in total usage and total spending. As model capability improved while token cost dropped, I expanded the use of my AI agents to more daily tasks—tracking headline news and specific topics I am interested in, summarizing market movements, researching academic papers, managing expenses, making travel plans and testing ad-hoc ideas. Why use a chatbot if you have dedicated AI agents that know your background and remember the conversation history?

The question is therefore not simply which model will hold the highest score next month. It is what happens when millions of users and businesses can afford to keep capable agents running for longer, on more tasks, with less concern about every token they consume. If the cost of agentic intelligence continues to fall while its capability rises, will efficiency save tokens—or will Jevons’ paradox help trigger an explosion in real-world AI use?

My verdict: the expansion of the AI Pareto frontier is benefiting users and the broader AI ecosystem. As US and Chinese labs compete, premium frontier models push the performance ceiling higher, while efficient lower-cost models make agentic workflows economical at scale. Together, they allow users to match each model to the value and difficulty of the task, rather than accept a single price-performance trade-off. Competition is accelerating the race, while the global circulation of research ideas, open weights and practical techniques is helping improvements spread quickly across the field. **The analysis made me more confident that the future of agentic AI is arriving faster than many expected.**



Appendix 1

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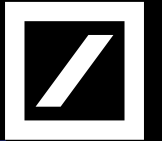
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This Month in Geopolitics: August 2026

Key events, themes and trends to watch in the month ahead

August 2026

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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At a Glance: Geopolitical events to watch in August 2026

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



In the spotlight for August 2026: US-Iran peace talks and US trade policy



US-Iran war (all of August)

Expect more volatility – but also efforts to resume dialogue – during August. Tensions escalated significantly in July, with renewed military strikes and the US reimposing sanctions on Iranian oil sales. Both sides appear unwilling to concede for fear of losing ground. While President Trump has suggested that the ceasefire is over, **we retain our assessment that both parties are incentivized to find a near-term resolution** as kinetic attacks will not drive a resolution. That said, we should prepare for more volatility in the coming months as Iran is unwilling to cede control of the Strait of Hormuz and neither side is prepared to lose face. The Yemen-based **Houthi blockade of Saudi Arabia**, announced on 20 July, further heightens risks by threatening crude oil flows via Saudi Arabia's Yanbu on the Red Sea (see map), a terminal which handles about 4 million barrels of crude oil and condensate exports per day.

The persistent security risks will likely continue to drive the Gulf states to reduce their vulnerability to the Strait of Hormuz. Shipping levels through the Strait will periodically be impacted by Iranian threats to navigation and political considerations. Even once dialogue resumes and fighting subsides, **traffic will likely take time to recover** given the challenges of de-mining, a lack of trust between the US and Iran, and Iranian threats. Infrastructure repair to ports, terminals and energy infrastructure, as well as the provision of security guarantees are other factors to watch. Full operational normalization, involving infrastructure repairs, [is not expected](#) until 2027 and beyond.

US trade policy (all of August)

When the US 10% global Section 122 tariffs expired **on 24 July**, the Trump administration moved quickly to fill the gap with Section 301 tariffs of 10-12.5% on more than 60 countries. In August, we should expect the Trump administration to **keep building its replacement “permanent tariff framework”** using Section 301 and 232 authorities. The USTR's Section 301 investigations into “structural excess capacity” in 16 manufacturing-heavy economies remains ongoing and could yield new tariffs in the coming weeks. The Trump administration is also pursuing several Section 232 national security investigations with scheduled tariff effective dates of 31 July and 29 Sept. This should keep pressure on countries to strike bilateral deals favorable to the US, spanning trade, investment, industrial and technology concessions.

To watch will be movement on unresolved country-specific issues, e.g., India's tariff talks, the EU's proposed 30% tariff, in addition to continued legal uncertainty as court appeals over the tariffs' legal basis play out. This will set a complex backdrop to the G20 finance ministers' meeting where trade protectionism is already shaping up to be a point of friction.

Key regional ports in the Persian Gulf



Source: Deutsche Bank. For more on the impact of the Iran conflict, see [Twenty Miles That Shook the World - Deutsche Bank Research Institute](#)

Key events, issues and themes to watch in August 2026

Listed in chronological order



US-Mexico-Canada trade talks (all of August)

The renewal talks are expected to be challenging and potentially protracted. 1 July marked the start of the US-Mexico-Canada (USMCA) trade deal's mandatory six-year joint review process whereby the three countries are required to assess whether to extend the agreement for another 16 years, review it or allow it to move into annual review mode. The US has declined to confirm the renewal of USMCA "in its current form".

In August, we expect further US-Mexico-Canada negotiations focused on resolving outstanding issues (e.g., automotive rules of origin, steel/aluminum, economic security, and issues around agriculture, labour and environment). At the G7 summit, Canada and the US agreed to a 30-day countdown for a new cross-border economic/security agreement, which was extended to 1 August.

In any case, Mexico and Canada are both subject to a 10% Section 301 tariff that took effect on 24 July. In addition, the US 50% tariffs on a range of Canadian goods announced on 20 July, and additional Section 232 investigations (lumber, copper, semiconductors, pharmaceuticals, trucks, critical minerals, etc.) could complicate talks further as Mexico and Canada are top exporters in several of these sectors. The Trump administration is also using the review to push non-trade demands on migration, drug trafficking and defence cooperation.

First weeks of new UK Prime Minister (all of August)

Andy Burnham became the UK's 7th Prime Minister since 2016 on 20 July. He enters office facing a sluggish economy, cost-of-living pressures, and a massive welfare bill. He has pledged this moment as a "circuit-breaker" for Britain and has quickly set out to focus on concrete cost of living measures. In August, we can expect Mr. Burnham to be busy with the early scaffolding for his "No. 10 North" project focused on devolution of power. Mr. Burnham has signaled his priorities will be cost-of-living issues, devolution, public control of essentials, social commitments and longer-term structural reform. UK parliament is typically in summer recess through August, which limits legislative action but gives Mr. Burnham room to set the tone for his premiership with an intention to move quickly on key commitments in the autumn.



Prime Minister Andy Burnham was sworn in on 20 July. Source: [House of Commons](#)

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China-US talks ahead of next Xi-Trump summit (all of August)

Bilateral dialogue to tee up deliverables for the 24 September summit will be a key watchpoint during August. Recent developments – including US allegations of [election interference](#) and warnings over [Chinese AI models](#) – reinforce our view that relations remain structurally competitive despite the recent stabilisation in ties. Investors should anticipate more, not fewer, bilateral economic restrictions over the medium term.

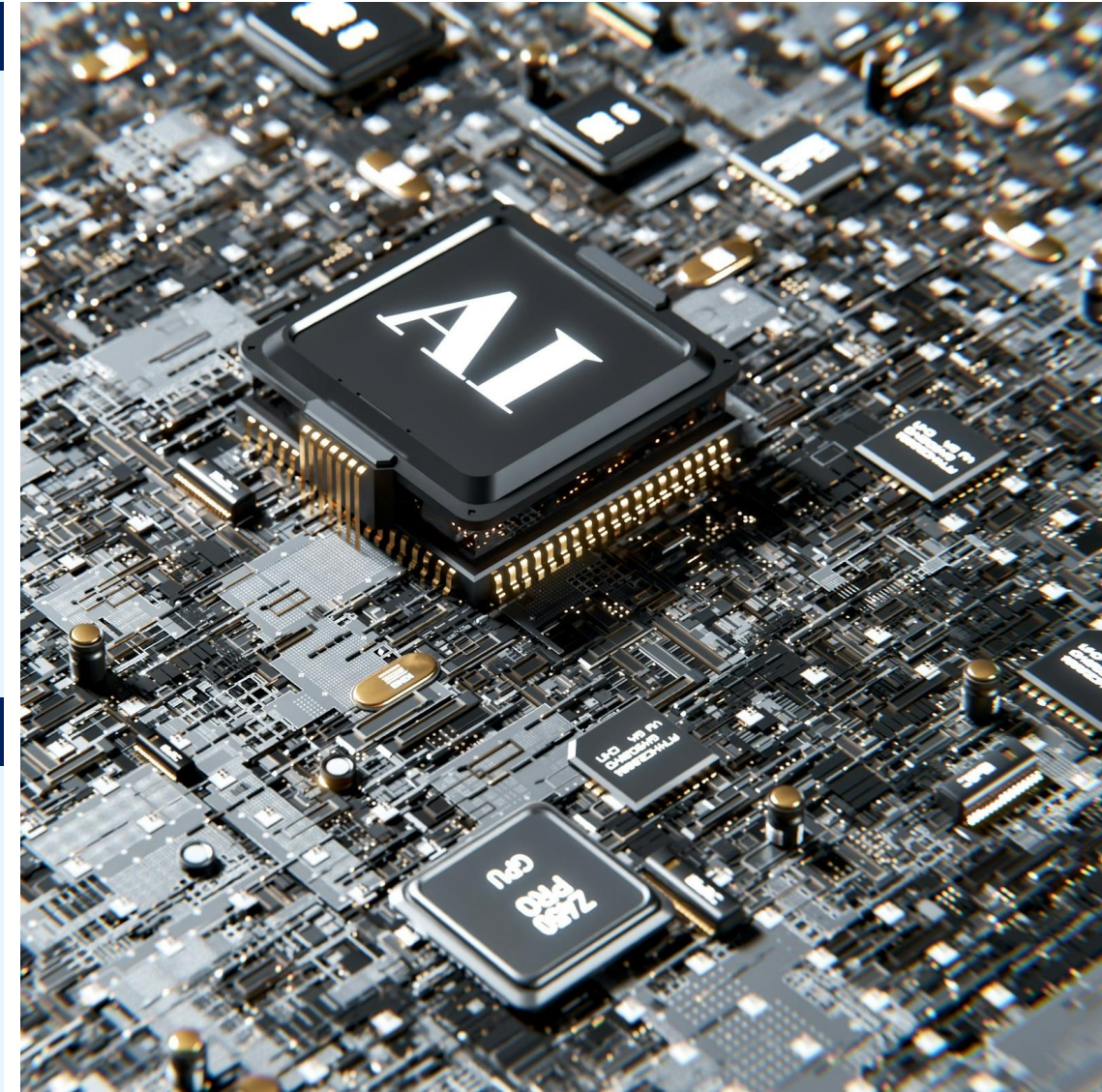
Nevertheless, there is currently no evidence to suggest these dynamics are likely to delay the next Xi-Trump summit, which will likely go ahead as planned on 24 September. During August, investors should watch for any sign of progress on talks on the Board of Trade (intended to boost non-sensitive trade), Board of Investment (to facilitate investment in non-sensitive areas) and an extension of the Busan agreement – which delayed a range of new economic restrictions until November 2026.

Both sides could also strengthen commitments around AI dialogue, with first talks [reportedly](#) expected to happen ahead of the summit. Dialogue on AI was a key deliverable from the May Xi-Trump summit, and both sides could seek to address persistent tensions around the technology, although a major breakthrough appears unlikely.

China's leaders gather for summer retreat in Beidaihe (early August)

The retreat provides an opportunity for Party leaders to exchange views on major policy issues. This year's meeting could potentially cover topics ranging from the conflict in Iran, relations with the EU and US, the implementation of the 15th Five-Year Plan (2026-2031) and early deliberations around the next Party Congress in late 2027.

The retreat primarily serves as an opportunity for internal alignment and consensus building on key domestic and external policy issues. There are no official readouts, but subsequent state media framing, leadership signals, and Party meetings in the autumn could offer clues on the direction of policy debate.



Source: [Igor Omilaeu](#) on [Unsplash](#)

Key events, issues and themes to watch in August 2026

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Abelardo de la Espriella inaugurated as Colombia's next president (7 August)

The president's early agenda is expected to include significant policy shifts, including a planned urban security coordination mechanism for major cities; a more pro-US and pro-Israel foreign policy stance; and a review of Colombia's diplomatic posture toward Cuba, Nicaragua, Venezuela and multilateral bodies. Key watchpoints over the next 2-3 months will include cabinet formation and congressional coalition-building. The incoming administration has also vowed to disrupt narcotics production and trafficking, and counter violence by armed groups operating in parts of the country – including in a bid to attract more foreign investment.

Kazakhstan legislative elections (23 August)

Kazakhstan will elect all 145 deputies to the newly-created Kurultai. This will be the first election to the new unicameral legislature established after March 2026 constitutional reforms replaced the old bicameral parliament. These elections are unlikely to materially impact governance in the country with President Tokayev remaining firmly in control of the political system. Kazakhstan's constitutional court ruled that president Tokayev may seek another term under the new 2026 constitution, effectively resetting his term count despite having been elected in 2022 to what was billed as a single, non-renewable seven-year term ending 2029. While the election itself is unlikely to produce policy surprises, it will be an indicator of the government's ability to manage the political transition under the new constitutional framework and maintain domestic stability.

Iceland referendum on resuming EU accession talks (29 August)

Iceland will hold a referendum asking voters whether the country should resume EU accession negotiations. Iceland had applied to join the EU in 2009, but suspended accession talks in 2013 after a change of government and later withdrew the application.

The current coalition government pledged to finally let voters decide, and accelerated the timeline partly due to shifting geopolitical dynamics, including US tariff hikes and Trump administration statements on acquiring Greenland. Polling is tight with surveys showing 52% in favor with 28% opposed. A vote to resume negotiations would not commit Iceland to EU membership but would mark a significant shift in the country's strategic orientation.



Abelardo de la Espriella will be inaugurated as Colombia's next president on 7 August. Source: [Wikimedia Commons](#)

Key events, issues and themes to watch in August 2026

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G20 Meeting of Finance Ministers and Central Bank Governors (31 Aug to 1 Sep)

This will be a key checkpoint in the US' G20 presidency, where Washington sets the agenda for global economic policy coordination among the world's largest economies. The US has identified as its priorities: modernizing financial regulation, strengthening understanding of excessive global imbalances, enhancing debt transparency and facilitating debt restructuring, endorsing digital assets and improving cross-border payments. Treasury Secretary Scott Bessent has sought a “streamlined, results-oriented” agenda.

The first G20 finance ministers meeting under the US presidency in April failed to produce a joint communiqué or chair statement and no joint press conference was held at the end. The current crisis in the Middle East amongst other geopolitical developments may crowd out purely technical finance discussions.

Washington's tighter focus may clash with other members who still want to discuss debt relief for developing countries, climate finance and inequality. With a US leadership openly skeptical of the G20's broader mission, to watch will be how much substantive coordination is feasible.



Source: [Giorgio Trovato](#) on [Unsplash](#)

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Situational awareness - do we have it?

August 3, 2026

Executive summary

There are two events in the last few weeks that look like non-events. The Situational Awareness hedge fund unwound a \$16bn book of leveraged AI trades with barely a ripple beyond a few days of volatility. Meanwhile, the Kospi fell more than 20% in 48 hours and then rallied 25% off the low — finishing the week almost flat. Taken individually, both look like markets doing their job: absorbing shocks and moving on.

We think that read is wrong, or at least incomplete. Both events were symptoms of the same underlying condition: leverage in ETFs, margin accounts and hedge fund books that was encouraged by more than a decade of low and stable rates. Indeed, reports that over 3% of Korea's adult population received a margin call over the last two weeks is deeply concerning if true.

This piece argues that 'small vol' events are going to affect markets with an increased frequency. That comes as the backdrop of normalised interest rates and geopolitical unrest has been augmented by a Fed that has moved away from the forward guidance that reduced some volatility in the past.

We also worry that the risks in household debt (and investment accounts on margin) is being ignored as the focus stays on the risks in growing sovereign debt. If this is right, 'small vol' episodes are likely to become more frequent and potentially have larger consequences.

In the end, we worry that investors, corporates and policymakers are focused on avoiding the next big financial crisis and, in doing so, may be underestimating the risk in small events that can cascade.

A tale of two 'small vol' events

Markets have spent the last few years braced for another 2008-style event. That vigilance has a blind spot. It trains attention on the big, obvious risks and away from the smaller, everyday ones that can cascade in the right circumstances.

The implosion of the Situational Awareness hedge fund is the first case in point. The fund sold its \$16bn public equity book after highly leveraged trades on AI companies went wrong. Markets absorbed the unwind with only a brief bout of volatility — no contagion, few headlines beyond the financial press.

The second was in Korea. At one point last week, the Kospi fell over 20% in barely 48 hours, before staging a 25% rebound from the low. The headline index ended the week close to flat and broader markets have shrugged it off. But a flat index conceals an extraordinary amount of forced selling and buying beneath the surface, much driven by margin calls on leveraged retail positions.

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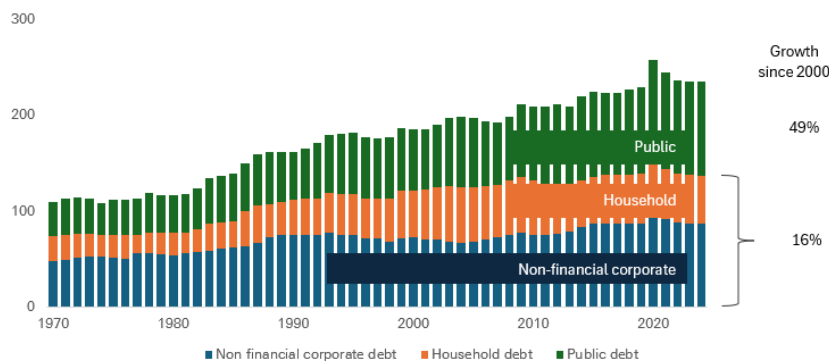
Why 'flat on the week' is the wrong takeaway

An index that round-trips 45 percentage points in days is not the same thing, economically, as one that simply sits still. Leveraged and margined positions are typically closed out mechanically and irreversibly on the way down, regardless of what happens on the way back up. The P&L may net out at the index level; the wealth destruction for individual, often first-time, leveraged investors does not.

We believe the potential for household leverage to cause problems is underestimated because the aggregate figures look okay. When we show the following chart, clients gravitate to the rise in sovereign debt, not the plateau in household leverage.

Figure 1: Household debt has lost the spotlight to public debt since 2000

Global debt as a % of global GDP



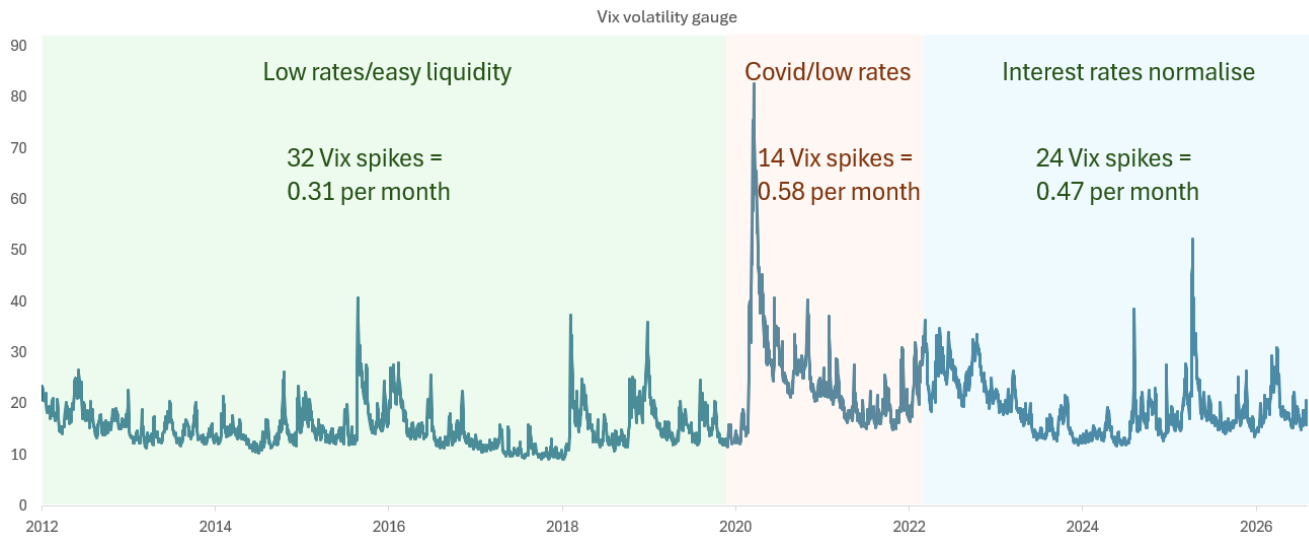
Source: IMF, Deutsche Bank Research

Small vol events are becoming more frequent

The following chart is created using our own judgement. But it shows that, by our reckoning, the number of spikes in the Vix since 2022 has been materially higher over the last few years than during last decade. The last four years are an important comparison period as this is when interest rates began to normalise.



Figure 2: A greater frequency of 'small vol' events appears to be the norm, at least for now



Source: Factset, Deutsche Bank Research

Is the Fed trying to bring volatility back on purpose?

One explanation for why policymakers may be tolerating episodes like these comes from Rob Armstrong (a former colleague) [here](#). He advanced a theory about Fed chair Kevin Warsh that essentially says, his withdrawal of forward guidance is a deliberate attempt to reintroduce 'uncertainty premia' into markets that have used lower volatility to take on more leverage than they otherwise would. The idea is that "Too little everyday volatility ultimately leads to financial crises."

The mechanism, on this reading, is that forward guidance suppresses everyday volatility – and stable prices and calm markets encourage more borrowing, both by governments and by market participants, than would occur under normal price discovery. Warsh cannot say this outright without spooking markets, hence the vaguer 'referee' rhetoric – but the policy actions, in Armstrong's view, point the same way.

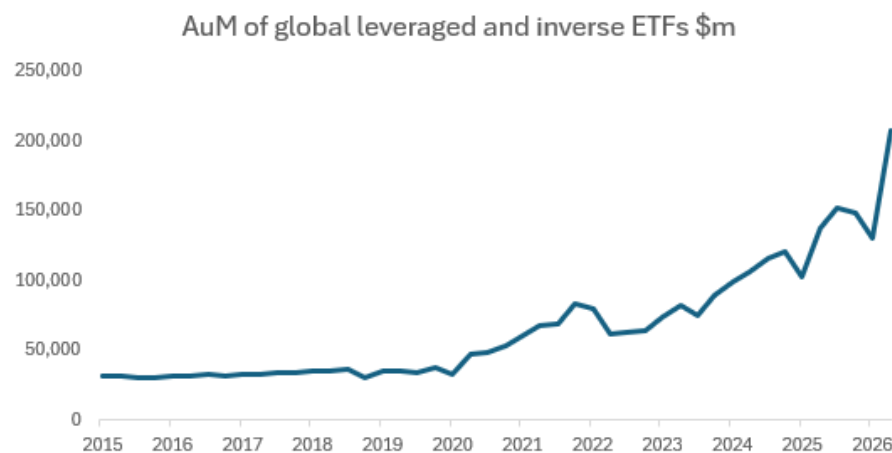
If policymakers genuinely believe more day-to-day volatility is healthy, investors should expect more 'small vol' episodes like Situational Awareness and the Kospi swing and should stop treating each one as isolated, ring-fenced events.

Leveraged ETFs and margin debt: Should we be afraid?

Leveraged and inverse ETFs, and the margin debt that often sits alongside them, are growing quickly across many major markets. They are a key mechanism that turns an ordinary correction into a cascade because they force mechanical, non-discretionary selling as prices move.



Figure 3: The semiconductor and broader tech rally has fuelled the popularity of leveraged ETFs



Source: Bloomberg Finance L.P., Deutsche Bank Research

In Korea specifically, the launch of single-stock leveraged ETFs tracking Samsung Electronics and SK Hynix in April turned two already-dominant, trillion-dollar-plus stocks into vehicles for retail leverage almost overnight. Retail investors accounted for a large proportion of holders. When the underlying stocks fell, those products amplified the move.

The risk is not confined to Korea. The same products are growing in the US and Europe. A market that is many times larger, and far more directly connected to consumer spending and credit availability, having its own 'Kospi moment' is a materially bigger problem for systemic risk than a niche hedge fund unwinding.

A footnote: why the name 'Situational Awareness' went viral

The fund's name borrowed from Leopold Aschenbrenner's 165-page 2024 essay, which argued AGI was plausible by 2027 and sketched a path from massive compute build-outs to an 'intelligence explosion' and eventual government-led consolidation of frontier AI ('The Project').

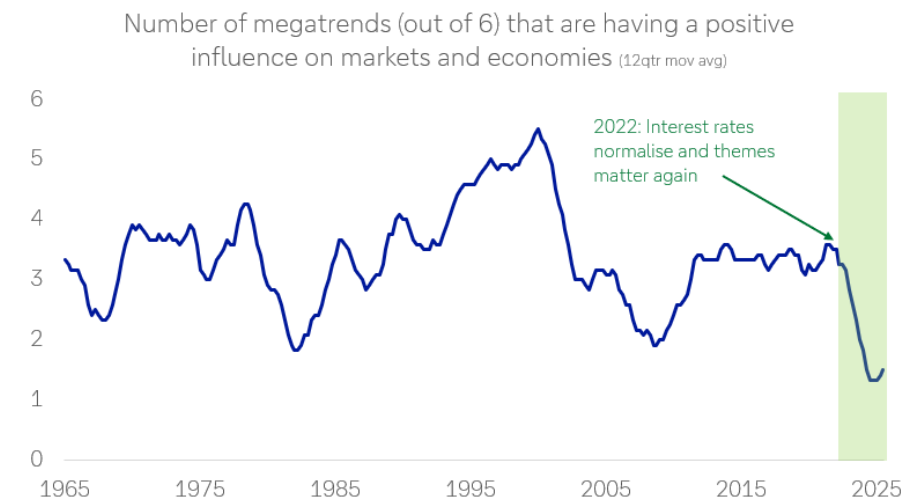
Part of a developing medium-term trend

Our megatrend framework tracks six long-run forces (technology, sovereign deficits, geopolitics & globalisation, domestic politics & social discontent, demography, and energy) and combines them into a single indicator of how supportive the backdrop is for markets and economies.

Overall, the number of megatrends working in favour of markets and economies has fallen to a level last seen only twice before – during the 1970s oil crises and around the 2008 financial crisis. Both were periods marked by exactly this kind of 'small vol' pattern: idiosyncratic-looking events that, in hindsight, were symptoms of a more fragile underlying system.



Figure 4: Prior periods of megatrends being this deeply negative have resulted in the greater frequency of volatility events



Source: Deutsche Bank Research

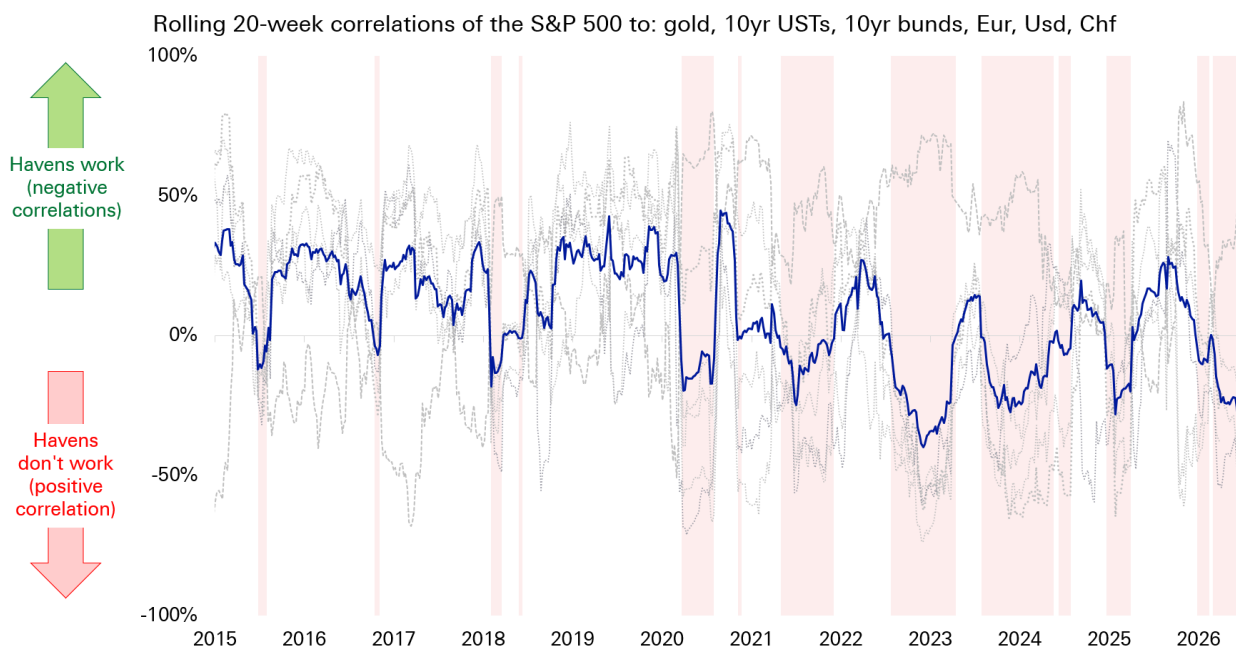
The single most important negative megatrend is sovereign deficits. Developed-market governments are refinancing and issuing new debt at rates well above those of the last decade. At the same time that our model flags unpredictable sovereign debt sell-offs, particularly those triggered by exogenous shocks and investor emotion, as a growing risk.

Havens are not there to catch you

Ordinarily, investors would look to traditional havens (gold, the dollar, the Swiss franc, the yen, US treasuries and bunds) to cushion volatility. However, our haven indicator shows that, since 2020, this combination of assets has failed to hedge risk (as proxied by the S&P 500) particularly during more shock events, including covid, the 2022 rate hikes, the 2025 tariff shock and the 2026 Iran war.



Figure 5: Traditional haven assets have not worked in the 2020s, particularly during shock events and periods of volatility



Source: Bloomberg Finance L.P., Deutsche Bank Research

If havens keep failing when they are needed most, the cushioning effect investors implicitly rely on to keep 'small vol' small is weaker than it used to be. That raises the odds that future 'small vol' episodes may cascade into something more.

What if this happened in the US or Europe?

It is tempting to treat the Kospi swing as a Korea-specific story — a function of retail culture, single-stock products and two dominant, correlated semiconductor names. But if a similar episode plays out in US or European, the results could be systemic.

The transmission mechanism would not be limited to portfolio losses. It would run through consumer spending, as households with margin-called leveraged positions cut back; through credit, as brokers and lenders reassess collateral and risk; and through confidence, as retail participation reverses. A leveraged retail shock in the US would be a materially larger systemic risk than a single hedge fund unwind.

Another example

Silicon Valley Bank is another example. On paper, SVB's failure should have been a contained, idiosyncratic event — a mid-sized bank with a concentrated depositor base and a duration mismatch. In practice, it triggered enough of a confidence panic that policymakers felt compelled to effectively guarantee all deposits across the banking system, a far larger intervention than the initial problem seemed to warrant.

The lesson is not that every small event becomes systemic but the ones that do are rarely obvious in advance. And when there is an increased frequency of these



events in a market with more leverage, less reliable havens, and policymakers newly tolerant of everyday volatility, there is an increased risk that 'small vol' events may catalyse larger problems.

Conclusion: are we situationally aware?

The market's instinct is to file Situational Awareness and the Kospi swing under 'noise' because neither, so far, has cascaded into a systemic event. That instinct may be dangerous, particularly if policymakers are willing to tolerate more everyday volatility.

Given our megatrend model indicates that the supportive backdrop for markets is near a multi-decade low, 'small vol' episodes can potentially be used as an indicator of how stress in the financial system is quietly building up.

There are plenty of other parts of the market where 'small vol' events may catalyse problems in areas that were built up with leverage during the periods of easy money. Just two examples are:

- Private equity: a large amount of TMT portfolio companies bought during the era of near-zero rates remain on PE firms' books, held longer than usual as exit markets stayed shut. A reckoning may still be ahead.
- Corporates: many listed companies still carry underperforming divisions bought during debt-fuelled M&A booms last decade. As a result, they have lower asset turnover metrics, a legacy of a decade in which cheap financing incentivised growth in revenue over earnings.

Appendix: Why Situational Awareness went viral

The original essay by Leopold Aschenbrenner in 2024 was 165 pages long. Below is a bullet point summary (generated by AI).

The Path to AGI by 2027: The document argues that Artificial General Intelligence (AGI) is strikingly plausible by 2027. This prediction is based on extrapolating "Order of Magnitude" (OOM) trendlines in three areas: physical compute scaling, algorithmic efficiencies, and "unhobbling" gains (transforming models from chatbots into active agents).

Massive Industrial Mobilisation: AI is shifting from a software boom to a massive industrial process. Leading AI labs will build training clusters costing \$100s of billions by 2028 and over \$1 trillion by 2030. By the end of the decade, these clusters could require power equivalent to 20% of total U.S. electricity production.

The Intelligence Explosion: Once AGI is achieved, it will likely be used to automate AI research itself. Because digital researchers can run in millions of copies and at much higher speeds than humans, a decade of algorithmic progress could be compressed into a single year, leading rapidly from AGI to superintelligence.

The Security Crisis: Current security at leading AI labs is described as "Swiss cheese," leaving key algorithmic secrets and model weights vulnerable to state-



actor theft, particularly by the CCP. The author warns that leaking these secrets in the next 12-24 months would be a catastrophic national security failure.

Superalignment Challenges: Reliably controlling AI systems that are vastly smarter than humans is an unsolved technical problem. Current alignment techniques (like Reinforcement Learning from Human Feedback) will break down because humans cannot supervise behaviour they do not understand. Solving this will require using AI to help automate alignment research.

"The Project" (U.S. Government Intervention): Given the decisive military and economic advantage of superintelligence, private startups cannot handle it alone. By 2027 or 2028, the U.S. government is expected to initiate a nationalized AGI effort—referred to as "The Project"—to consolidate resources, secure model weights, and ensure the free world prevails over authoritarian powers.

Geopolitical Stakes: Superintelligence will provide a military edge comparable to nuclear weapons. A lead of even a few months could be decisive, making a healthy lead for democratic allies essential for safety and global stability.



Appendix 1

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Global events matter, but don't, but do

July 30, 2026

Executive summary

There is a common assumption that markets look through global events such as war and geopolitics. But the effects of these and other events extend far more broadly into investor portfolios than just the headline indices. Our latest data indicates just how much most people oversimplify the reaction of investors and markets to global events.

Indeed, we find that investors do not look through global events. On the contrary, they are highly polarised about them. On our analysis, the vast majority of investors think global events will either be a “risk” or an “opportunity” for markets. That is very different from almost all other broad themes where a higher proportion of people simply think there is “no effect”.

Our dbDataInsights team asked 500 people from the US and UK if and why they think global events affect their investment portfolios and what they are doing about it. We then used AI to analyse all this unstructured data. In this piece, we compare the results to specific datapoints that indicate investors are actively reshaping their portfolios to either capture opportunities or hedge the risks that they see in global events.

In the end, we see that positive and negative views on global events can coexist. The negative respondents focus on the immediate effects: oil prices rise, inflation rises, markets fall. The positive respondents focus on longer-term effects: markets adapt, companies innovate, energy systems diversify, and portfolios positioned in the right sectors benefit. Both views may be internally coherent; they simply operate on different time horizons.

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Why global events are making a comeback

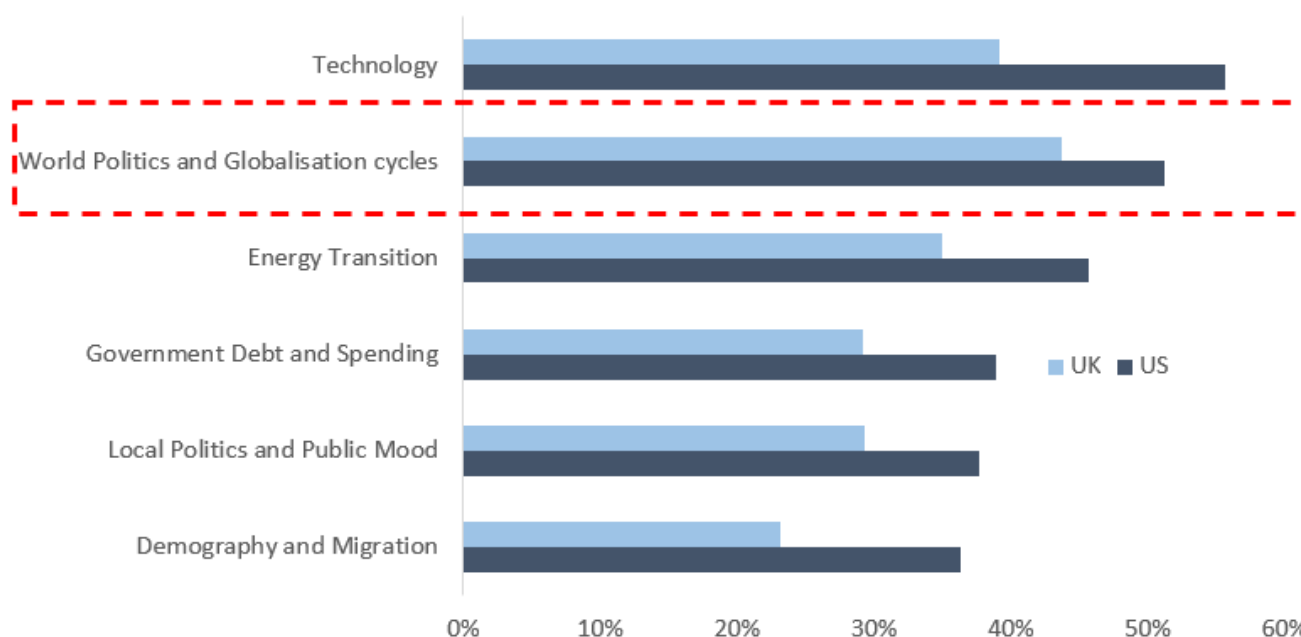
Global events are more than just war and geopolitics. They extend to incidents that affect energy prices, inflation, fiscal sustainability, and technology. Indeed, these forces are some of the strongest that affect global economies and we discuss them in detail in our piece on [global megatrends](#).

Our thesis is that, for much of the post-financial-crisis period, low interest rates drowned out the other effects of global events. Low rates raised valuations, pushed investors into risk assets, and made it easy to explain market moves through central bank policy. Now that rates have normalised, global events are more visible as market drivers again.

Look through or not, investors are changing their portfolios

Our data indicates that investors are reshaping their portfolios as a result of global events. These investors view them not as merely one-off shocks but things that are becoming persistent forces.

Figure 1: Over the last three years, have the following thematic megatrends resulted in you making changes to your investment portfolio?



Source: Deutsche Bank dbDataInsights

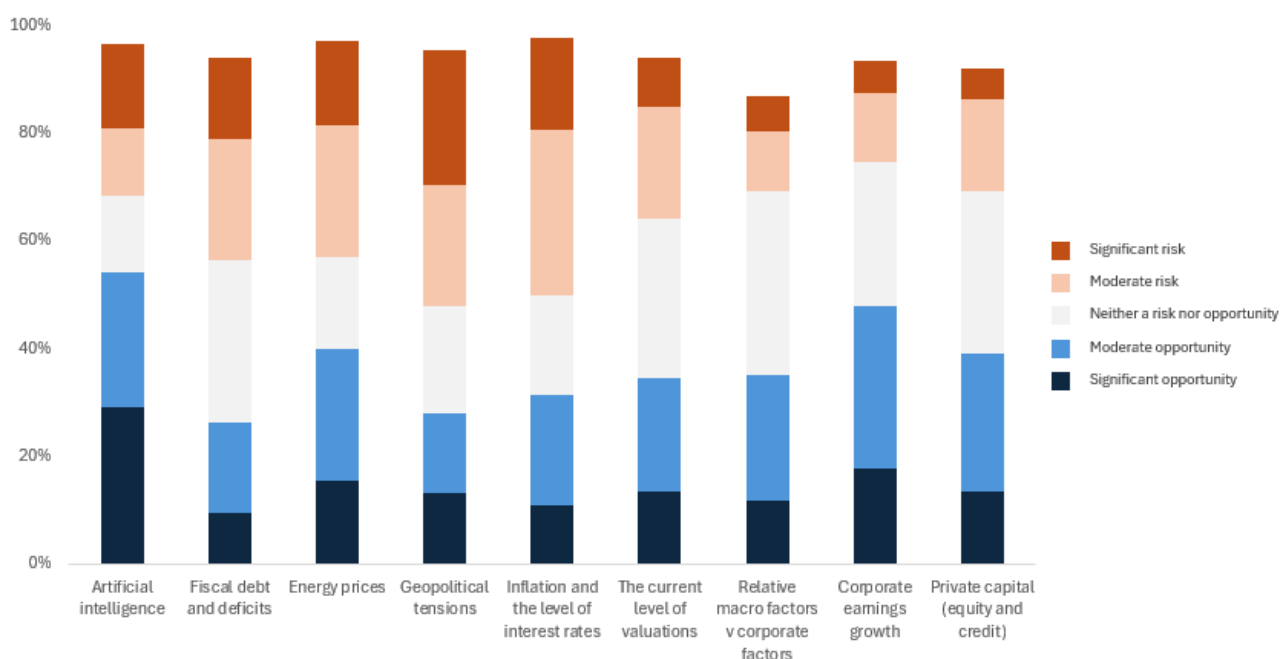
The investor base is therefore not split between people who “understand” global events and people who do not. Rather, it is split between investors who emphasise different time horizons and different asset exposures. A broad equity investor may fear a Middle East shock because it threatens inflation and sentiment. An energy, gold, or defence investor may welcome the same shock because their holdings could benefit. A long-term diversified investor may view the event as noise.



Investors are highly polarised about global events

Retail investors are not passive observers as many may believe. The following table shows that our surveyed investors have strong beliefs both ways about global events and their impact on portfolios. Most importantly, when we look at geopolitics, energy prices, and inflation, the proportion of people who think there is no effect is very low compared with the other themes.

Figure 2: For each investment theme, is it a risk or an opportunity for investors in 2026? (US)



Source: Deutsche Bank dbDataInsights

Aside from the shock effect, it is understandable that the same global event can produce very different portfolio expectations. A war in the Middle East can be viewed as a risk because it raises oil prices, hurts consumers, and depresses equities. But it can also be viewed as an opportunity if an investor owns energy stocks, gold, defence shares, or believes that markets will rebound after the shock. Indeed, in our survey, several positive-impact respondents explicitly mentioned these assets and actions.

Energy is a good example. Short-term energy disruption is painful for consumers and companies. Yet, energy disruption can have positive medium-term effects if it pushes economies toward greater energy security, diversification of supply, and lower energy intensity. Indeed, since 1990 the amount of GDP generated per kilogram of oil equivalent has quadrupled in terms of purchasing power parity. In other words, economies have become much more energy efficient over time which helps explain why the current closure of the Strait of Hormuz has not yet had the disastrous effects of the oil shortages during the 1970s.



This is consistent with our megatrend framework which shows geopolitics and globalisation do not have simple linear impacts on economies. Trade in goods can be reorganised through supply chains, and trade in ideas can continue to accelerate through connectivity. In other words, geopolitical deterioration is not always a pure negative for markets. It changes the winners and losers or increases dispersion.

The most important topics

The following table shows the top 3 themes that were observed in our survey of why people think global events are either positive, negative, or neutral for their portfolios.

Figure 3: Top 3 types of global events that matter to people

Global events that make people more positive, negative, or neutral about their portfolio			
US	Negative	War and geopolitical conflict, especially Iran, Middle East, Ukraine/Russia and escalation risk	28%
US	Negative	Inflation and cost-of-living pressure reducing returns and company earnings	14%
US	Negative	Oil, gas and energy price shocks feeding through to markets	13%
UK	Negative	War and international conflict, especially Iran, Middle East, Ukraine/Russia and wider unrest	26%
UK	Negative	Inflation, rising prices and higher living or business costs	20%
UK	Negative	Energy, oil and fuel price shocks, including Middle East supply disruption	17%
US	No impact	Diversification across assets or holdings expected to absorb shocks	25%
US	No impact	Long-term investing, with short-term events expected to smooth out over time	18%
US	No impact	Low-risk, fixed income, cash-like or insulated investments	18%
UK	No impact	Long-term investing, with market ups and downs expected to smooth out	25%
UK	No impact	Uncertainty, lack of knowledge or unclear rationale	22%
UK	No impact	Diversification or resilient portfolio construction	18%
US	Positive	Global events creating growth opportunities, new markets or higher returns	20%
US	Positive	Technology, AI, digital transformation, healthcare, renewable energy or innovation	16%
US	Positive	Economic recovery, stabilisation and improving market confidence	15%
UK	Positive	Economic recovery, stronger markets and improving investor confidence	20%
UK	Positive	General optimism or confidence that things will improve	16%
UK	Positive	Global events creating investment opportunities, new markets or growth sectors	15%

Source: Deutsche Bank Research

Energy prices are the everyday face of geopolitics

Energy prices are the mechanism through which many investors experience global events most directly. In the written responses, distant conflicts become financially relevant because they raise oil, gas, petrol, utilities, shipping costs, food prices, and inflation. This is especially visible in the UK negative responses, where energy costs and inflation appear repeatedly.

Energy prices are viewed as a risk by 40% of the US investors surveyed and 55% of the UK investors. They are also viewed as an opportunity by 40% of the US investors and 29% of UK the investors. The polarisation here is obvious, and the country difference matters. The US has a large domestic energy sector, meaning some investors may see higher energy prices as positive for certain holdings. The UK investor base, by contrast, appears more likely to interpret higher energy prices through the household cost-of-living and company-cost channels.



Why some investors think global events do not matter either way

These investors are not necessarily saying that global events do not matter for the world. They are saying global events do not matter enough for their portfolios.

The main reasons both the US and UK investors in our survey cite are diversification, long-term investing, defensive holdings, local exposure, and financial discipline. They also frequently think that portfolio outcomes are driven more by company fundamentals, dividends, valuations, financial advice, local exposure, or long-term compounding, than geopolitical events. In the written responses, this often appears as “markets go up and down,” “over time they smooth out,” “my portfolio is diversified,” or “I invest for the long term.”

There is also a behavioural component. Some investors may choose “no impact” because they do not feel confident predicting global events. Several responses in the ‘no impact’ columns are uncertain, vague, or non-committal. This does not mean respondents believe global events are irrelevant. It may mean they do not know how to connect those events to their specific investments.

The “no impact” camp is therefore divided into two groups. The first is strategically insulated: diversified, long-term, defensive, or locally exposed. The second is cognitively uncertain: unsure, disengaged, or unable to explain the link.

Why some investors think global events will hurt their portfolios

The ‘negative impact’ responses are dominated by concerns about geopolitics. In the US, the most important theme is war and geopolitical conflict, particularly Iran, the Middle East, Ukraine/Russia, and escalation risk. In the UK, negative responses are similar but are even more explicitly tied to the inflationary consequences of global events. They frequently mention energy prices, oil, gas, the cost of living, and the way foreign conflicts can feed into domestic household and business costs.

When investors are negative about global events, their written responses suggest that many see a clear macro chain:

1. Global conflict disrupts energy, trade, or confidence.
2. Energy and commodity prices rise.
3. Inflation and interest-rate pressure increase.
4. Company margins and consumer spending weaken.
5. Stock markets fall and portfolio values decline.

This chain appears again and again in the responses, even when written informally. Investors may not use terms such as “supply shock,” “margin compression,” or “risk premium,” but they are effectively describing those concepts.

The UK-US difference is notable. Our surveyed US investors appear somewhat more willing to see opportunities within global volatility, while the UK investors are more likely to view the same themes through the lens of inflation, energy



dependence, and broader economic vulnerability. That may explain why “World Politics and Globalisation cycles” is the highest-ranked cause of portfolio change among UK respondents.

Political uncertainty also has a strong presence in the written responses. In the US, this focusses on President Trump, tariffs, foreign policy, and the perceived unpredictability in government. In the UK, people also mention Trump and US policy, but usually as part of a wider concern about global instability. This implies that investors do not separate domestic politics from geopolitics as cleanly as analysts often do. Leadership, war, tariffs, inflation, and markets are therefore perceived as part of one connected risk cluster.

Why some investors think global events will help their portfolios

The positive camp contains three broad types of investors. First, the optimists who believe technology and growth will dominate. Second, the sector opportunists who own assets that benefit from disruption. Third, the tactical buyers who treat volatility as a chance to buy assets cheaply.

Notably, many investors say they apply a clear investment logic. Essentially, ‘disruption creates opportunity’. They cite the effect of global events on an economic recovery, technology, AI, energy transition, lower prices after market sell-offs, defence, oil, gold, emerging markets, and international trade recovery.

The most important positive theme in the US responses is growth opportunity: global events can open new markets, increase demand, create new investment opportunities, and raise returns. Technology and AI are also central. Many respondents expect AI, digital transformation, health care, renewable energy, and innovation-led sectors to do well.

This US-UK divide appears again in the written responses. US positive respondents are frequently enthusiastic about AI, tech, innovation, market growth, and sector opportunities. UK respondents are also positive on technology and innovation, but their positive responses more often include general optimism, economic recovery, and rebounds after conflict.

There is also an opportunistic trading mindset. Some respondents say market dips allow them to buy, or that volatility creates opportunities.

Indirect effects that are implicit in investor logic

The most concerning megatrend in the world today is fiscal debt and deficits – and it is an important part of the broader story about global events. Fiscal issues are less prominent in the written responses from investors compared with war, oil, inflation, and President Trump. Yet the implicit themes run through many identified factors. Indeed, 38% of US and UK respondents view fiscal debt and deficits as a risk. At the same time, around a quarter of people in both countries view them as an opportunity.

So, although investors talk frequently about inflation, rates, taxes, government borrowing, and weaker growth, they do not always connect those concerns explicitly to sovereign deficits. But fiscal anxiety appears indirectly embedded in worries about inflation, interest rates, debt, taxes, and government policy.



Government debt and spending has already prompted portfolio changes by 39% of the US investors and 29% of the UK investors over the last three years. That is lower than technology, world politics, and energy transition, but still significant. It suggests that fiscal concerns are not absent, they are simply less emotionally vivid than war or energy prices.

Thus, investors who are positive on AI may be implicitly assuming that technology can overpower fiscal and geopolitical headwinds. Investors who are negative on global events may be implicitly assuming that debt, conflict, inflation, and policy instability will overpower technological progress. The debate is not simply about whether global events matter. It is about which global force dominates.

Conclusion: The polarisation of opinion on global events means we should brace for volatility and dispersion

In the end, the surveyed investors overwhelmingly think global events matter because they reach into their portfolio through inflation, oil, politics, growth, and technology. Investors think the world does not matter when they believe their portfolio construction, time horizon, or asset selection can neutralise those forces.

The investor base is therefore not split between people who “understand” global events and people who do not. Rather, it is split between investors who emphasise different time horizons and different asset exposures.

Cultural differences are important. The surveyed US investors are most likely to have changed portfolios because of technology, followed by world politics, while the UK investors are most likely to have changed portfolios because of world politics and globalisation cycles, and then technology.

This suggests that US investors are more likely to frame the future around innovation and technology while UK investors see investing more through the lens of global political cycles.

In investment terms, this means global events should be understood less as a single directional call on equities and more as a source of dispersion. In other words, they just change the relative winners and losers.



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What a "Strong and Rich Japan" could mean

July 24, 2026

Executive summary

- Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. Japan's new economic blueprint seeks to put the country on a path of investment-driven growth to meet the challenges of a new era.
- Japan is not alone in its bid to prioritize industry and rearm, but given a higher starting point for government debt, it will have to find room to spend while maintaining fiscal sustainability. This will inform Japan's approach in very significant ways.
- First, Japan is likely to remain focused on supporting nominal GDP growth, aiming to keep it above government funding costs to create deficit space. Second, Japan will look to actively mobilize domestic savings to fund investments and manage yield pressure. Each of these shifts could have implications for the currency.
- Japan's excess savings have arguably been kept in two places: in cash and abroad. The government is looking to get both moving. Households have around 50% of their \$15tn of savings in cash and deposits. Meanwhile, GPIF has 50% of its \$1.8tn portfolio abroad. If industry, energy, defence need to come home, so might the capital to fund them.
- Over the past few months, Japanese policy makers have been very focused on stabilizing the currency. However, if fiscal capacity is becoming the most important policy criteria, incentives could shift from FX to yield management.
- The ultimate impact on USD/JPY will depend on the levers the government chooses to pull to manage yields. If GPIF is mandated to bring money back into domestic assets, this could be very bullish for the JPY. However, if the BoJ is coopted to support bonds through renewed JGB purchases, this could be very negative for the JPY.
- Suppressing yield volatility could come alongside bigger moves in FX from here, depending on the approach the government takes. This favours higher JPY vol.

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Echoes of Meiji

Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. In 1868, pre-modern Japan was staring at the threat of foreign involvement and colonization, having seen the Opium Wars unfold in China. Japan had signed unequal trade treaties that opened Japanese ports to free trade imperialism, prevented Japan from setting tariffs, and exempted foreigners from local laws. Japan's response changed history: under the Meiji Restoration, Japan returned rule to the emperor, strengthened the state, and pursued extremely rapid industrialization.

One could argue that Japan again faces existential challenges given fractures in globalization, intensifying great power competition, insecurity around energy and critical minerals, the need to participate in AI innovation, and external dependence for defence. Japan's answer under PM Takaichi: to return to its roots as an industrial nation driven by state and fiscal policy. The slogan of the 1868 Meiji Restoration was "Fukoku Kyohei" which translates into "Enrich the Country, Strengthen the Armed Forces" or more simply: rich country, strong army. It is perhaps telling that Takaichi's government is introducing an economic blueprint that calls for a "Strong and Rich Japan". ([Nikkei Asia](#)).

Invest, invest, invest

Japan's Cabinet has now passed the [Honebuto](#), or the government's new Basic Policy on Economic and Fiscal Management. The document calls for moving beyond "conventional thinking", "putting an end to the trend of underinvestment", and "responding to the era of intense global industrial policy competition". The cornerstone of Japan's proposed policy shift seems to be Y370trillion of public-private partnership investments with the government having identified 17 key strategic areas from AI and quantum, to defence, aviation, shipbuilding, critical minerals, and others. These investments are seen as crucial to boosting potential growth. Indeed, the decline in Japan's potential growth rate has not principally been about demographics as is commonly assumed, but about the collapse in the capital stock ([Figure 1](#)). Japan went through decades of very low investment after the 1990 asset bubble burst. It now wants to return to being an investment-driven economy.

Finding the funding: two big shifts

Japan is not alone in its bid to reindustrialize and rearm, but compared to peers, Japan arguably has less fiscal space. Japan's debt to GDP hovers above 200%, while Germany and Korea are under 50%, making it easier for them to pursue significant expansion in their fiscal deficits. While Germany is moving from austerity to expansion, Japan will have to find spending room while continuing to chart a fiscally sustainable path. This will hugely inform Japan's policy approach. There are two important shifts Japan appears to be making to enable fiscal spending.

First, changing the fiscal target. The Honebuto has confirmed a shift in Japan's core fiscal target away from the primary balance. The primary balance will now no longer be a mechanical target; it will be managed over many years, and be allowed to deteriorate temporarily as required. The new fiscal target is now the "stable reduction of the total debt-to-GDP ratio for national and local governments." The second notable and enabling shift is the pursuit of an "asset management nation" strategy. Japan is looking to more actively mobilize domestic savings to fund

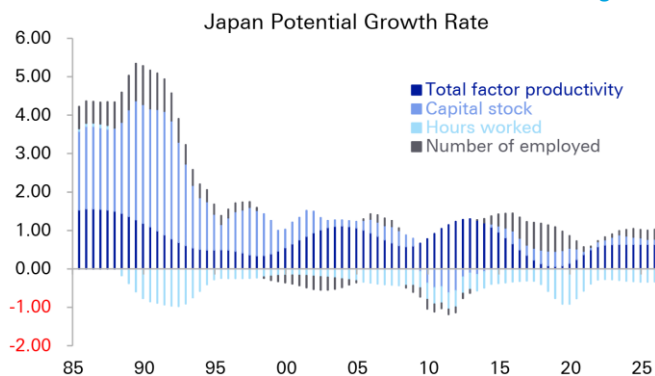


investments and help keep government financing costs manageable. Each of these shifts could have significant long-term implications for the currency.

The new fiscal approach means keeping r below g

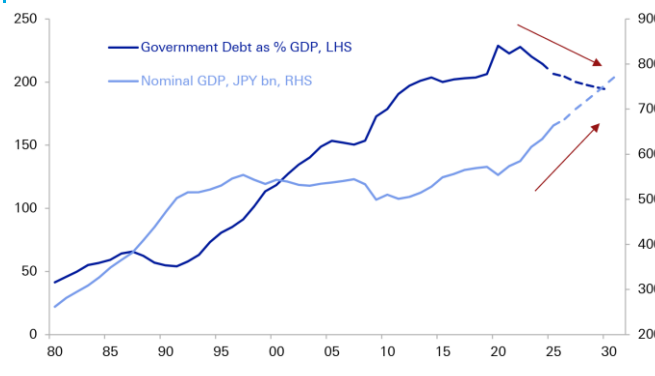
We think debt sustainability arithmetic has become key to understanding Japan's economic policy. Recall that: $\Delta d_t \approx (r-g)d_{t-1} + pd_t$ where d is the debt-to-GDP ratio, pd is the annual primary deficit, g is the nominal GDP growth rate, and r is the average interest rate paid on outstanding debt. The formula implies that Japan can run primary deficits and still achieve a declining debt-to-GDP ratio, as long as r is below g . In other words, Japan's cost of funding needs to be below nominal GDP growth for Japan to have fiscal deficit space. One might consider Abenomics' bid to return Japan to inflation as crucial to putting it on a path to fiscal health. As Figure 2 illustrates, the peaking in Japan's debt-GDP ratios has gone hand-in-hand with the revival in Japan's nominal GDP.

Figure 1: Japanese potential growth has declined on decades of underinvestment – this could now change



Source: Deutsche Bank Research, Bank of Japan

Figure 2: Rising nominal GDP has helped bring down debt-to-GDP



Source: Deutsche Bank Research, Haver Analytics

The extent of fiscal deficit space will crucially depend on the interplay between r and g with Figure 3 illustrating the potential space created by different $(r-g)$ scenarios. We note that the Honebuto outlined a 3% nominal GDP target (g) for Japan till 2040, made up of a 2% inflation target, and 1% real growth target, with aims of increasing it. This perhaps sheds light on why the government has become sensitive to the 3% level on 10Y yields. To be sure, Japan's overall cost of funding remains much lower for now at near 1% given just a fraction of debt is refinanced every year, but there are points of vulnerability. While the weighted average maturity of debt outstanding is around [9 years](#), BoJ holds close to 50% of outstanding JGBs against which they pay interest on excess reserves to banks. This means that almost half the consolidated liability base is in fact linked to front-end floating rates. Therefore, not only are long-run government bond yields above 3% not likely to be sustainable, front-end rate hikes also contribute to consolidated fiscal costs. Very roughly, every 100bps of rate hikes costs ¥5tn or around 0.7% of GDP.

The $(r-g)$ formula not only explains the importance of understanding the drivers of ' r ', but also explains the sensitivity to ' g '. It helps one appreciate why policymakers remain so worried about upsetting the balance on hard-won nominal GDP growth. Indeed, Japan tends to react very cautiously and dovishly



to any signs of growth risks (tariffs, Iran war) and has preferred to let inflation run hot and durable before tightening.

Japanese policy makers face a fine balancing act in optimizing this formula: dovish monetary policy may help to support g , but is also driving up r via inflation expectations reflected in long-run bond yields. Conversely, hawkish monetary policy could hurt ' g ' and also raise the cost of ' r ' via reserve payments. In light of this, the focus on trying to manage long-end yields makes more sense.

How might Japan do this? Changes to supply-side of debt management have been tried via reduced long-end issuance; the government may now be trying to more actively influence demand.

An Asset Management Nation

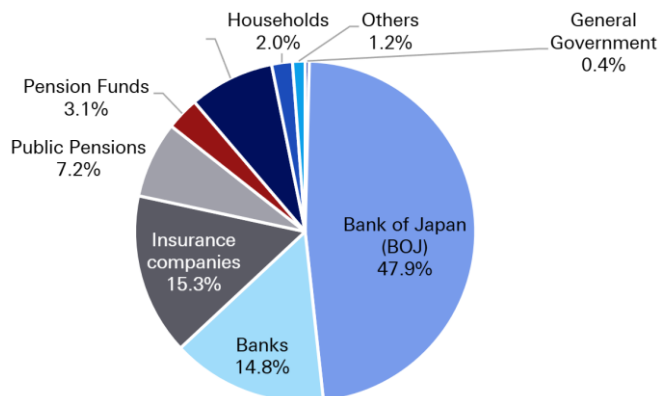
Very simply: Japan needs to find more buyers of JGBs (Figure 4). It is natural to look at where its own domestic capital has been invested. The answer points to two big places: in cash and overseas. Japan has close to USD5tn in portfolio assets held abroad across central bank, household, pension fund, insurance, and bank balance sheets, more than the size of GDP. Indeed, our research has long shown that one of the best metrics of fiscal space is a country's external position, and Japan may need to draw on it. Intuitively, the bid for strategic autonomy around the world should coincide with a growing "nationalism of capital." If industry, energy, defence all need to come home, so might the capital to fund it. We have explored this in earlier research [here](#). More of Japan's money could now be needed at home. Finance Minister Katayama has been explicit that Japan's ambitions to create a growth-oriented economy will coincide with a "strong stock market and positive interest rates" in which domestic citizens should reap benefits. The other perspective is they will help provide the money.

Figure 3: Keep g above r is key to Japan's fiscal space. Japan does not want to return to the lost decades

Scenario	Nominal GDP	Nominal Rates	Debt Stabilizing Primary Balance	Scenario Description
1	4.5%	1.75%	-5.5%	2025 Regime
2	3.0%	2.0%	-2.2%	Japan manages rise in yields to keep them below g
3	0.0%	1.5%	1.5%	Return to Lost Decades (1995-2010)

Source: Deutsche Bank Research, Bloomberg Finance LP

Figure 4: Japan will need to find more buyers of JGBs



Source: Deutsche Bank Research, Ministry of Finance

Where the money could come from

Pension fund and household assets appear to be the two big areas of focus for markets. The government pension fund, GPIF, currently has 50% of its \$1.8tn [portfolio](#) abroad. Meanwhile, households have around 50% of their \$15tn (!) of savings in cash and deposits (Figure 5).



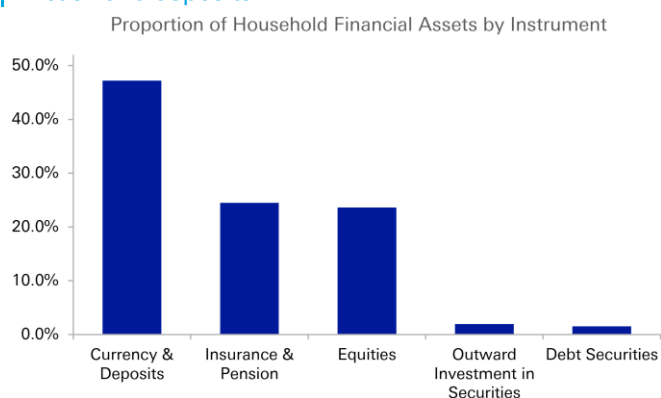
Movement of household deposits into JGBs could be encouraged through a few channels. JGBs may be brought under the NISA tax-exempt investment scheme; the retail JGB offering may be expanded or made more attractive. There were also references in the Honebuto to advancing on-chain finance, with the power of tokenization potentially being tapped to encourage financialization of savings. Mobilizing household savings could be material for yield management given their sheer size. For context, the 1000tn yen of household deposits are almost the size of the entire JGB market. It is not however likely to have any FX impact as it would mostly be an internal domestic flow.

In contrast, a shift from GPIF to invest more in domestic assets could be very material for FX, with PM Takaichi noting this is a key initiative. GPIF's [current policy mix](#) prescribes 25% investment in domestic equities and debt respectively, with +/-6% bands around it. If GPIF simply shifted to the top end of current permissible ranges from current levels (as of March 2026), this could bring around \$200bn back to Japan across domestic equities (\$130bn) and domestic bonds (\$75bn).

The bigger shift to play for however is an early policy review of GPIF's overall asset mix that revises up the domestic asset targets themselves. In [Figure 6](#) we illustrate scope for inflows at different levels of target allocation to domestic bonds alone. Hypothetically, if the target share of domestic bonds doubled to 50%, over USD400bn could flow into JGBs from GPIF. While this may seem extreme before Abenomics in 2012, GPIF had a 67% target allocation to domestic bonds in their policy mix. One could reasonably expect any flows to occur with a glide path, but the scale alone could be significant enough to impact the currency.

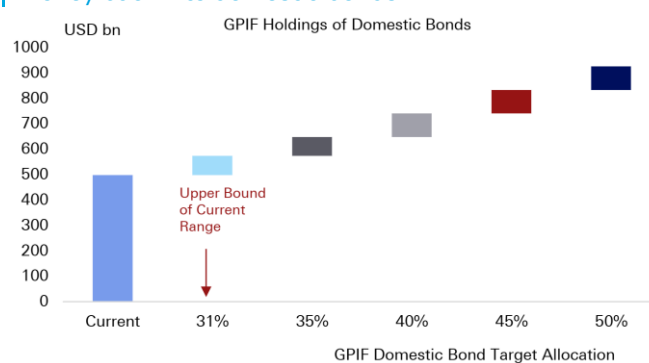
We note that GPIF has a current return objective of a 1.9% real return above nominal wage growth. The extent of any domestic reallocation will depend on how much Japan's domestic assets can deliver that return, or whether government pension returns are made subservient to financing costs for now. Another group that may be called upon to play a bond absorption role could be local banks. This is a source of financial repression has been used extensively around the world and could return where deficit financing demands it.

Figure 5: Japanese households hold half their savings in cash and deposits



Source: Deutsche Bank Research, Haver Analytics

Figure 6: GPIF could be encouraged to bring more money back into domestic bonds



Source: Deutsche Bank Research, GPIF



A shift to yield management could lead to bigger FX moves

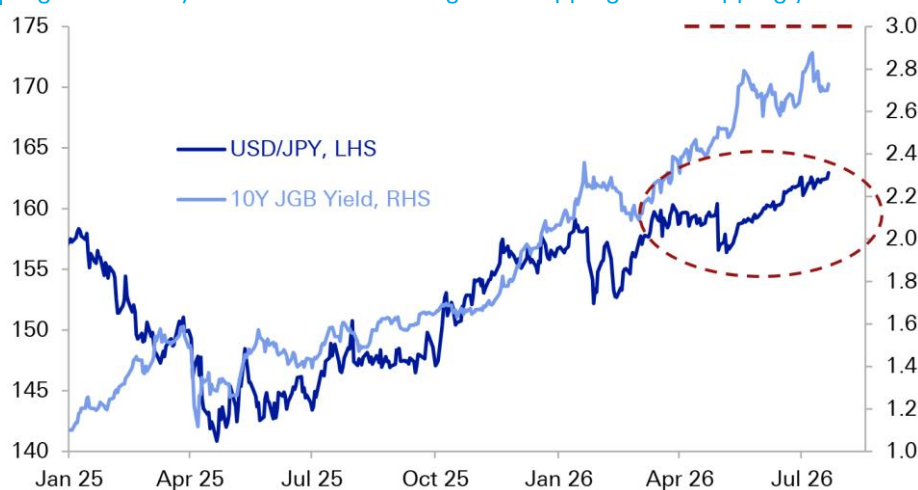
Over the past few months, Japanese policy makers have arguably been solving for stabilizing the currency. Household angst and imported inflation risks have likely encouraged this. Policymakers have engaged in verbal, rate check and actual intervention this year. And, Japan has in fact been successful in keeping USD/JPY near 160 amid the US-Iran war, a major global energy shock, and volatility in Fed pricing.

If, however, fiscal capacity is becoming the biggest and more existential policy criteria for Japan, then incentives may be shifting from FX management to yield management: from capping USD/JPY to capping 10Y yields and borrowing costs.

Suppressing interest rate volatility may come at the cost of amplifying it in FX, depending on the specific levers that the government chooses to pull. If GPIF is mandated to bring a larger share of their assets back into domestic assets, this would be very bullish for the JPY. We have previously identified GPIF as the biggest weapon in Japan's external arsenal. Equally, if BoJ is encouraged to accelerate tightening to manage inflation risk premium in long-end yields this too could be bullish JPY. If, however, the BoJ is [coopted](#) to support yield management through a more active role in the bond market, potentially resuming JGB purchases, this would be negative for the JPY. Similarly, if BoJ is encouraged to keep monetary policy dovish to prioritize (r-g) considerations this too would be bearish JPY.

Ultimately, the direction of travel for USD/JPY will depend on which measures the government favours. Clearer and more imminent signs of GPIF repatriation would favour the stronger yen tail, while pressure on BoJ to manage rates and yields again could open the weaker tail. Efforts to suppress yield volatility could come alongside bigger moves in FX from here. This favours higher JPY vols. We note USD/JPY vols are near multi-year lows with a 1Y USD/JPY straddle breaking even roughly below 150 or around 170.

Figure 7: Policy focus could be shifting from capping FX to capping yields



Source: Deutsche Bank Research, Bloomberg Finance LP



Appendix 1

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Time billionaires want more risk. Is AI to blame?

July 22, 2026

Executive summary

Are you a time billionaire? Some say “yes” if they expect to have more than 1bn seconds (or 31.5 years) in front of them. And as the number of younger investors increases, and their time horizons span 30, 40, or 50 years, they want two things: 1) More risk than they have today, 2) AI to manage their money and trade for them.

So, is AI convincing investors, particularly young people, to increase the current level of their investment risk? In this report, we use proprietary data from our dbDataInsights team to break down exactly who wants more investment risk and over what timeframe. We highlight that both Americans and Britons under the age of 55 are particularly keen to increase (relative to today) both their risk tolerance and their dependence on AI to manage their portfolio. The younger the age group, the more they say they will increase their risk over the coming 1, 3, and 5 years.

We then discuss various factors that are contributing to the increased desire for both risk and AI advice. Behavioural psychology feeds directly into the equation, particularly as ‘social’ investing has become a trend. ‘Aspiration inflation’ is also very tangible but we point out that young investors (and their risk appetite) are the product of the environment that has been created for them. In the end, there are pros and cons.

Several conflicts and conclusions come out of our work. The first is that it is intriguing that consumer sentiment is so low at the same time that the average person wants to take on more investment risk. A second is that this desire for risk at a household level comes as experts increasingly point to intensifying global systemic risks. A third is that the willingness to turn to AI for investment advice comes as the social winds against AI more broadly are growing in strength.

Authors

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Research Analyst

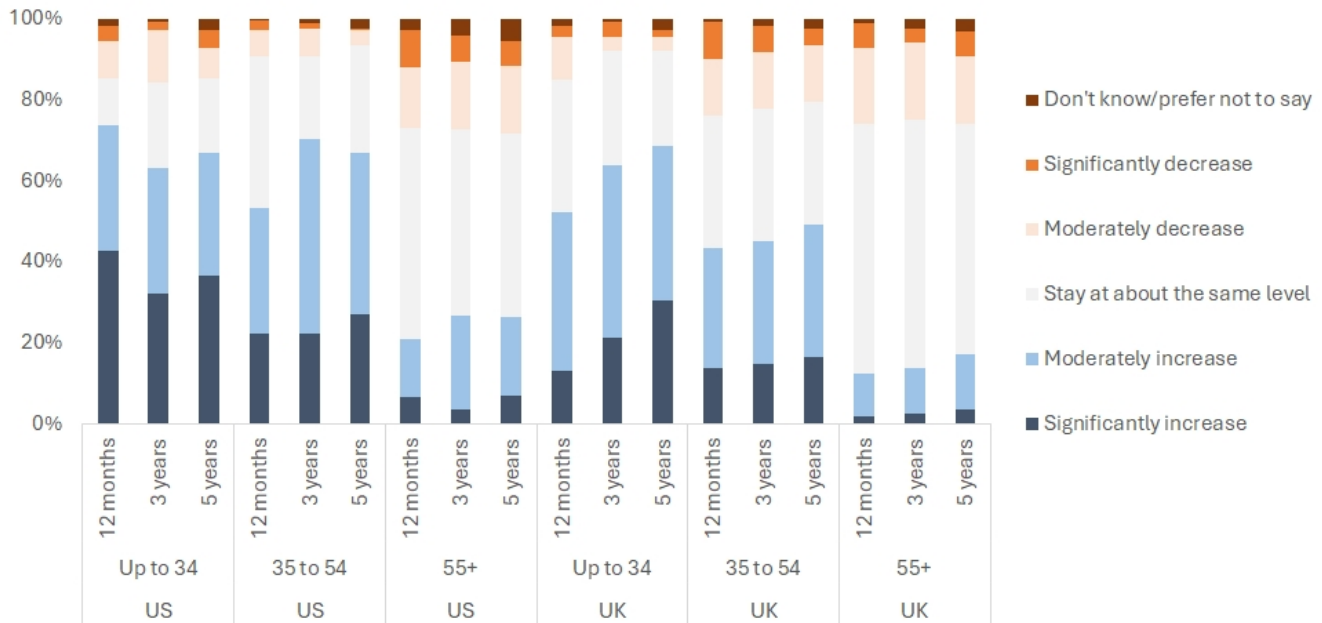
Galina Pozdnyakova
Research Analyst



An appetite for risk

At one level, it seems strange that six years into a decade that has been so volatile, people want to take more, not less, risk with their investments. Indeed, the following chart shows that all age groups (up to the 55+ group) want to substantially increase their investment risk over a 1-, 3-, and 5-year time frame.

Figure 1: Over each of the following timeframes, do you expect your level of risk-taking in relation to your main investment portfolio to...



Source : dbDataInsights

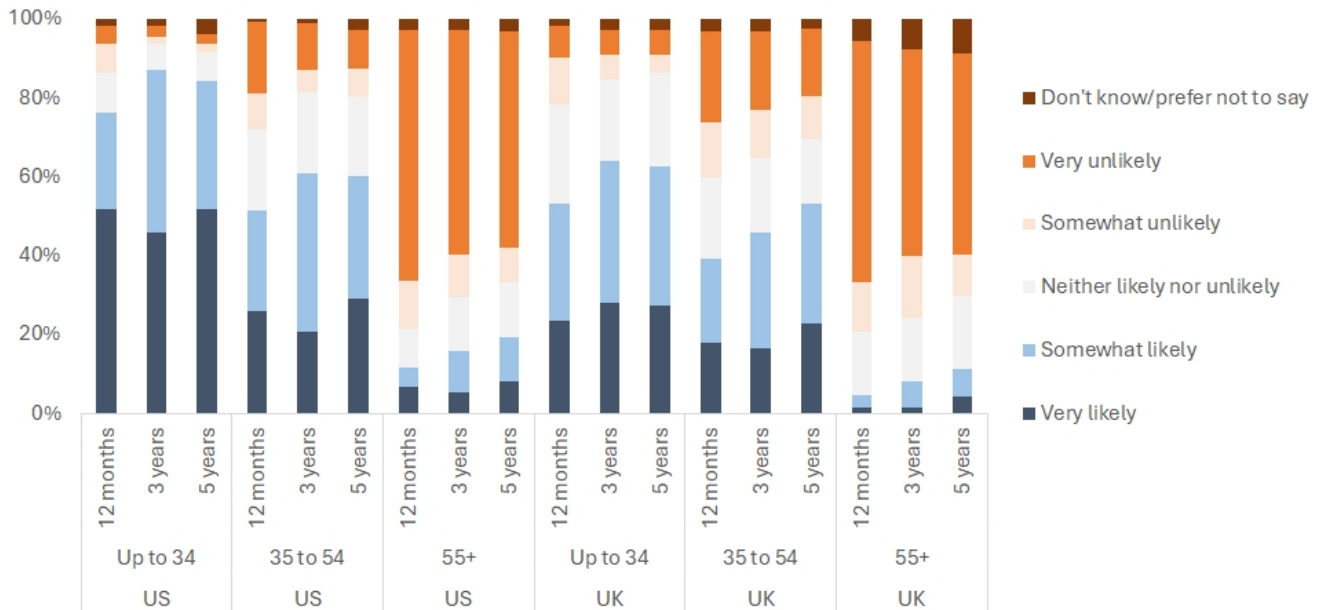
A stark observation is the age gradient. In short, the younger you are, the more likely you are to want to take even more investment risk in the future. The effect is most pronounced in the US but the trend is the same in the UK.

Is AI encouraging risk?

People are not only using AI to find investment advice, but also their shift towards it is accelerating. As the following chart shows, people under the age of 55 expect to show a strong desire to move to an AI managed investment method over the next 1, 3, and 5 years. This phenomenon is particularly aggressive in the US but still pronounced in the UK.



Figure 2: Within the following timeframes, what is the likelihood you will switch from your current investment management method (self-managed or human advisor) to a Generative AI-managed solution?



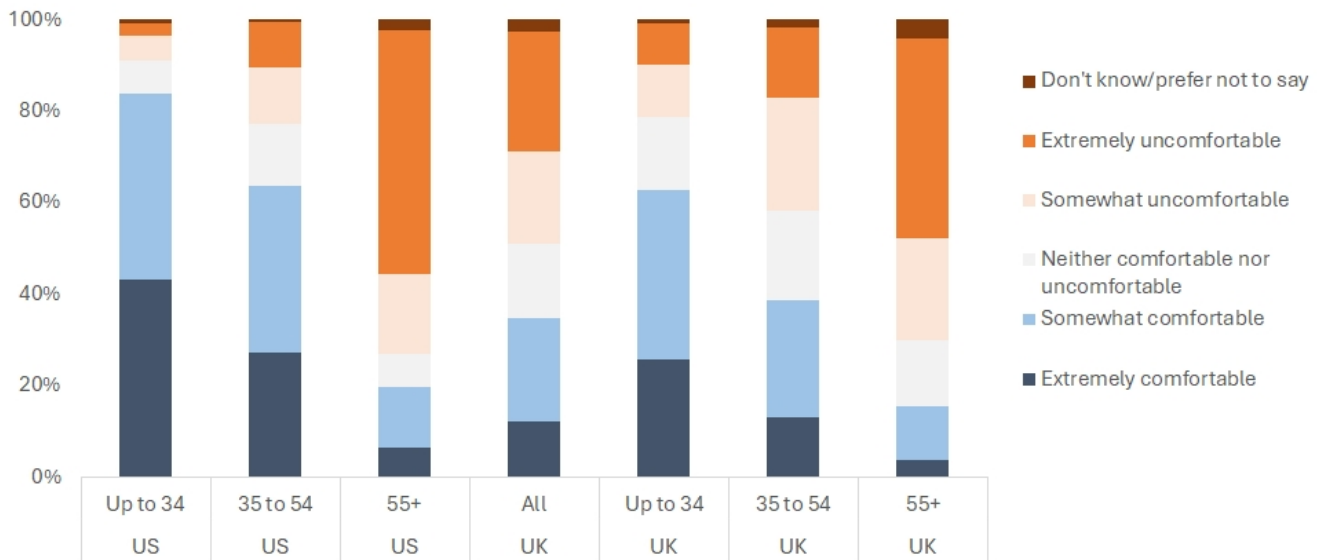
Source : dbDataInsights

Will AI trade for you?

Regardless of how AI is or is not encouraging risk, people are certainly increasingly comfortable with AI automatically making trades in their investment account. For example, Americans up to 34 years old are happy giving AI responsibility for automatically executing trades on their account by a ratio of 10:1. In the UK, the same age group is happy with AI by a ratio of 3:1. The ratio falls as people age.



Figure 3: How comfortable would you be with a generative AI automatically executing investment trades on your behalf, if you had set the parameters (e.g., buy/sell limits, asset allocation)?



Source : dbDataInsights

Does AI give confidence to time billionaires?

AI may amplify risk appetite, particularly for young people, because it reduces the friction around decision-making. Everyday investors can now ask an AI tool to explain asset allocation, compare funds, summarise macro risks or model hypothetical scenarios. Even if the output is imperfect, the interaction can create a sense of fluency. That matters because people are more likely to take action when a complex problem appears legible.

Some research supports this direction of travel. The World Economic Forum has previously highlighted that Gen-Z and millennial investors are embracing AI and digital platforms for financial advice and education, and that younger investors place greater importance on intuitive, technology-enabled wealth-management systems. The WEF has noted that 45% of Gen-Z and millennials began investing in early adulthood, compared with 15% of Gen X and baby boomers.

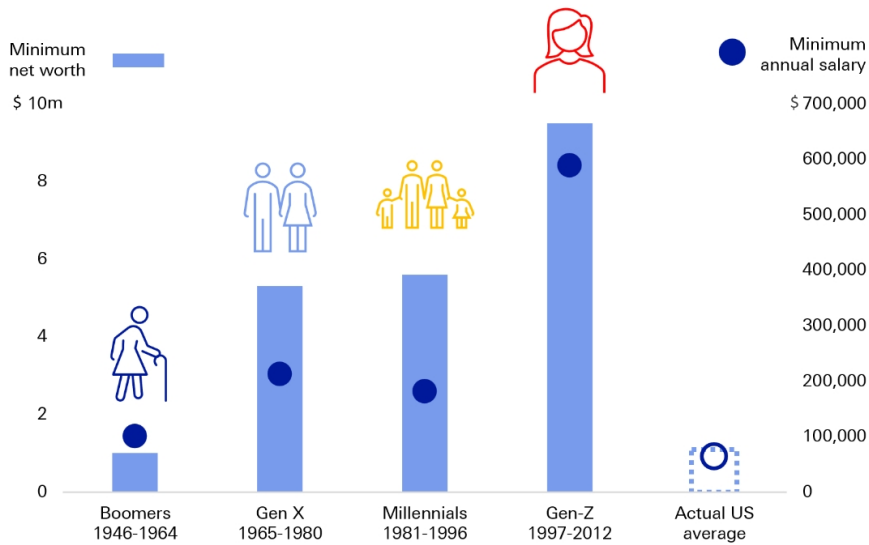
What else is driving greater risk taking?

Aspiration inflation

Some theories suggest the ease of 'aspirational connections' and 'dream scrolling' on social media has fuelled sky-high expectations for Gen-Z. In fact, one survey shows that Gen-Z thinks they need twice as much wealth and triple the annual salary as the previous generation to achieve "financial success".



Figure 4: Survey: "What does financial success look like?"



Source : Empower "Secret to Success" study: [Secret to Success](#) Empower, November 2004, Deutsche Bank. (Note: Average salary as of 2024, average net worth as of 2022)

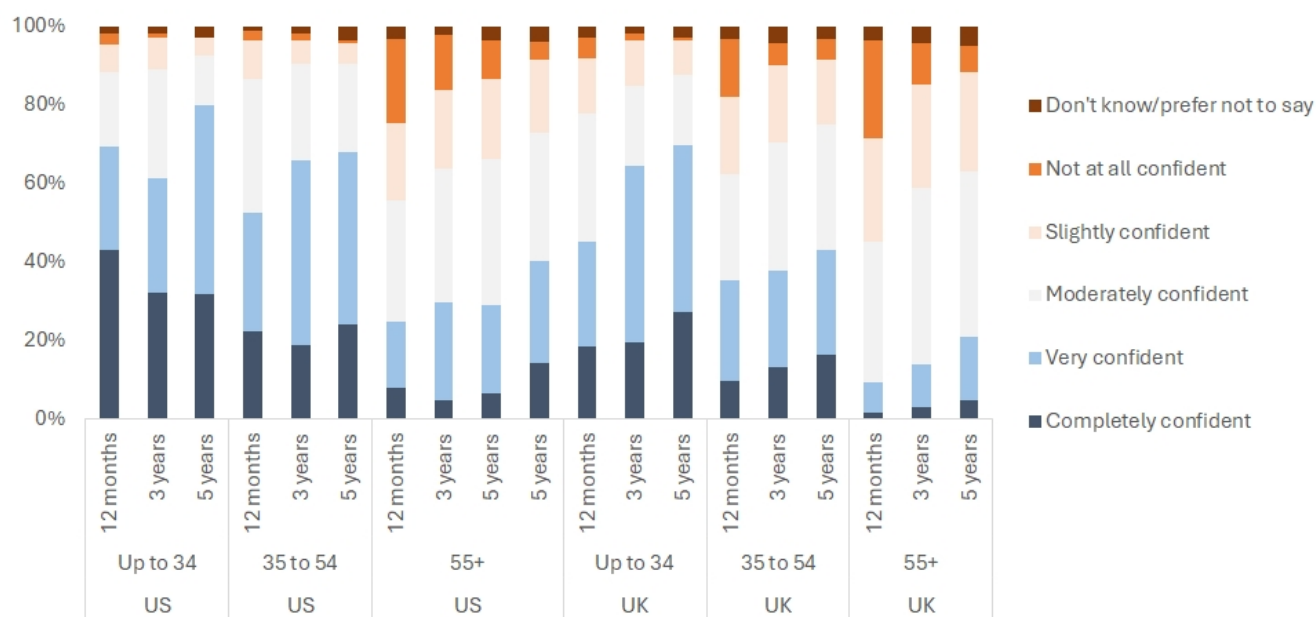
Of course, the flip side of the same coin is that younger people are increasingly worried about housing affordability, education costs, uncertain retirement prospects and perceptions of stagnant real income growth. As a result, this may encourage them to see traditional saving as insufficient. If higher-risk investments can appear less like speculation and more like necessity then risk-taking is not a rejection of prudence, it is an attempt to close an opportunity gap.

Bullish return expectations

The chart below shows that people in all age groups are confident that their investment returns will meet or exceed their expectations. Again, age groups matter. Younger people are overwhelmingly confident of strong returns across 1, 3, and 5 years.



Figure 5: Over the following time frames, how confident are you that your main investment portfolio will meet or exceed your expectations?



Source : dbDataInsights

Rallying against the risk of doing nothing

Portfolio risk appetite is not inversely related to global risk perception. In a risky world, households may feel that the cost of remaining conservative is also rising. If inflation, technological disruption and asset-price dispersion are perceived as threats, risk-taking can become a hedge against being left behind.

Young investors are a product of their environment

Many younger investors have been socialised as they came of age in a market culture shaped by mobile trading, social media, crypto assets, zero-commission brokerage, online financial influencers and rapid cycles of boom-and-bust narratives. Investment decisions are discussed, compared, gamified and sometimes celebrated in digital communities. This can make risk-taking feel less isolated and less exceptional. In turn, this amplifies psychological issues of social proof, herding, and fear of missing out. In the end, the emotional cost of failure is lower.

Anthropological work on risk argues that societies do not merely discover risks; they classify, narrate and prioritise them. Mary Douglas's cultural theory of risk argued that communities highlight different hazards depending on their social organisation, values and forms of trust. What counts as a salient risk is, in part, culturally produced.

Studies also emphasise that risk-taking is often mediated by rituals, expertise and tools that make uncertainty feel manageable. In modern finance, models, apps, dashboards and AI systems can serve a comparable psychological function. They do not eliminate uncertainty, but they make uncertainty appear structured and therefore more actionable.



The increased ease at which people can access risky products raises the idea of 'prospect theory'. This highlights that people become risk-seeking when they perceive themselves to be in the domain of losses. They then can be more likely to gamble to return to their reference point rather than accepting a smaller certain loss. An investor who is underwater relative to a recent peak, or who feels they have 'missed out' relative to peers who bought earlier, is, under this framework, more likely to reach for a higher-variance trade to close that gap.

Other psychologists have highlighted how sensation seeking can correlate with risk-taking in gambling, extreme sports, and financial speculation. It tends to be higher in younger populations — a pattern consistent with our survey findings.

Do consequences feel lower than in the past?

Risk-taking is not simply a function of appetite; it is also a function of perceived downside protection. Behavioural researchers analyse how humans recalibrate their risk-taking in response to changes — real or perceived — in the safety of their environment.

Consider the Peltzman effect. This describes how people can take greater risks if they know they are protected from downside risk. Evidence for this has been argued by researchers in various domains:

- Some drivers were found to be involved in more accidents when their cars had greater safety features
- The introduction of HANS safety devices in Nascar was argued by some to be related to an increase in more aggressive racing
- Skydiving/wingsuit users readily admit that the increase of the quality of equipment has made them push the limits of the sports

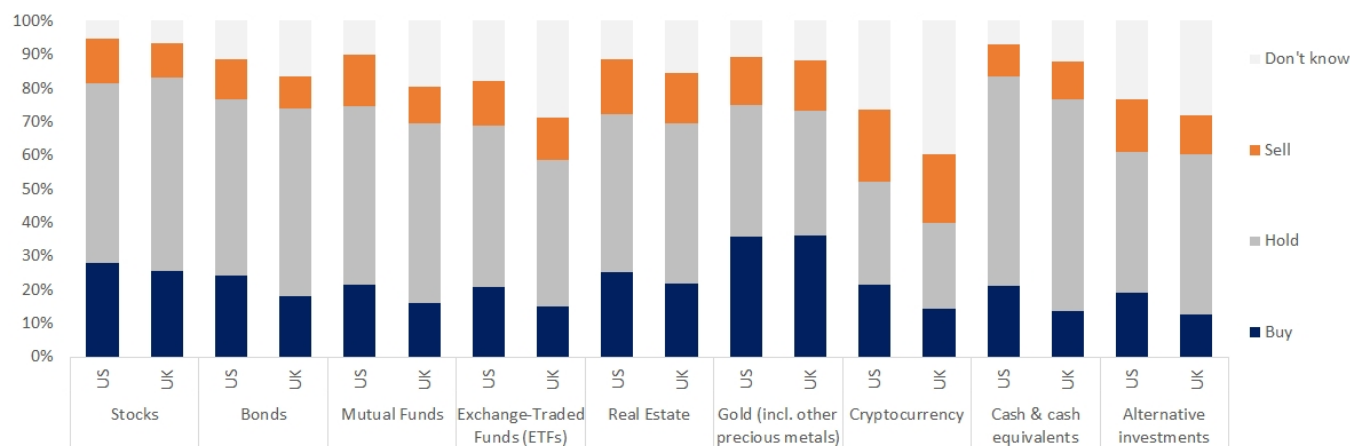
Application to finance and investing

Post-2008, investors have become used to policymakers increasing their response to shock events. At the same time, financial innovation has created the perception in some that risk is either lower or can be managed better. Consider that every major US market decline since the 2008 financial crisis has been followed by a relatively timely recovery to new highs. A generation of investors has now only ever experienced dips as buying opportunities — a very different experience from the multi-year drawdowns and stagnant markets of the 1970s or the 2000s.

Hence, the number of people who say they treat uncertainty as an opportunity to take on traditional risk assets. The following chart shows that people think that going into traditional 'risk' assets is the best strategy during periods of uncertainty. In fact, the average American wants to buy, rather than sell, equities during uncertain periods by a ratio of 2:1. In the UK that ratio rises to 2.5:1.



Figure 6: In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?



Source : dbDataInsights

Of course, investors may agree that the world is risky while also believing that AI, technology stocks, private markets, crypto assets or thematic funds offer the best path to future security.

In addition to general faith in policymakers, several other factors may be encouraging risk:

- Deposit insurance, broker-dealer protections, and post-2008 regulatory reforms have reduced certain tail risks that were live concerns a generation ago, such as a brokerage or bank outright failing with a customer's assets inside it.
- Diversified, low-cost index products have made it structurally harder to lose everything through a single-stock mistake compared with earlier eras of concentrated, high-fee retail investing.
- Information is more abundant and more immediate, which may create a (not necessarily accurate) sense that fewer unknowns remain and that outcomes are more predictable.
- The practical consequences of experimentation have been reduced by fractional shares, low-cost funds, commission-free trading and the ability to start with small sums. A young investor can now test higher-risk assets without the same minimum account sizes or advice costs that once limited participation.

Implications for wealth managers and policymakers

Younger investors are unlikely to be persuaded by generic warnings about risk. As such, AI should be framed as a support mechanism rather than a substitute for suitability, behavioural coaching and long-term planning.

For policymakers, the issue is not whether AI-driven investment engagement is good or bad. It is how to ensure that increased access is matched by adequate disclosure, financial literacy and safeguards against misleading personalisation. If AI makes investing easier, it should also make risk more understandable.



If the current environment of ever greater risk taking continues, there are several key concerns that may exacerbate the downside risks:

- Leverage positioning: US margin debt has pushed to over 4.5% of GDP. That is well above prior peaks in 2021, 2007 and 2000. Retail investors can easily access leverage through brokers, ETFs, or other specialised products.
- Speculation has broadened: New avenues for risk taking include the ease of options trading (particularly same-day and binary options), leveraged ETFs, and meme assets, prediction markets and sports gambling.
- Safety buffers are shrinking: Household savings rates are falling in the US and other key countries



Conclusion

The common thread across not only financial decisions, but also auto safety, extreme sports, medicine, and gambling is that humans do not hold risk-taking constant when their environment changes. They recalibrate around a target level of risk, and that target appears to rise when protective factors (real or perceived) are introduced. Applied to investing, this suggests the relevant question is not just, 'Why are people taking more risk?' but, 'What changed in their perceived environment that permitted it?'

AI may be one driver of this desire for risk but there are others, too. Yet, it is concerning that perceived consequences may have declined faster than the actual potential consequences. AI can widen this gap if it gives users the impression that risks are fully understood or controllable. Our data points to pros and cons: AI may democratise investment engagement, but it may also compress the distance between curiosity and action.

The younger cohorts in our survey may have greater time horizons, but they also face housing, employment and debt pressures that can constrain their ability to absorb volatility. The next generation of advice must therefore be both more digital and more behavioural.



Appendix 1

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El Nino: The next supply shock?

July 20, 2026

Current forecasts point to one of the strongest El Nino events on record in Q4. That matters because El Nino events can act as a multi-dimensional supply shock, with a causal link to higher food and energy prices, alongside broader supply-chain disruption. This would be problematic at any point in time, but it's a particular issue today because of existing supply-chain stresses and the backdrop of above-target inflation. So a very strong El Nino event would be another negative supply shock, at a point when the global economy has limited room to absorb one.

This paper explores what an El Nino is, and why a very strong one later this year would be so concerning. It also takes a look at what happened on previous occasions, and the channels through which it affects the broader economy and financial markets.

We also consider the potential consequences this time round. Much will be weather dependent, but the precedents from the 1970s show how an El Nino event can coalesce with other shocks to keep inflation persistently above target, just as we've seen in recent years. Moreover, this is particularly damaging for many EM economies, where food can make up as much as a third of consumer expenditure.

Authors

Henry Allen

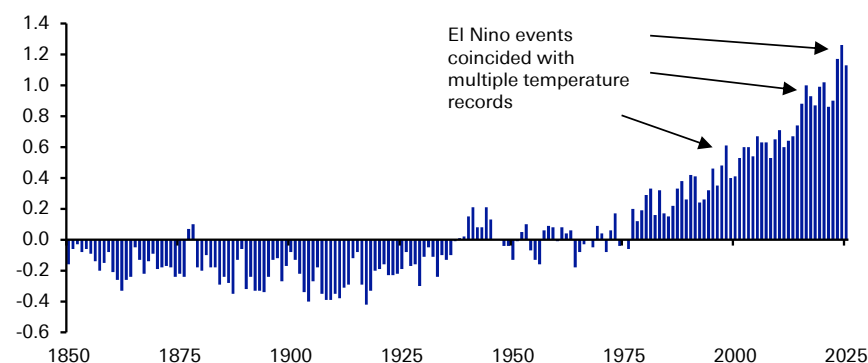
Macro Strategist



What is an El Nino?

- An El Nino event is the warm phase of the El Nino-Southern Oscillation (ENSO), which is a variation in winds and sea surface temperatures in the Pacific Ocean. This oscillates between warming phases (called El Nino) and cooling phases (called La Nina).
- The warming El Nino phase occurs when there are unusually warm sea surface temperatures in the eastern Pacific. These warmer waters mean that the Pacific jet stream moves south relative to normal, which creates changes in weather patterns and ecosystems.
- We can see the effect on global temperatures too. Even as temperatures have been rising anyway over recent decades, the record years have often coincided with El Nino events, including 1998, 2016, and 2024.
- Unfortunately, El Nino events are correlated with a higher frequency of natural disasters, with more people affected by disasters in El Nino years relative to neutral years. Flooding is one that sees a particular increase, but it is also associated with temperature changes too.
- The effects of an El Nino vary considerably by region. In the US, it normally coincides with wetter conditions in the southern US along the Gulf Coast. In stronger episodes, this can extend to California and the Southwest too. But in other regions it is more likely to cause drought, like Australia, whilst India sees weaker monsoon rains.
- In practical terms the main effect of an El Nino is disruption to food production. But there are other effects too. El Ninos have been correlated with the spread of climate-sensitive diseases, and one study found that the El Nino in 2015 contributed to the spread of the Zika virus since it created conditions beneficial for mosquito-borne transmission.¹ Furthermore, natural disasters like flooding increase the frequency of poor hygiene conditions or undernutrition, so there is an impact there too, and El Ninos have been linked to many famines and even conflicts.
- For markets, an El Nino is capable of hitting agricultural production, hydropower, shipping, and even has effects on public health and political stability. In other words, it operates as a multi-channel supply shock.

Figure 1: Global land and ocean average temperature departure with respect to 1901-2000 average (degrees celsius)



Source : National Centers for Environmental Information, Deutsche Bank

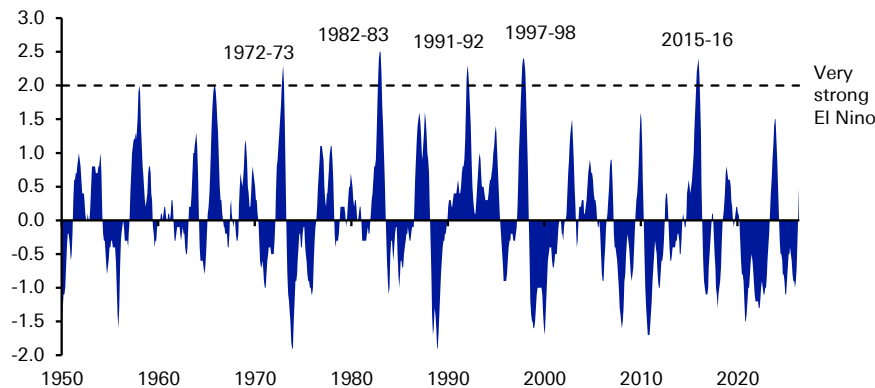
1 Caminade C, et al. Global risk model for vector-borne transmission of Zika virus reveals the role of El Niño 2015. Proc Natl Acad Sci U S A. 2017 Jan 3;114(1):119-124.



What's particularly bad about this one?

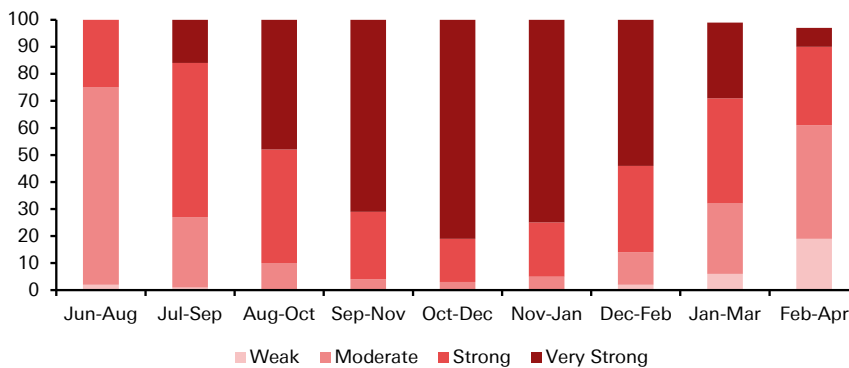
El Nino cycles typically repeat every few years with irregular frequency. However, today's episode is particularly concerning because forecasts suggest it will be a very strong one, ranking among the most serious in recent history.

Figure 2: Relative Oceanic Niño Index



Among others, the US Climate Prediction Center have said this could be one of the strongest in over 75 years, with an 81% chance of it reaching the "very strong" threshold over Oct-Dec. It even has a 73% chance of reaching the "strong" threshold of 1.5C as soon as the July-September period.

Figure 3: US Climate Prediction Center forecasts (%) for El Nino



A very strong El Nino means it's more likely to shift rainfall and temperature patterns across multiple regions simultaneously. And from a macro perspective, the risks are non-linear. For instance, a modest food shock can be absorbed via inventories, imports or substitution. But a large shock propagating across multiple countries can't be absorbed as easily, with second-round inflation effects more likely to result.

The timing is particularly bad right now, because the global economy is already facing significant supply-chain stress. Most notably, the closure of the Strait of Hormuz has led to a spike in oil and gas prices this year. And even if the El Nino is temporary, the resulting inflation shock will push inflation higher at a time when it's



already above target in many countries.

These risks are already being considered by policymakers in decision making, particularly in emerging markets. Indeed, this year has seen central banks in India, Peru and Colombia all reference El Nino in official publications.

What happened on previous occasions?

2023-24: Supply-chain disruption and record temperatures occurred with the last El Nino.

- The most recent El Nino offered a clear example of what can happen, and this didn't even hit the "very strong" threshold.
- 2024 was the warmest year on record, and the episode saw severe droughts in parts of the world, including parts of South America, along with severe flooding in East Africa and South America, consistent with the typical redistribution of rainfall during El Nino.
- Meanwhile, the Panama Canal had its 3rd driest year on record in 2023, meaning that authorities had to cut shipping passing through. The disruption to commercial shipping demonstrated how El Nino is more than just a food price story.

2014-16: The last "very strong" El Nino had broader food security and public health implications

- Just as in 2024, the El Nino of 2016 also saw the biggest upward global temperature anomaly to date.
- Ethiopia suffered its worst drought in decades, leaving millions of people in need of food assistance. Moreover, the Pacific hurricane seasons saw 16 hurricanes in each of 2014 and 2015, the joint-highest number of record.
- One study found that El Nino was a contributing factor to the spread of the Zika virus, since it created conditions beneficial for mosquito-borne transmission.

1997-98: The "very strong" El Nino also had major climactic and public health implications

- 1998 was another record year for global temperatures, and included many extremes around the world. For instance, San Francisco saw its wettest rainfall season since 1889-90, whilst Peru saw 16 times its average rainfall.
- Meanwhile, after major flooding in east Africa, there was an outbreak of Rift Valley fever in Kenya, Somalia and Tanzania. The outbreak also led to major losses of livestock too.

1982-83: The El Nino occurs when high inflation is still lingering

- By 1982-83, inflation had fallen substantially from its earlier heights, aided by monetary tightening like that seen under Paul Volcker at the Fed. However, the El Nino didn't help matters, as it caused major climactic disruption once again, including flooding in the United States, and droughts in Indonesia and Australia.

1972-73: The El Nino played a role in raising inflation before the first oil shock



- Although the oil shock of 1973 was the primary factor that drove the stagflation of the 1970s, several other shocks including the El Nino had already occurred beforehand. This was significant because collectively, they served to push up inflation and inflation expectations. For instance, US fiscal spending had already risen in the late-1960s, thanks to the Johnson Administration's Great Society programmes and the Vietnam War. Then in 1971, President Nixon ended the dollar's link to gold, ending the system of fixed exchange rates that had prevailed after WWII, and in February 1973 there was a dollar devaluation. Amidst all these inflationary pressures, the El Nino event further drove up food prices.

Even further back

- Back in the mid-17th century, there was a period when El Nino events happened about twice as often on average. This was an unusually turbulent period of global history that's been termed the "General Crisis". Some historians have proposed that climactic changes such as El Nino played a role in fomenting instability.

How does an El Nino affect the economy and global markets: The main transmission channels

A useful way to think about El Nino is a set of different supply shocks via multiple channels: food prices; energy prices; supply chains; EM balance sheets; and growing risk premia resulting from the heightened risk of social/political unrest. For individual economies, the impact depends on their exposure to food in the consumer basket, and the extent of intervention from policymakers, be that via food subsidies or tighter monetary policy to counteract inflation.

1. Food: The clearest inflation channel

- **Direct supply hit:** El Nino shifts rainfall patterns, raising the odds of drought in parts of Asia and Australia, and flooding in parts of the Americas. This causes damage to crops, kills livestock, and leads to shipping disruption.
- **Emerging market exposure:** The impact of a food shock varies considerably across different economies. In the US for instance, food-at-home makes up just 8% of the consumer basket for CPI inflation. By contrast, in several EM economies, over a third of household expenditure is on food, and an IMF paper² found that the inflation response to El Nino was biggest in economies where food made up a higher share of the consumer basket. So emerging markets like India and Indonesia are among those more severely affected. Moreover, food inflation is a problematic category because it's a necessity, and consumers can't easily substitute away from it, as they can with luxury goods.
- **Second-round effects:** Beyond the immediate impact, El Nino also has second round effects on inflation. For instance, governments may cushion the impact of higher food prices with subsidies. That might reduce the domestic hit temporarily, but by further boosting demand and raising fiscal deficits, it pushes the adjustment elsewhere and risks creating further inflation overall.

2. Energy and Hydropower: The less-obvious inflation channel

2 IMF Working Paper: Fair Weather or Foul? The Macroeconomic Effects of El Niño (2015), Cashin, Mohaddes and Raissi.

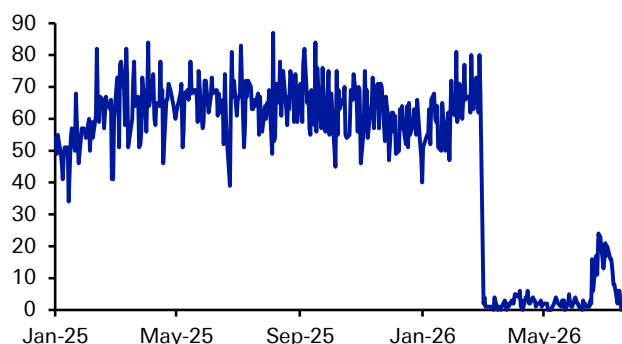


- **Hydropower shortfall:** Lower rainfall in some regions means there's less electricity generated from hydro-electric dams. Moreover, the droughts mean this cut to electricity generation is happening just as heat is raising demand for cooling, and more water is required for irrigation. So there's less supply of energy, just as demand goes up.
- **Fuel-switching channel:** With hydropower output down, this then forces a substitution into other forms of energy, such as gas, coal or oil-fired generation, again raising demand for those other fuels and lifting power prices.

3. Shipping/supply chains

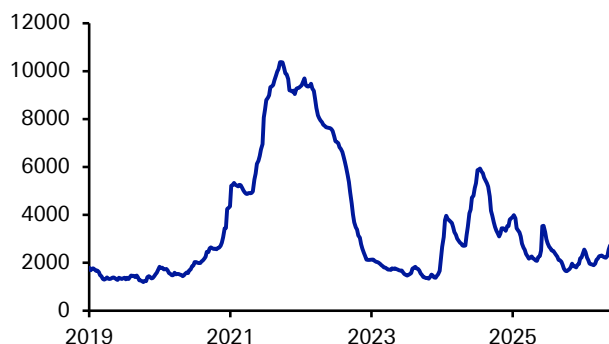
- **Climate effects on supply chains:** The last El Nino in 2023-24 showed how it can impact supply chains, quite aside from a direct impact on the food supply. For instance, Panama had its third-driest year on record in 2023, with rainfall around 30% below average, forcing authorities to restrict the number of ships passing through. This is a point covering around 5% of global maritime trade.
- **Interaction with other chokepoints:** Any supply-chain issues are particularly troublesome today, because the Strait of Hormuz has seen traffic almost completely stop thanks to the US-Iran conflict. So the more transit points that are impacted, the more difficult it is for other routes to absorb any disruption. Indeed, some measures of shipping rates have already doubled since the start of the year
- **Macro Transmission:** If key routes are restricted or closed, then that raises shipping rates further, increases delivery times, and means firms want to hold more inventories against future disruption. So it's another negative supply shock, on top of that seen from higher food and energy prices.

Figure 4: Strait of Hormuz tanker vessel crossings



Source : Bloomberg Finance LP, Deutsche Bank

Figure 5: World Container Index Composite Container Freight Benchmark Rate (USD/40 Foot Box) Drewry



Source : Bloomberg Finance LP, Deutsche Bank

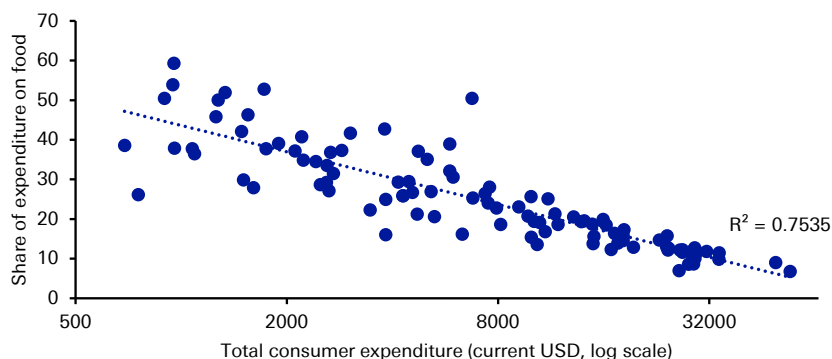
4. EM vulnerability

- **Climate impact:** By virtue of their geographic position, emerging markets are more likely to face the impact of flooding or drought associated with El Nino.
- **Household exposure:** Emerging market economies have more exposure to food prices, so any price shock hits them harder, with food making up over a third of consumer expenditure in several countries.



- **Policy challenges:** This vulnerability makes the policy response to an El Nino event particularly acute. Central banks might raise rates to keep inflation expectations in check and defend the currency, but that doesn't solve the actual supply shock itself. Moreover, if fiscal policy is used to cushion the impact, the shock is then shifted onto public balance sheets, and sovereign risk can become an issue.

Figure 6: Total consumer expenditure vs share of expenditure spent on food, 2023



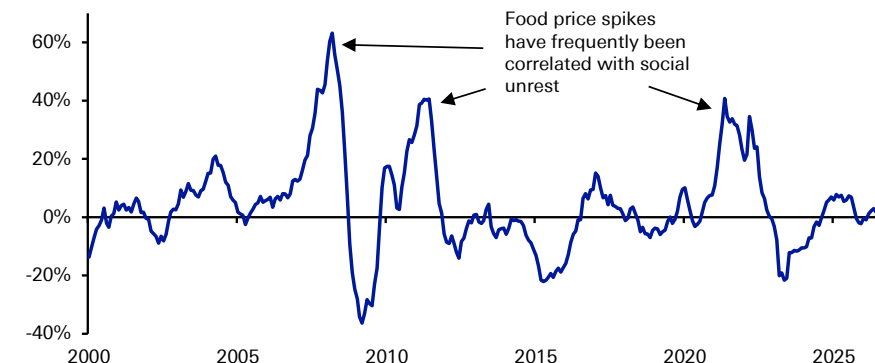
Source : USDA Economic Research Service, Our World in Data, Deutsche Bank

5. Social/political unrest

- **Higher food prices raise the risk of unrest:** Throughout history, higher food prices have often been correlated with political unrest. Clearly it takes many factors to spark unrest, but the squeeze on disposable incomes means that any existing dissatisfaction is magnified. If an El Nino leads to higher food prices, that in turn makes unrest more likely.
- **Recent precedents:** In 2010-11, higher food prices were regarded as a contributing factor behind the Arab Spring. A few years earlier in 2007-08, there was a wave of instability across many EM countries, including Tunisia, Egypt, Cameroon, Ivory Coast, and Haiti. And more recently, the 2022 price shock was a contributor to protests in Ecuador, Sierra Leone, Haiti and Panama.
- **Historical precedents:** Looking further back, several of history's most important revolutions took place against the backdrop of higher food prices. The French Revolution of 1789 came after a succession of harvest failures. The 1848 revolutions in Europe happened at the end of the so-called "Hungry Forties", when there was a severe famine after the failure of potato crops. And in Russia in 1917, food shortages played a key role in fomenting discontent, with supplies hampered in part by the reallocation of farm workers into the military for WWI.



Figure 7: UN Food and Agriculture World Food Price (year-on-year % change)



Source : Bloomberg Finance LP, Deutsche Bank

What does it mean for economies and markets?

A very strong El Nino event would be another negative supply shock when the global economy has limited room to absorb one. The combination of higher food and energy prices will mechanically raise inflation, and broader supply-chain disruption risks creating further price pressures when the Strait of Hormuz has already been blocked.

This is particularly problematic, because experience shows that a prolonged period of above-target inflation raises the risk of inflation expectations becoming unanchored. That in turn can create a self-fulfilling wage-price spiral, as if firms and individuals expect high inflation, then they will set their prices and bargain for wages accordingly.

In several respects, this is reminiscent of the experience in the 1970s. Back then, a series of seemingly unconnected supply shocks (including an El Nino) occurred that collectively kept inflation high. Even though each might have been temporary on paper, implying policy should look through and not over-react, the reality was that the shocks began to feel permanent because they kept on happening. So inflation expectations rose and policy eventually had to react.

Emerging markets are particularly vulnerable as well, with the risk greatest in places where food is a larger part of the consumer basket, and where fiscal space is more limited to accommodate any shock. Moreover, we know from recent history that higher food prices are also linked to higher instances of social unrest, creating further economic uncertainty.

Finally, there is a clear human cost to severe El Nino events. The frequency of natural disasters goes up and there is a greater risk of famine. Moreover, public health issues are often correlated with El Nino events, with several channels in operation, including the heightened risk of cholera after flooding, or the added risk of malaria from conditions more favourable for mosquitos.

Given the forecasts for the strength of this El Nino, this will be an important event to look out for over the months ahead.



Appendix 1

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Open-source AI 101: the battle for the future of AI

July 21, 2026

Open-source versus proprietary AI models in a nutshell

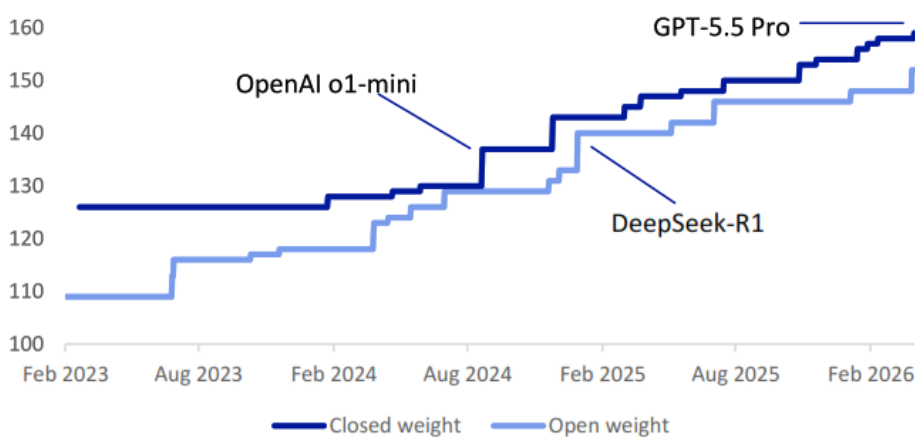
AI-related stocks tumbled at the end of last week on a second DeepSeek moment, as the latest Chinese open-weight AI model reminded investors that cheap, light alternatives are snapping at the heels of the state-of-the-art proprietary models.

In this brief report for generalists, we explain this latest technology format war:

1. Open-what? A new twist on age-old format wars
2. What are open AI models?
3. Why are they disrupting the markets?
4. Why would a company give its model away?
5. What comes next?

The emergence of DeepSeek's breakthrough R1 model in January last year wiped \$600bn off the market value of Nvidia in one day on concerns about whether vast hyperscaler capital expenditure was justified in the light of a "good enough" model developed in China using older chips and trained for a fraction of the cost.

Figure 1: Epoch Capabilities Index score: open-weight models narrow the gap with closed-weight proprietary models (data to end April 2026)



Source : Epoch AI, Deutsche Bank Research

Authors

Adrian Cox
Research Analyst



A similar release on Friday revived those concerns, with the Magnificent Seven (mostly hyperscaler) stocks shedding 1.8 percent and the Philadelphia Stock Exchange Semiconductor Index turning down 1.6 percent, bringing its loss for the week to more than 10 percent. Alphabet's Google, Microsoft, Amazon's AWS, Meta and Oracle are set to spend about \$700bn building AI capacity this year, up by about 70 percent from last year.

Even so, while powerful open models raise questions about the business models of proprietary model makers and semiconductor demand, they may be bullish for AI adoption, applications and enterprise customisation. Greater competition, more efficient models and lower prices may supercharge a new phase of growth.

1. Open-what? A new twist on age-old format wars

Open-source software is nothing new.

- A [study](#) by Harvard Business School in 2024 found that about 96 percent of commercial programs include some code created or distributed for free by public-facing tech forums. Without it, enterprises would pay 3.5 times more, or roughly \$8.8 trillion, to build their software and platforms, it found.

But open AI models are contentious.

- Some critics saying that once released they may be less secure and more likely to be misused. (We mainly use the term "open models" for readability and explain the distinction between open-source and open-weight in the definition section.)
- Some open models have also been [accused](#) of being trained through "distillation", using the outputs of more capable proprietary models to teach a new model, which may breach providers' terms and magnify errors, biases or safety weaknesses in the original models. Critics also wonder out loud about whether some open models have been geared more towards scoring highly on benchmarks – so-called [benchmaxxing](#) – than for solving real-world problems.

Meanwhile, concerns have been rising about whether proprietary models can be [depended on](#).

- Anthropic's most powerful Claude models were recently de facto banned for several weeks on security concerns. Critics also say calls by proprietary model makers for safety regulation would in fact protect incumbents.

The debate is reminiscent of past format battles.

- Every great technology needs a rival. In the "war of the currents" of the late 1880s, Thomas Edison defended his direct-current (DC) empire by portraying the alternating-current (AC) system promoted by George Westinghouse and Nikola Tesla as lethally dangerous—even supporting demonstrations in which animals were electrocuted with AC.
- And remember VHS versus Betamax video cassette players in the late 1970s and early 1980s? VHS did not win by being the better technology: JVC licensed it broadly, manufacturers produced cheaper machines, recording times were longer and a self-reinforcing ecosystem developed around it. Open-source AI could follow the same path. It does not need to beat every proprietary model on every benchmark. It needs to be good enough, inexpensive and ubiquitous.



However, while the contest between open and proprietary AI is often presented as a winner-takes-all format war, the most likely outcome is a combination of the two.

- At the consumer level, it may resemble Apple and Google's Android, a controlled premium ecosystem coexisting with a larger, more customisable ecosystem. And at the industry level, it may resemble the IBM personal computer, where widespread compatibility commoditised the foundational technology while shifting significant value into chips, software, services and applications built around it. Many organisations will use both.

2. What are open AI models?

There are three distribution models. For users, it boils down to control and – to coin a phrase – with greater control comes greater responsibility.

Proprietary, or closed, models are generally the plug and play option, accessed through the developer's application, an application programming interface (API), or a managed cloud services.

- For a rough analogy, think of Apple's iOS operating system or Microsoft's Windows, where the company controls the underlying technology and the terms on which others can use it.
- With proprietary AI models, the developer keeps the model weights, training methods and most details of the training process private. It controls pricing, updates and safety restrictions, and can alter or withdraw access. Prominent examples include OpenAI's flagship GPT models and Anthropic's Claude models.

Open-weight models are open-source-lite.

- Their trained parameters – the billions or trillions of numerical values that encode much of the model's learnt behaviour – are typically released for users to download, subject to an applicable licence. You can run the model on your own infrastructure, adapt or fine-tune it, and technically alter or remove its safeguards.
- But the release may not include the original training data, the complete training code or enough information for you to reproduce the model from scratch. Many of the models referred to casually as open-source – including DeepSeek's R1 and Meta's Llama models, as well as many released by France's Mistral – are in fact more accurately described as open-weight.

Genuinely open-source AI systems are much rarer.

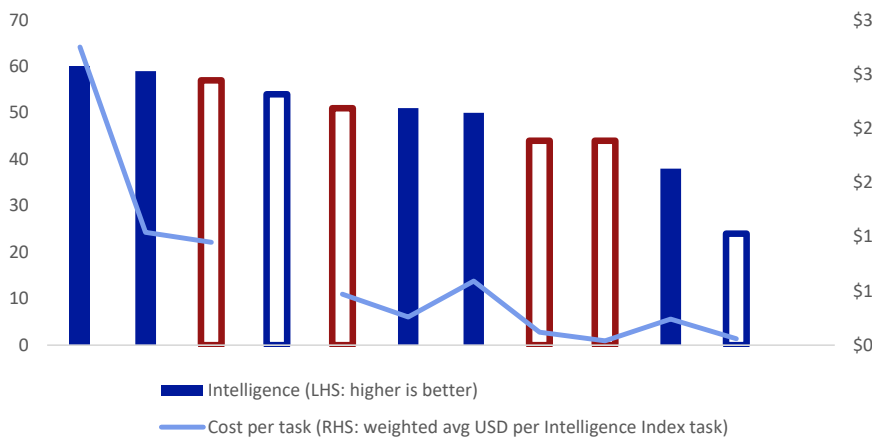
- Under the [Open Source Initiative](#), they should give users the freedom to use, study, modify and redistribute the system, supported by access to its parameters, training and inference code and detailed information about the training data. The grand-daddy of open-source is Linux, a free operating system created in 1991 and found in nearly all cloud infrastructure and the majority of the world's servers and mobile devices.



Consequently, open-source models often offer:

1. **Lower costs, particularly at scale or for specialised tasks.** If you run the model yourself, you can avoid the developer's per-token API charges and can select smaller or compressed models requiring fewer calculations. **The downside?** Downloadable does not mean free: you still have to pay for compute, electricity, engineering, monitoring and maintenance. Open models may also not be as universally capable as the leading proprietary systems and may lack their polished integration with wider products and services.
2. **Greater customisation and control.** You can fine-tune a model for your industry, language or task, rather than relying on a one-size-fits-all service. **The downside?** Although open models are cheaper over the long-run, you need to invest a lot upfront in infrastructure and expertise. You have to take more responsibility for testing, guardrails, cybersecurity, updates and reliability—though managed cloud providers can fill some of the gaps.
3. **Local deployment.** You can install some models on your own servers, or even on "edge devices" such as cars and phones, rather than relying on a connection to the cloud network run by the provider. That makes them potentially faster (less "latency") and more secure as you don't need to share your data. Once you download it, it's yours, and API access cannot be suspended without warning. That is key for some regulated organisations where data can't leave the premises. **The downside?** Not every model can run economically on local hardware, and responsibility for maintaining performance and protecting the system falls on your shoulders.

Figure 2: Intelligence index of selected AI models vs cost per tasks: US (blue) and proprietary (solid colour) models are no longer far ahead of Chinese (red) and open-weight (white with border) models



Source : Artificial Analysis Intelligence Index, Deutsche Bank Research. Data as of July 20, 2026



3. Why are they disrupting the markets?

Open models challenge four assumptions about proprietary models embedded in AI valuations.

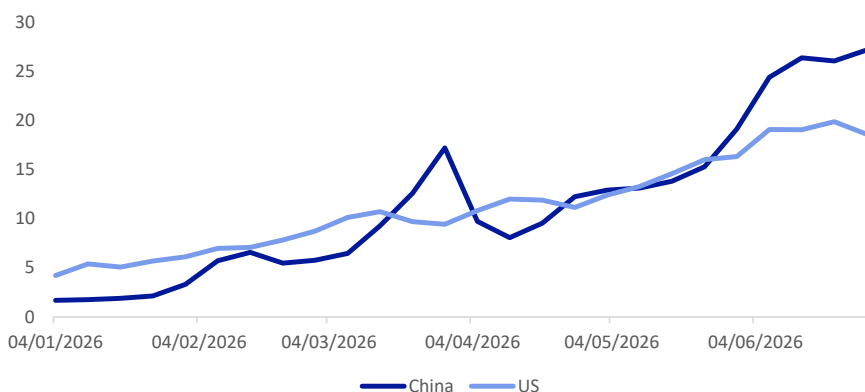
First, scarcity. If capable models can be copied, modified and hosted by many providers, AI models might begin to look like a commoditised software infrastructure rather than a uniquely monetisable product.

Secondly, pricing power. AI agents and “tokenmaxxing” employees are guzzling up vast numbers of tokens, the small units of data that AI models process. As pilots scale up and AI subsidies come to an end, the leading makers of comprehensive models are starting to charge for each token rather than a flat rate. For cost-conscious companies, open models that are good enough for most purposes are looking increasingly attractive.

Thirdly, capital investment. If you can get something like the same performance out of an open model with lighter training and inference (running) horsepower, can the projected spending on chips, networking equipment, electricity and data centres be justified?

Fourth, US technological moat. Chinese open models weaken the perceived durability of US supremacy. The Trump administration says it is in “a [race](#) to achieve global dominance in AI” and has driven policies to strengthen domestic production and adoption and stifle Chinese access to cutting-edge chips and other technology. Chinese models have overtaken US models this year in terms of the number of tokens processed on each by the OpenRouter developer platform.

Figure 3: OpenRouter traffic: trillion tokens going through model platform this year, broken down by country of model



Source : OpenRouter, Deutsche Bank Research

For more, see our recent report “[AI IPOs 101: why they’ll change the boom forever](#)”.



4. Why would a company give its model away?

Open models may be commercially viable because producers do not need to monetise the weights themselves. It's about losing the battle to win the war.

1. Monetising complementary products

- Developers can charge for convenient hosted APIs, premium subscriptions, faster inference, enterprise support, fine-tuning, security and guaranteed service levels. The equivalent of in-app purchases for a free app gives less tech-engaged customers the best of both worlds.

2. Building an ecosystem and standard

- A widely adopted model attracts developers, tools, optimisations, fine-tuned variants and research. All of that can improve the model, lower the cost of acquiring new customers, and make its interface an industry standard.

3. Commoditising competitors' profit pool

- This is a classic case of [commoditising the complement](#): giving away something for free (or very little) to stop your rival from earning excess returns, while increasing demand for (and monetising) your own surrounding infrastructure. So cloud providers that produce or encourage open models can benefit from selling compute to host open models; consumer platforms can benefit from better recommendations and advertising; and application companies can benefit from lower input costs.

4. Boosting reputation, talent and capital

- In a world where you could fit all of the elite AI researchers (about [5,000](#)) into a mid-sized concert hall, and where companies are racing to attract capital to fund exponentially increasing compute costs, a leading open model demonstrates technical competence, attracts researchers and polishes the case for fundraising or an initial public offering.

5. Increasing geopolitical influence

- Open models can spread a country's technical standards, software ecosystem and cultural standards internationally. For China, capable open models reduce its dependence on US APIs and present it as the provider of choice to emerging markets. France has championed Mistral's open models as a European alternative. The US is not sitting idle either. OpenAI itself released smaller, open-weight gpt-oss models as a defensive move last August, and Thinking Machines launched Inkling, its own open model, last week.



5. What comes next?

History shows that formats can coexist where each offers something different.

- The contest between doesn't have to be an either/or. With so much at stake, OpenAI's honeymoon of a near-monopoly on generative AI in the 18 months or so after the launch of ChatGPT in November 2022 was never going to last.
- This is in line with the oft-quoted Jevons Paradox, where more efficient steam engines increased, not decreased, demand for steam power – and hence coal. Similarly, demand for compute will likely continue to surge as the market bifurcates into a handful of cutting-edge proprietary models on the one hand and a myriad of smaller open-source models on the other – just like trains and cars both found their market and continue to live side by side. No one ever said: "That's enough intelligence for now".

But both will exert a positive gravitational force on each other.

- Open models will force proprietary models to keep a lid on prices and remain flexible; proprietary models will set the benchmark for industrial-grade quality, safety and fully integrated products. The competitive moat will increasingly move away from raw weights and towards the balance of proprietary data, product design, memory, security and efficiency.

The greatest unknown in this uneasy coexistence is the direction of AI regulation.

- A growing backlash against AI in the US and the West significantly raises the chance of dramatic policy that could significantly affect its trajectory. While a handful of key US model makers presents a large target for legislators, they are also able to speak with one voice in a way that is not possible for fragmented, often foreign, open-source providers.
- For example, Anthropic released an [Advanced AI Framework](#) last month that called for obligations on frontier AI developers to test their models for catastrophic risks, engage external evaluators and disclose incidents on an ongoing basis. Their message echoes warnings from scientists and economists ranging from job losses to extinction.
- But critics say onerous regulations could favour incumbents with deep enough pockets to cover the costs and would stifle innovation, putting national security at risk. Former White House AI tsar David Sacks posted over the weekend on social media that there was "no reason to limit American models on tasks that Chinese models handle without issue. We're only making ourselves less competitive".



Recent relevant reports

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[AI Q&A: Why have market vibes suddenly shifted on AI?](#) (Feb 16, 2026)

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Appendix 1

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Deutsche Bank
Research

The Rise of the RMB Financing Ecosystem

From Panda and Dim Sum Bonds to Deeper Capital Markets

July 2026

Perry Kojodjojo | Chen Kan

Deutsche Bank Research Institute

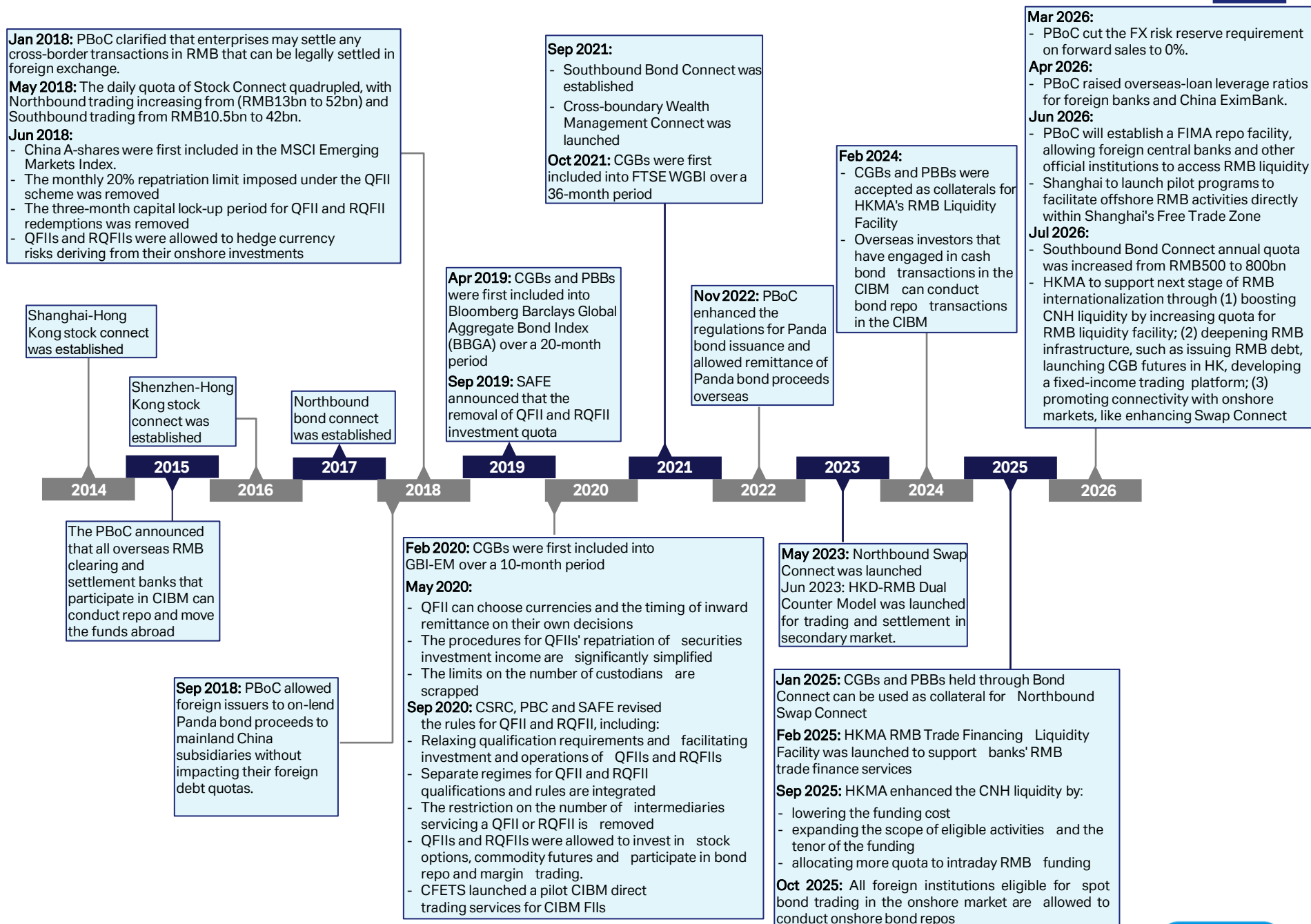
IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Research Institute



- **RMB's global ambition.** China is actively shifting RMB from one used for trade to a global investment and financing currency. The Panda (CNY bonds issued by foreign entities within China) and Dim Sum (CNH bonds issued in Hong Kong) bond markets are the core pillars of this strategy, creating the necessary ecosystem for the RMB's international expansion.
- **Panda bonds - Unlocking onshore China for global use.** The onshore Panda bond market has matured into a strategic funding tool for foreign issuers. Thanks to regulatory easing on fund usage, over half of the proceeds are now used offshore, making it a flexible funding instrument and cementing the role of foreign participants who now account for over 30% of issuance.
- **Dim Sum bonds - A deepening offshore hub.** The offshore Dim Sum market is experiencing a surge in sophistication and diversity, signaling its evolution from a policy-driven market into a mature offshore funding hub. It is now driven by new issuers like tech firms, is offering longer-term bonds, and is set to receive a massive demand boost from new regulations allowing mainland insurers to invest in the market.
- **Next frontier - a focus on building a complete financial ecosystem.** While both markets are growing, challenges remain, including broadening the issuer base beyond financial institutions. The key to addressing these challenges is building a complete RMB financing ecosystem, supported by deep secondary-market liquidity, reliable benchmark yield curves for price discovery, and comprehensive risk-management tools. Ultimately, this is a gradual but foundational journey to establish the RMB as a more widely funded, invested, and held international currency.
- **Expanding European participation in RMB funding markets.** While European entities dominate foreign issuance in both Panda and Dim Sum markets, this participation is highly concentrated among a few financial sector repeat borrowers, with corporates largely absent. Barriers include smaller deal sizes, a narrower investor base, and lack of market familiarity. As RMB internationalization progresses, we anticipate broader international issuer participation, particularly from European corporates given strengthening EU-China ties.

RMB internationalization timeline



RMB Internationalization: A New Chapter Driven by Diverging Bond Markets



China's policy stance on RMB internationalization has shifted from cautious to explicit, with the 15th Five-Year Plan (2026-30) elevating it to a strategic priority. This shift is supported by a series of concrete actions in early 2026 amidst a global de-dollarization trend. However, progress remains uneven. While there is strong official commitment, evidenced by record transaction volumes on China's Cross-Border Interbank Payment System (CIPS) and a surge in RMB goods trade settlement to a projected 30% by 2025, the currency's broader global footprint is still modest. The RMB holds only a 1-4% share across global payments, FX reserves, and debt securities, which is disproportionately low compared to China's approximately 19% share of global GDP.

A fundamental change has occurred in the nature of cross-border RMB usage. The center of gravity has moved from trade to investment flows, with capital and financial account transactions now constituting around 75% of total RMB settlements. **This means the next phase of internationalization requires building a deeper and more accessible financial ecosystem for raising and deploying RMB capital globally.** This includes creating more robust channels for funding, such as the onshore Panda and offshore Dim Sum bond markets, alongside more sophisticated hedging tools and a broader array of attractive RMB-denominated assets.

Since 2025, these two key bond markets have entered a new, more dynamic chapter, accelerating on different tracks and developing distinct characteristics. **The onshore Panda bond market is experiencing a record year, with issuance in the first half of 2026 exceeding RMB160bn.** This boom is driven by fundamental improvements to its architecture, including streamlined issuance procedures and more flexible rules that allow issuers to remit proceeds overseas. This has transformed Panda bonds from a ring-fenced currency pot into a source of globally usable RMB funding.

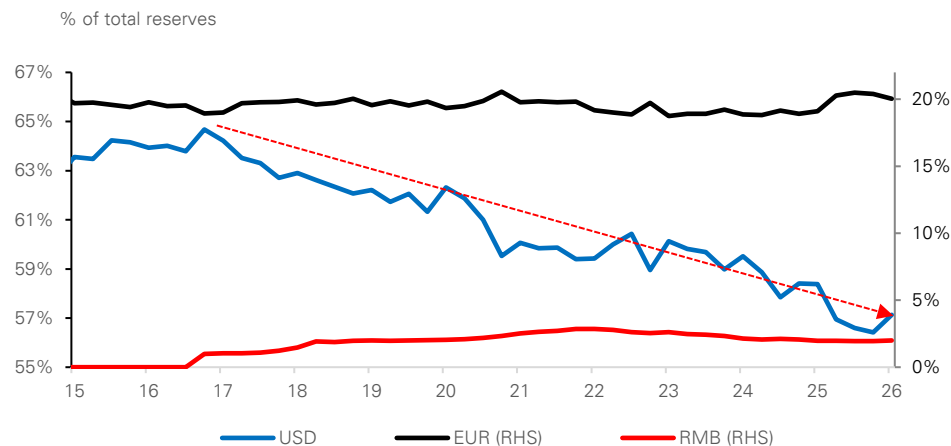
In contrast, **the offshore Dim Sum market's strong growth is driven by demand diversification.** It is poised for a significant boost as mainland insurers gain access via the Southbound Bond Connect, whose quota has been expanded to RMB 800bn, creating a new source of demand for an increasingly foreign-led issuer base.

The markets are also evolving with unique "DNA." **The Panda market's issuer base remains dominated by traditional sectors like financials, while the Dim Sum market is diversifying and embracing tech.** Issuance from IT giants has risen sharply in the Dim Sum market as they seek cheap, long-tenor funding for their AI and cloud capex cycles. This reflects a difference in their duration profiles as well. The Panda market remains overwhelmingly short-term, with nearly all issuance at three years or less due to bank-dominated demand and a flat yield curve. Conversely, the Dim Sum market is showing clear maturity with a growing share of bonds in the 5-year and longer tenors, which better aligns with the investment horizons of the tech sector.

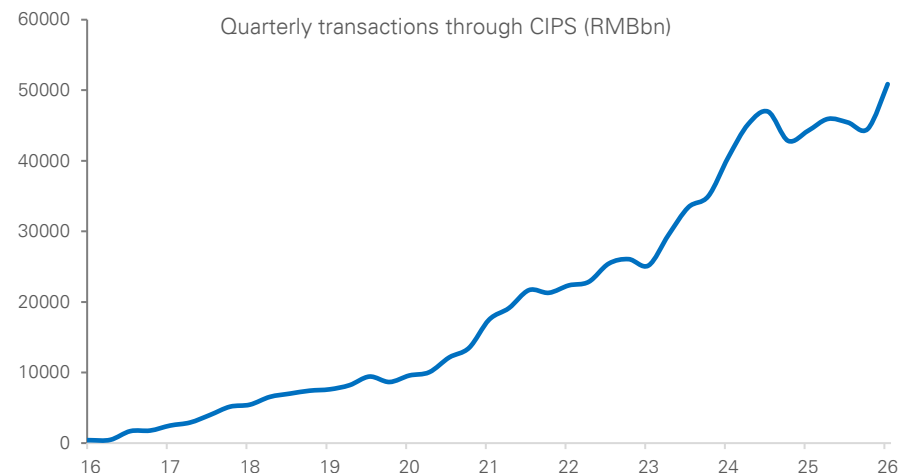
A strategic push amid ongoing de-dollarisation theme



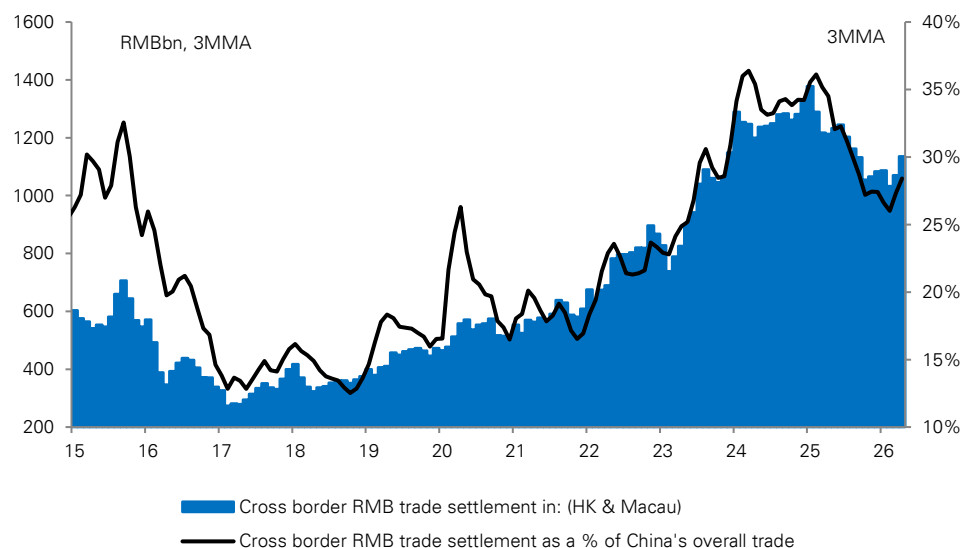
USD's declining share of reserves points to ongoing de-dollarization



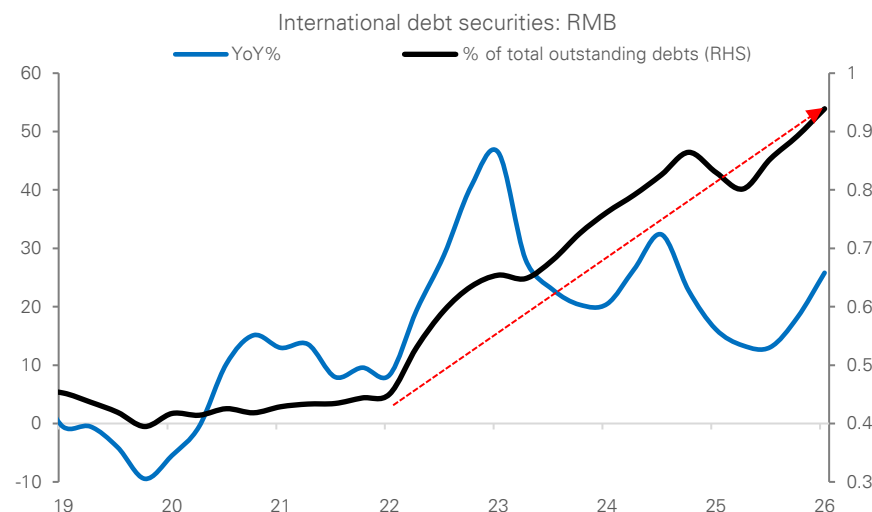
RMB transactions through CIPS hit a record high



A more active use of RMB in trade settlement



RMB-denominated debt issuance has risen notably particularly as China pushes to develop the Panda and Dim Sum Bond markets

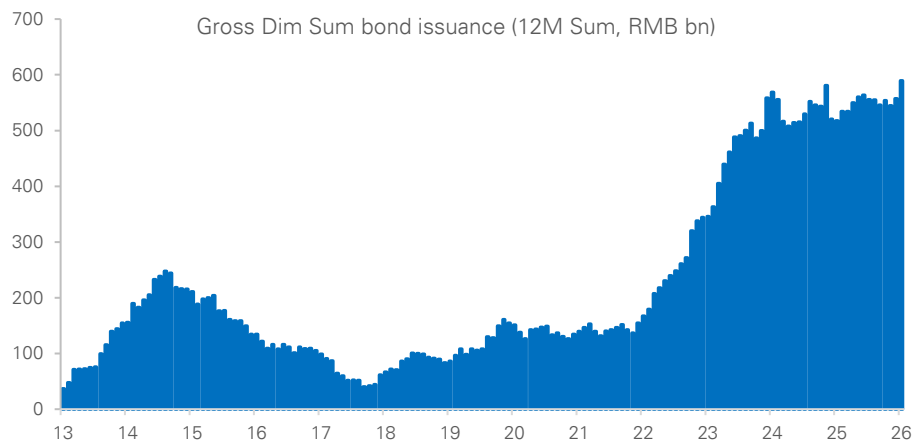


Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

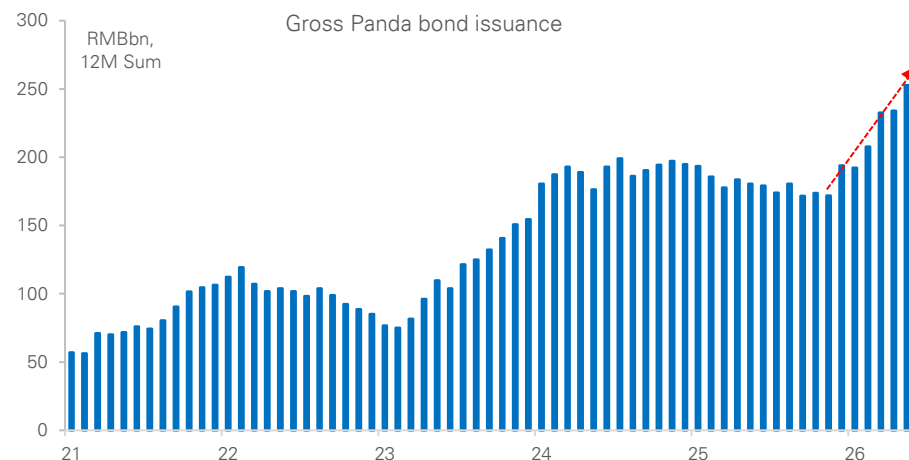
The divergence in focus - key trends to note



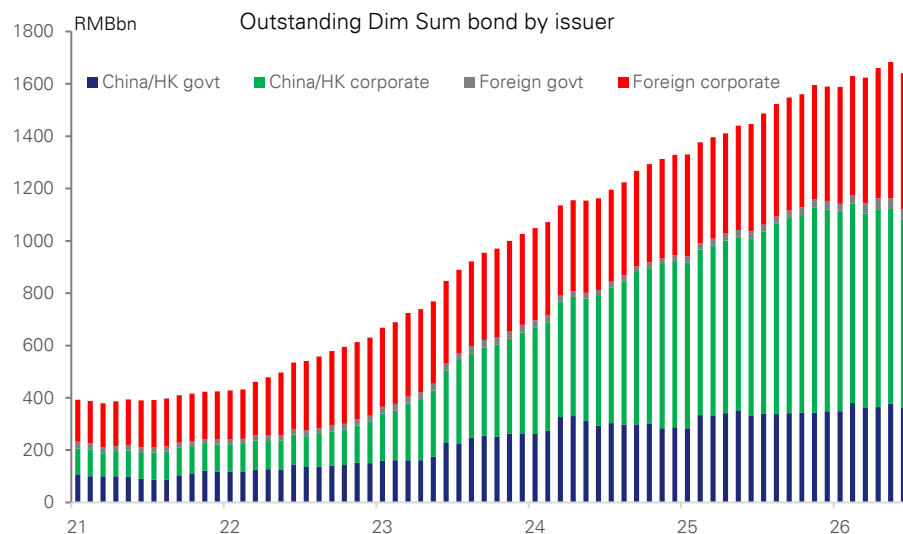
Steady growth in the Dim Sum bond market poised to deepen



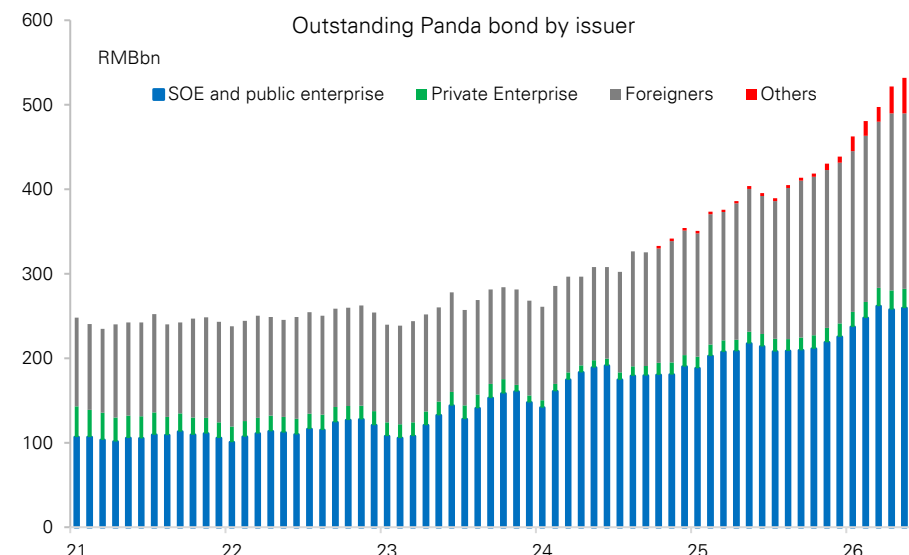
Policy easing unlocks new growth phase in the Panda bond market



Becoming the Go-To offshore RMB hub not just for government entities but also corporates



A hybrid market set to attract more foreign issuers



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

Panda Bonds - Maturing into a funding tool



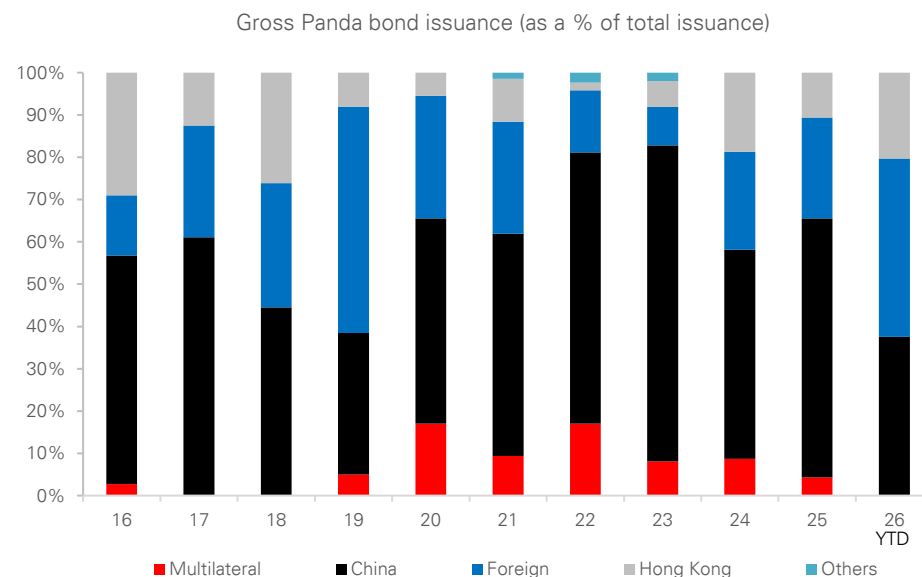
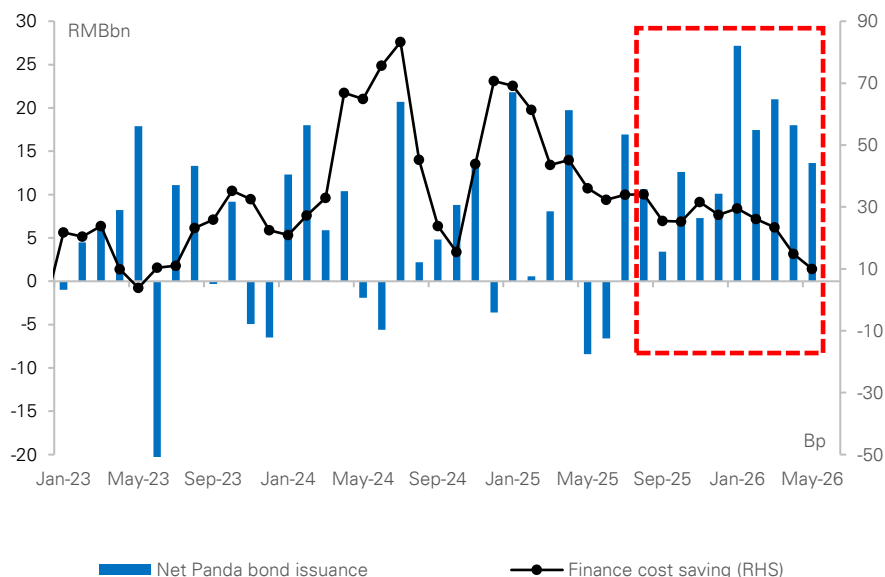
The Panda bond market's record issuance year, even without a clear cost advantage, confirms its shift from a tactical choice to a strategic funding tool. This maturation is driven by two key structural changes:

- **From onshore tool to global funding.** Regulatory easing has been a game-changer. Over half of Panda bond proceeds (53%) are now used offshore, making it a flexible, globally-integrated RMB funding instrument.
- **From niche to mainstay.** Foreign issuers are now a cornerstone of the market, consistently accounting for over 30% of issuance and showing early signs of diversification beyond just financial institutions.

While the market remains concentrated in short-term debt (<3 years) and dominated by domestic bank investors, the modest return of foreign buyers is a positive signal for future market depth.

Issuance remains resilient as market moves beyond cost saving

Foreign participation becomes a cornerstone of the market

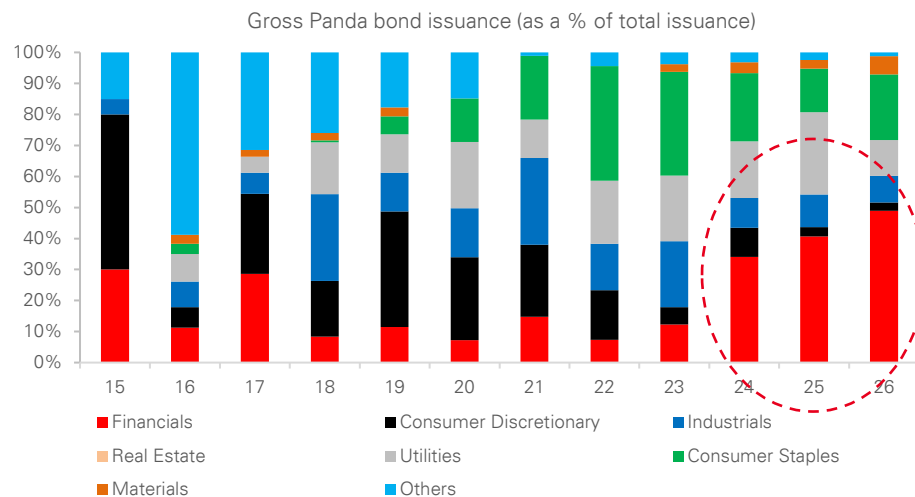


Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

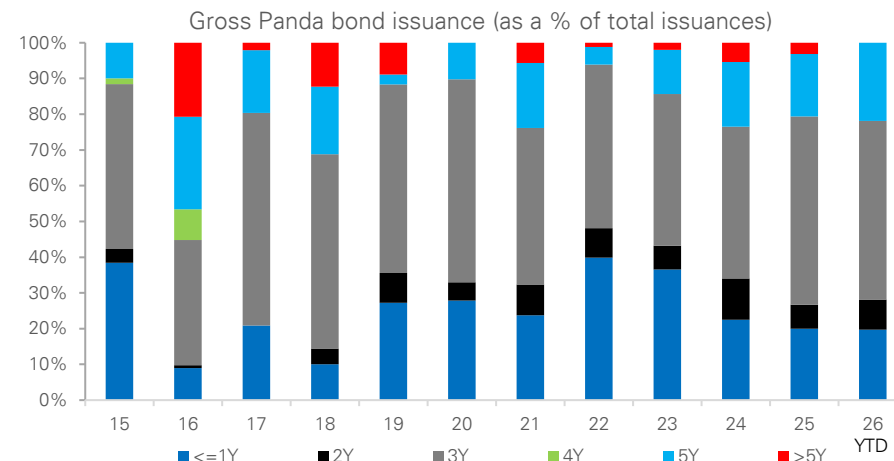
Key trends shaping the Panda market



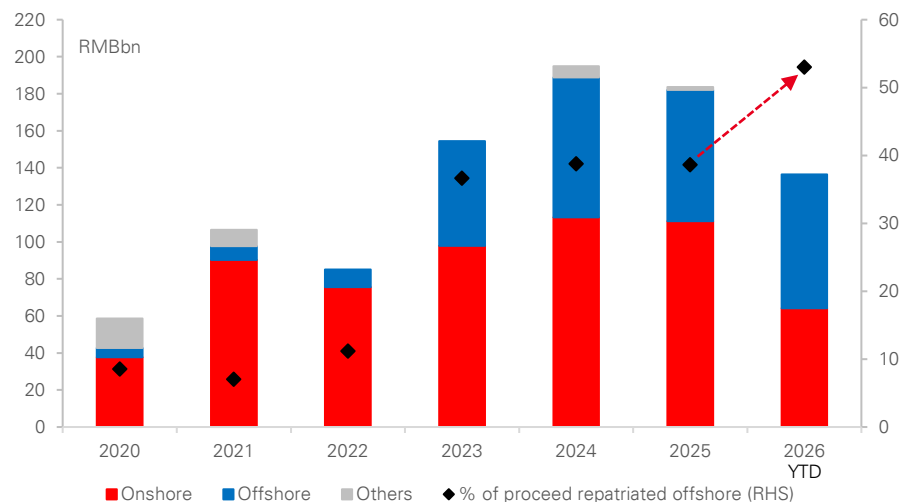
Financials continue to pave the way for broader issuer growth



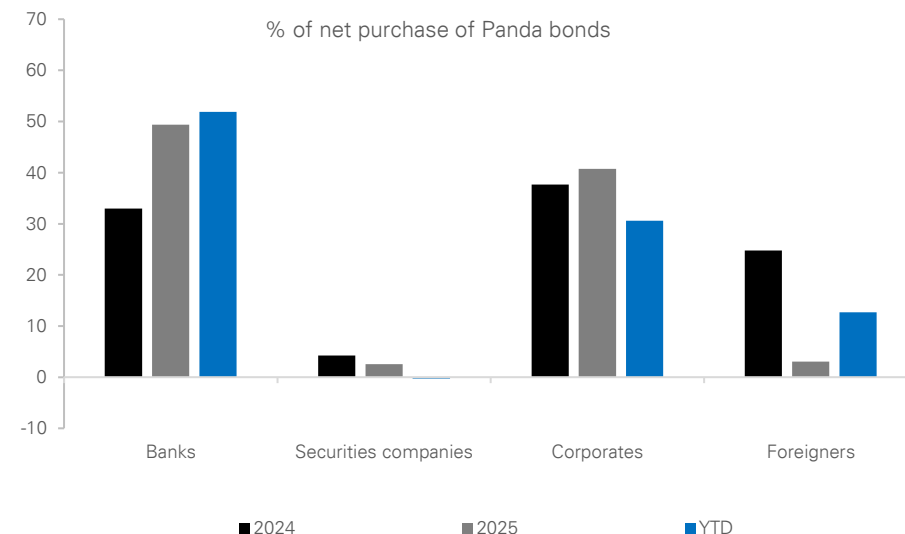
Short tenors still dominate, highlighting room for market deepening



Flexibility increases as offshore use becomes the default



Foreign investor demand returns, signalling renewed confidence



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

Dim Sum market is becoming deeper and more diversified

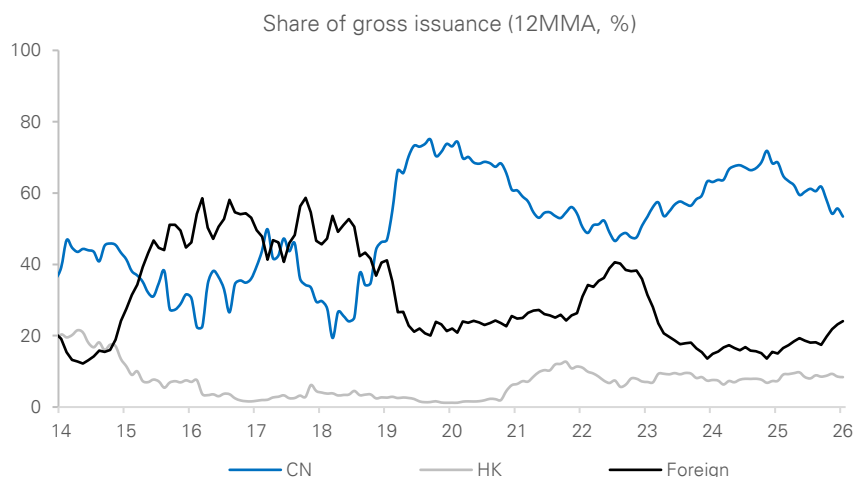


The offshore Dim Sum market is undergoing a structural transformation, becoming deeper and more sophisticated. Three key trends define its new trajectory.

- **Issuer base revolution.** The market is now driven by new participants. Foreign entities have become a cornerstone, accounting for over 25% of issuance. Domestically, high-growth Tech/IT companies have decisively replaced real estate as the primary issuers, reflecting growing demand for long-duration RMB funding to support AI, cloud and digital infrastructure investment.
- **Market maturation.** Issuance tenors are extending significantly. The growing share of bonds with 5+ year maturities, and the emergence of a nascent 15Y+ market, signal rising investor confidence and more sophisticated funding strategies.
- **Demand catalysts unlocked.** Recent policy changes are a game-changer for demand. Allowing mainland insurers to buy Dim Sum bonds via Southbound Bond Connect, alongside an increased quota (to RMB800bn) and wider product scope, is set to unlock a massive and sustainable new pool of capital.

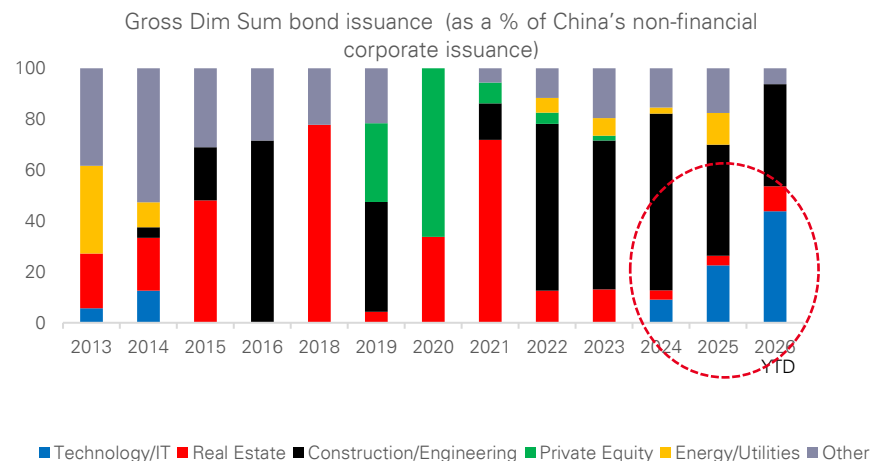
In short, the Dim Sum market is evolving from a niche offshore CNH pool into a more mature, diversified, and institutionally supported market with a powerful new demand base on the horizon. Together, these developments are helping to establish the foundations of a deeper and more liquid offshore RMB capital market.

Foreign participation becomes a cornerstone of the market



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

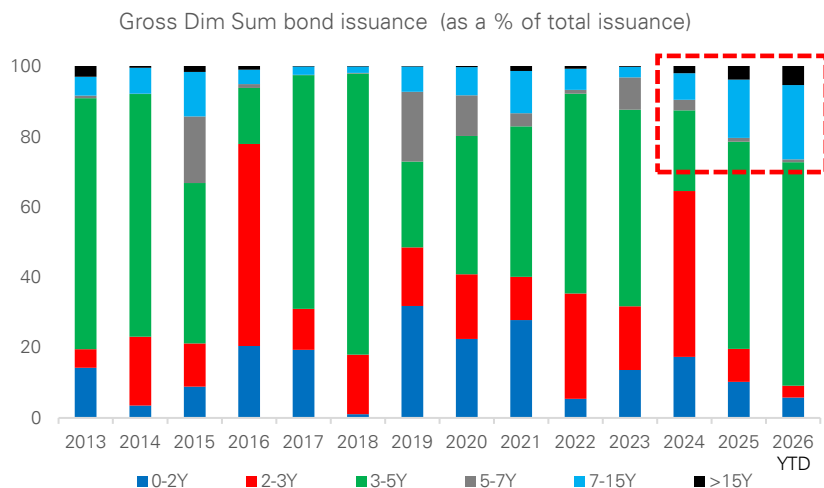
Tech firms replace property-linked sectors among domestic issuers



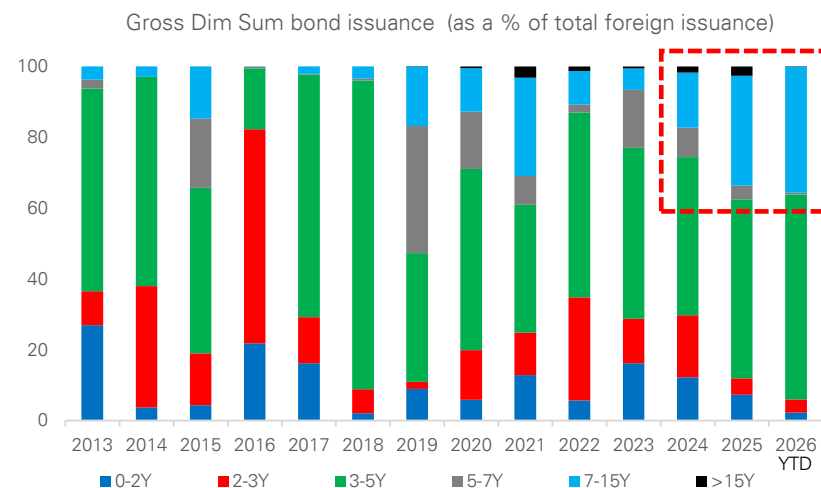
Market depth and demand continue to improve for Dim Sum bonds



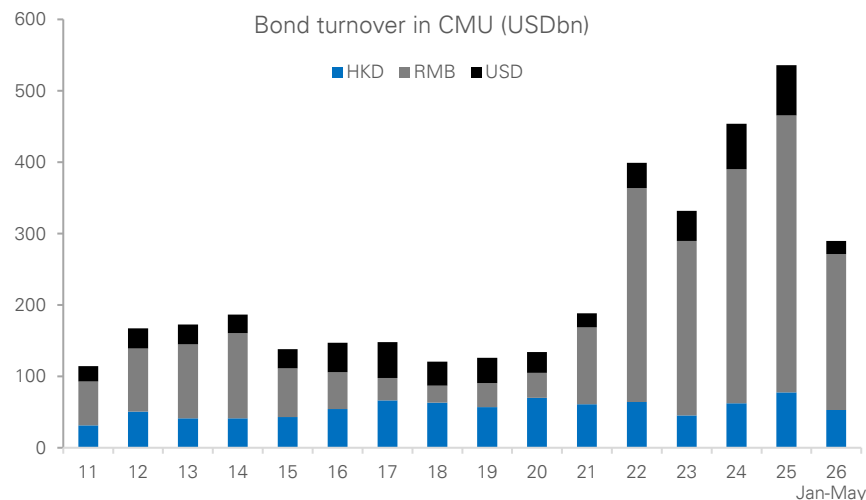
Extended issuance tenor for Dim Sum bonds...



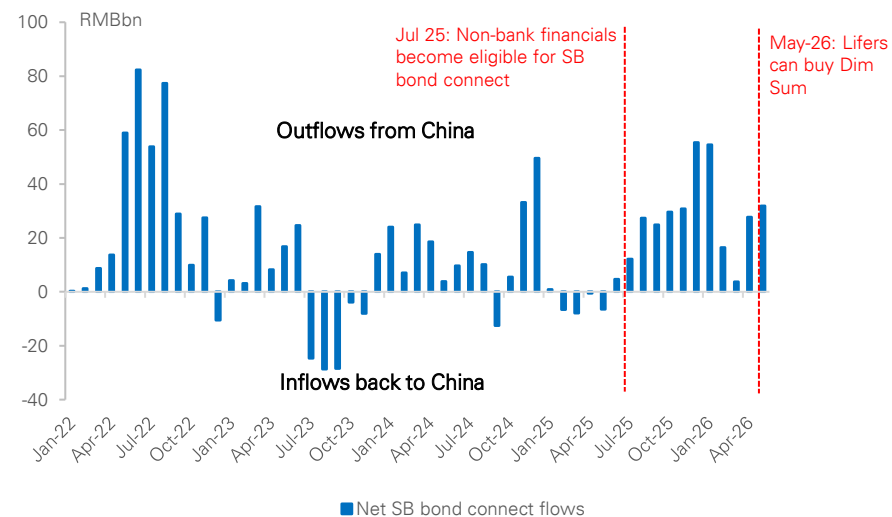
...led by foreign issuers



RMB bonds dominate Hong Kong bond market



Supportive demand-side regulation should underpin Dim Sum demand



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind



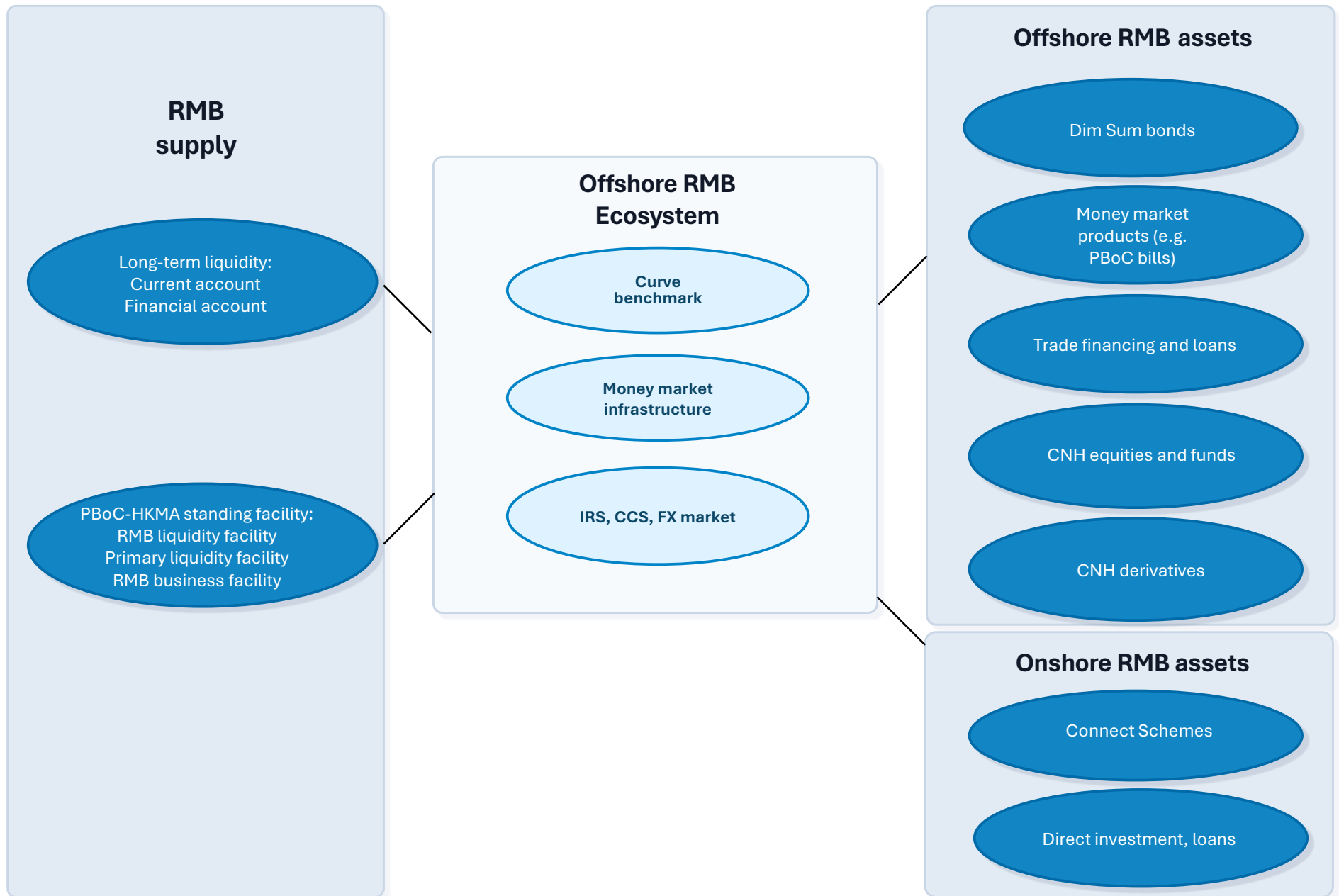
Over the next five years, **Panda and Dim Sum bonds are set to be the two core pillars driving the RMB's evolution from a trade-settlement currency into a global investment and financing currency.** Together, these markets are creating a comprehensive ecosystem that provides the essential assets for international players to fund, invest, and manage RMB holdings.

The Panda bond market, where foreign entities issue RMB-denominated debt inside mainland China, is successfully transitioning from a policy initiative to a viable funding channel. Regulatory reforms have improved access for a diverse range of international issuers, leading to growing market confidence and projected annual volume growth of 20-30%. However, significant hurdles remain, including a narrow investor base, a lack of robust tools for hedging interest rate risk, and complex issuance rules that need further standardization.

Concurrently, **the Dim Sum bond market, consisting of RMB bonds issued outside mainland China (primarily in Hong Kong), is solidifying its position as the primary offshore RMB hub.** The focus here is on building a reliable benchmark yield curve through regular government issuance and upgrading market infrastructure with enhancements to Bond Connect and innovations like digital bonds. This market's growth, however, remains highly sensitive to offshore RMB liquidity, currency expectations, and overall investor demand. Its long-term success depends on creating a deep and sustainable ecosystem rather than just growing issuance volume.

Ultimately, **the development of these bond markets is a gradual but foundational journey.** Much like the "Samurai" market was for the internationalization of the Japanese Yen, Panda and Dim Sum bonds are creating the necessary building blocks for the RMB's global future. While this evolution is subject to broader factors such as capital account policies, market reforms, and the geopolitical landscape, the strategic goal is clear: to establish a robust foundation for the RMB to be more widely funded, invested, and held across the world.

Offshore RMB Ecosystem: from RMB supply to RMB asset formation



Source: Deutsche Bank Research

Expanding European participation in RMB funding markets



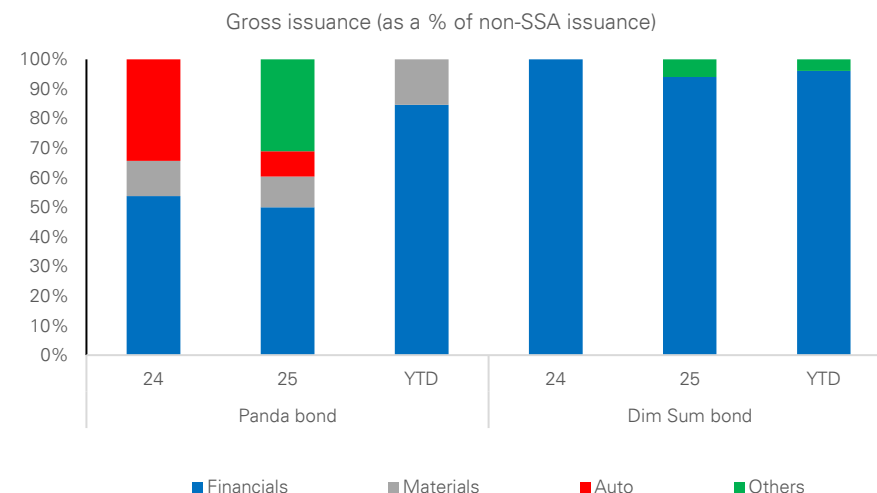
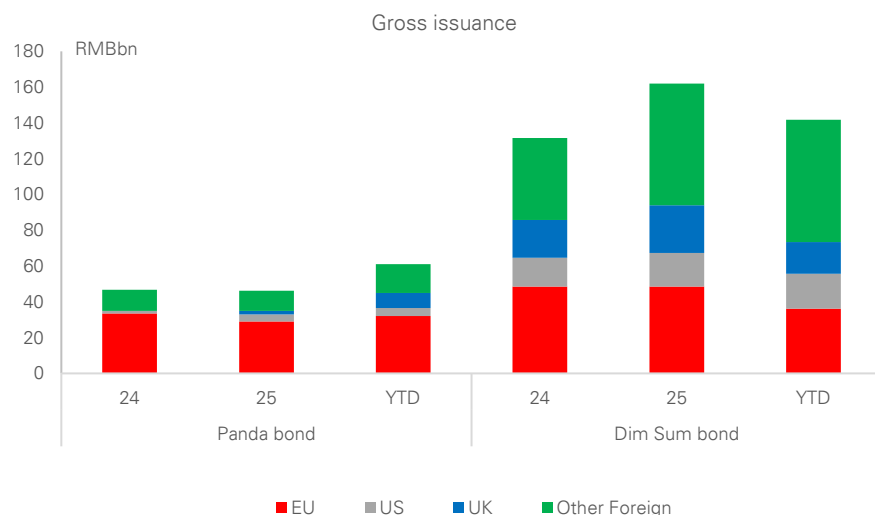
European entities are major players in the RMB bond market, representing over half of foreign Panda and a third of Dim Sum issuance. However, the issuer base is highly concentrated, with financial institutions dominating (85-95%) and repeat borrowers accounting for over 90% of activity. The key challenge lies in attracting new issuers, not retaining existing ones.

Primary hurdles include smaller deal sizes compared to major markets, a narrower investor base, and lack of market familiarity. These factors often make the upfront investment in RMB funding infrastructure, documentation, and investor relationships economically unfeasible.

Looking ahead, we anticipate these barriers will ease with RMB internationalization. A deeper offshore RMB ecosystem will support larger benchmark issuance, broaden the investor base, and improve liquidity. Further development of CNH derivatives markets will simplify currency risk hedging and RMB proceeds swaps. Additionally, China's low interest rates and growing China-related business exposure could enhance RMB funding appeal for international corporates. As the cost-benefit equation improves, we expect future market development to be driven by a broader range of international issuers, particularly European corporates, rather than larger issuances from existing borrowers.

EU issuers account for a large share of foreign RMB bond issuance...

...but the issuances are overwhelmingly concentrated in financial institutions

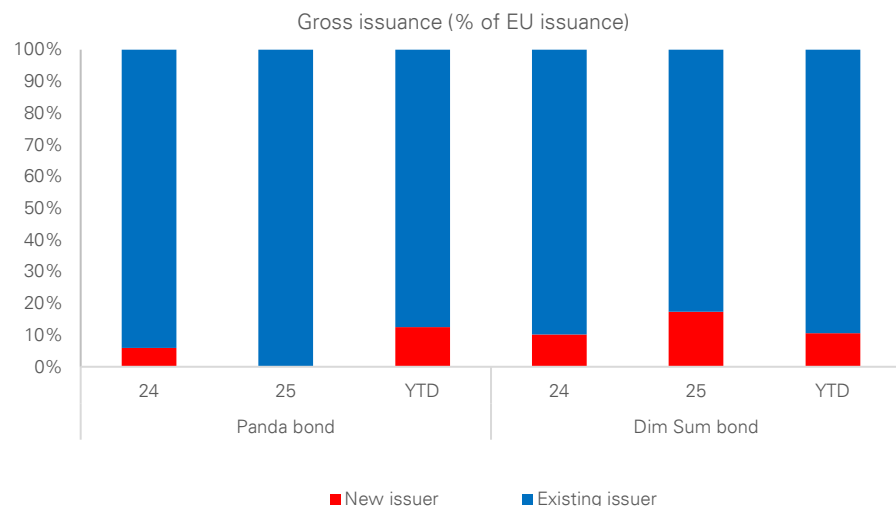


Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind, BIS

Structural tailwinds point to a broader EU issuer universe



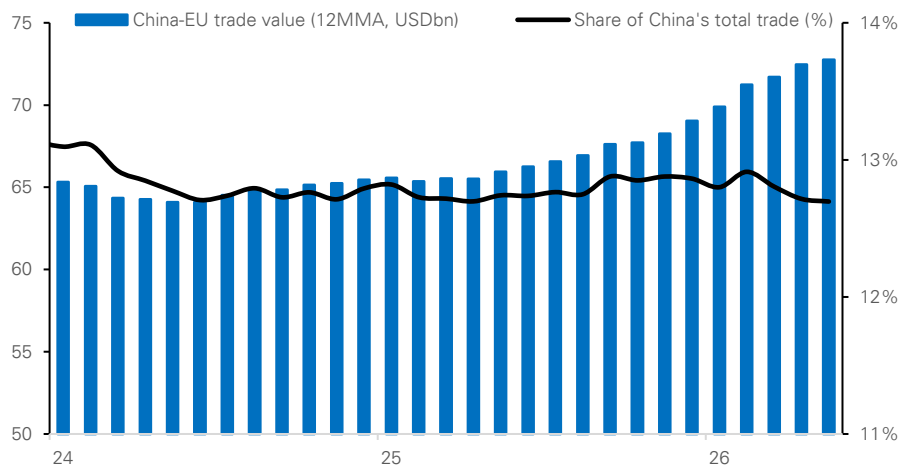
EU RMB bond issuance remains dominated by repeat borrowers



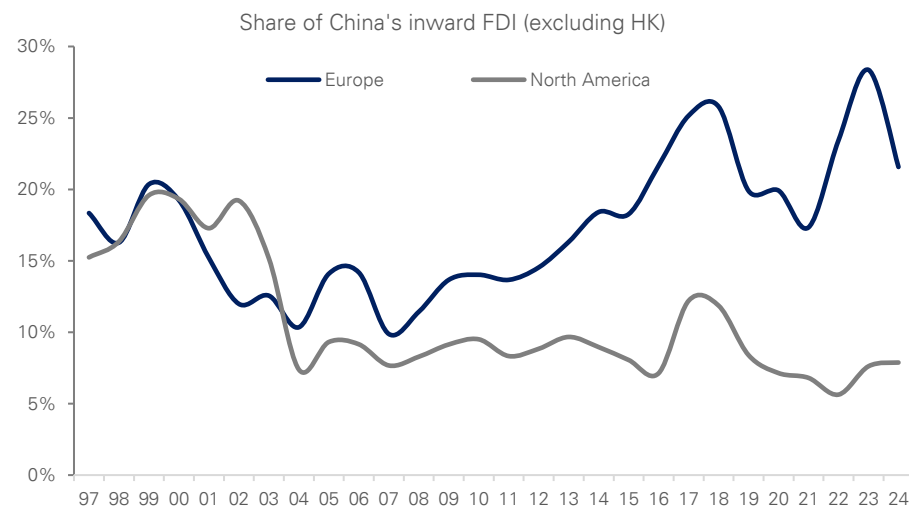
Global debt markets are gradually diversifying away from USD



EU is China's biggest trading partner...



...the most important source of FDI



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind, BIS



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Food For Thought: Sowing The Seeds of Food Inflation

July 15, 2026

Ongoing tensions in the Middle East have raised the prospect of indirect effects on inflation - particularly in relation to food prices. Indeed, much has been said about the rise in energy prices and its spill over into other commodities such as fertiliser, CO₂, and other key ingredients that sit within the agricultural pipeline. To be sure, both the size and duration of the shock matter for food prices.

In this note, we do four things.

- **First**, we highlight what's happened to key commodity prices in the wake of the Iran shock.
- **Second**, we set out a framework, estimating the potential impact of commodity price moves on food inflation for both the euro area and the UK.
- **Third**, we stress test our estimates by looking at the duration of the shock and whether pass-through of cost increases change when the shock is deemed 'transitory' or 'persistent'.
- **Finally**, we highlight some of the broader implications for inflation and where risks lie around our estimates.

Our analysis points to renewed upside risks to food prices stemming from recent oil and fertiliser market developments. The effects appear larger in the UK than in the Euro area.

Importantly, we find that pass-through isn't immediate, and in fact, can build over a long period of time given the mapping of commodity prices into food retail prices. What also matters is whether shocks prove persistent or transitory. Notwithstanding the MoU agreed to by the US and Iran, we estimate that the net effects from oil and urea prices will lead to modest inflationary pressure in the quarters ahead – something we will be watching for closely. That said, considerable uncertainty surrounds both the scale and timing of these effects, with a range of other supply-side factors also likely to influence outcomes. We will be monitoring market developments closely and will calibrate our projections accordingly.

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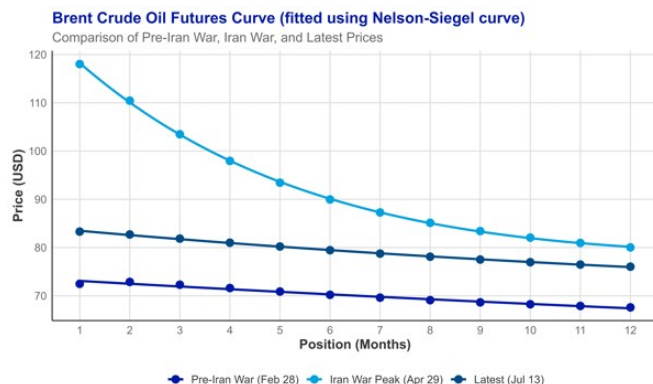


A recap on recent price movements

Recent geopolitical developments have been associated with large swings in fertiliser and oil prices. Oil prices initially rose as markets priced supply-disruption concerns. Subsequently, however, price rises retraced meaningfully as these concerns eased with continuing peace deal negotiations. To add context to these movements, we compare spot and futures prices before the episode, after the initial Iran shock, and using the latest available market data. For the futures curves, we fit a Nielsen-Siegel curve to trace out the shape of the curve across contract positions.

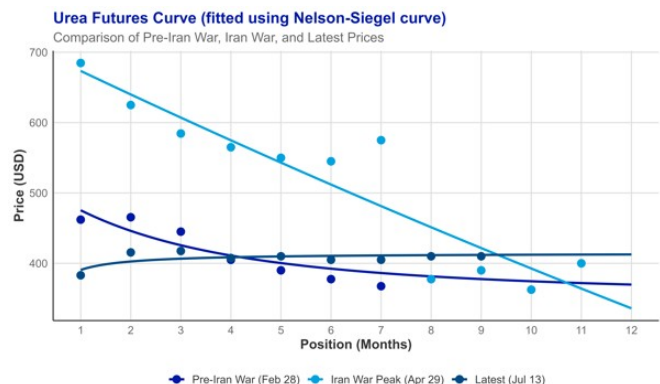
For fertiliser prices, we use the World Bank urea price series as a proxy for global nitrogen fertiliser costs. Urea is one of the most widely traded nitrogen fertilisers and is closely linked to the production process for agricultural inputs. We note, however, that this captures only one important fertiliser-cost channel rather than the complete range of farm input costs.

Figure 1: Brent futures pricing downward sloping...



Source : Deutsche Bank, Bloomberg Finance LP

Figure 2: ...same as urea futures pricing



Source : Deutsche Bank, Bloomberg Finance LP

Before the Iran War, the spot price for Brent crude oil was trading at about USD 72.50/barrel. This then spiked to USD 118/barrel during the peak of the Iran/US conflict, before settling at around USD 83/barrel at current market prices (figure 1). Interestingly, the futures curve is currently flatter than the steeper curve observed at the start of the war, which could imply a more gradual decline in prices in the next few months. The urea spot price also increased sharply at the beginning of the Iran war. In contrast, these prices have settled at a much lower level (figure 2).

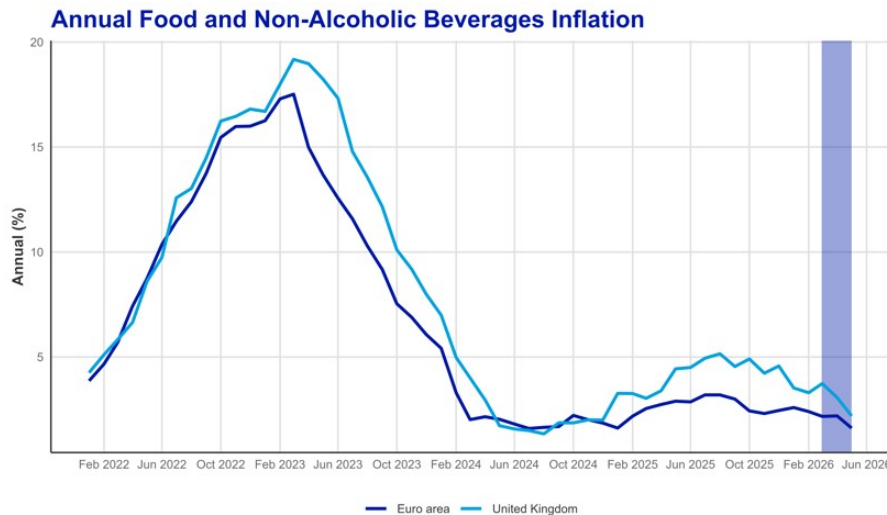
Perhaps unsurprisingly, in terms of recent food price developments, we have yet to see any tangible impact on retail prices.

Euro area food inflation has softened more than expected this year, declining from 2.5% y-o-y in Dec-25 to 1.6% in Jun-26 based on the flash estimate. This marks the lowest reading for food HICP since mid-2021, with the bulk of the adjustment occurring over the past two months. Focusing on food stuff, the largest disinflationary impulses have come from dairy products, sugar and sweets, and coffee. In 2025, these categories were among the main drivers of food inflation, accounting for roughly 35% of annual food inflation despite representing only 18% of the HICP food basket. In recent months, their contribution has fallen to just over 20%. Meat prices have also eased somewhat from their 2025 peak, although they remain the largest contributor to food inflation.



In the UK, we see a similar story. Heavy discounting has led to a drop-off in food prices. Food and non-alcoholic beverages CPI slowed to 2.2% y-o-y – reaching its lowest rate in 17 months. The slowdown in food prices can be seen across both processed and unprocessed food. The latest PPI data for food points to further disinflation in the pipeline, with producer prices across food, beverages and tobacco slowing to 1.1% y-o-y in May-26.

Figure 3: Food (and non-alcoholic beverages) inflation in the EA and the UK has been on a downtrend this year



Source : Deutsche Bank, Macrobond

How does energy and fertiliser costs pass-through into food inflation? There are two channels in which oil and fertiliser prices work their way into CPI.

Channel 1: the direct energy cost. Oil enters food at pretty much every node in the food chain. For example, manufacturers need diesel for tractors and irrigation pumps, marine freight and trucking, processing heat, packaging (plastics) and refrigeration. Energy-related inputs account for around 5–10% of farm production costs, although the share can be significantly higher in energy-intensive sectors such as horticulture and poultry.

Channel 2: fertiliser cost. Urea is essentially manufactured from natural gas. The higher gas prices go, the higher fertiliser prices go too. The response can effectively be summarised into two segments: an immediate margin squeeze, then a supply response as farmers cut application rates or switch acreage to lower Nitrogen crops (soybeans over maize), which lifts prices with a lag of a full crop cycle.

The transmission from input costs to retail prices, as a result, is heavily staggered. The upstream leg is fast: ammonia and urea reprice off gas within weeks. Put simply, an energy shock is visible in fertiliser markets almost immediately. The middle leg is slow and seasonal — application decisions are made pre-season, so a nitrogen cut driven by a spring price spike doesn't show up until (as early) as the following harvest. Ultimately, the final leg is both lagged and heavily damped.



Estimating pass-through to food prices

There are several ways to estimate the effect of oil and fertilizer price shocks to food inflation. In this case, we opt for a simple approach that allows the "data to speak": the local projection method ([Jorda, 2005](#)). Essentially what we do is trace out the impact of a shock over time, in this case oil and fertilizer price shocks, by running a separate regression for each forecast horizon (1-month ahead, 2-months ahead and so on...).

A key challenge is identifying the commodity-price shock itself. Oil and fertiliser prices move for many different reasons, including global demand, exchange rates, trade conditions, inflationary pressures and supply disruptions. To reduce the risk of conflating commodity-specific surprises with broader macroeconomic developments, we use a Romer and Romer-style approach ([Romer & Romer, 2004](#)).

The above approach involves regressing the shocks of interest on a set of potentially confounding variables, including the USD effective exchange rate, monthly dummy variables to capture seasonality, lags of the shocks of interest to account for persistence, global trade volumes and global CPI inflation. We then extract the residuals from this regression. These residuals could then be interpreted as movements in oil and urea prices that are not explained by our macroeconomic controls.

The shocks of interest that we use are the month-on-month percentage changes in Brent crude oil spot prices and the World Bank urea commodity price series.

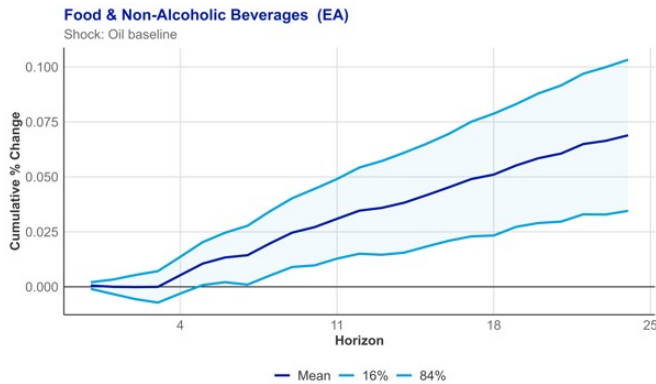
For the local projections, we use the HICP Food & Non-alcoholic beverages series for the Euro area and the UK Consumer Price Index Food and Non-alcoholic beverages. The regressions include the shock of interest (oil & urea), and three lags of the price variables. We then estimate the cumulative percent impact on the price level from a percentage point surprise in both oil and urea costs.

Baseline results

The baseline estimates suggest that oil and fertiliser price surprises have a measurable effect on food prices. The estimated response varies across the two economies and commodities, with the UK showing larger pass-through than the euro area in our specifications. Our estimates, however, become relatively more uncertain the further out we go as shown by the widening error bands.

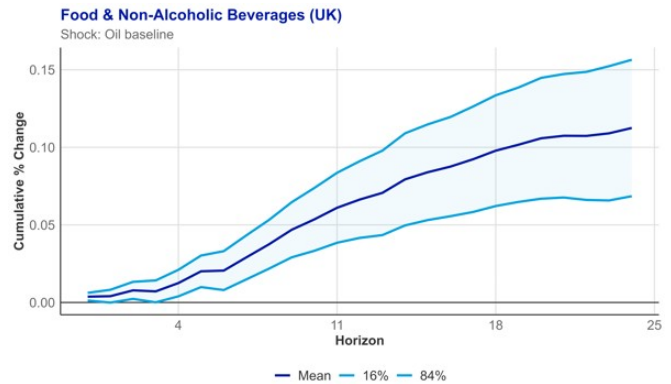


Figure 4: A 1pp shock in oil price leads to nearly a 7bps increase in food HICP (EA)...



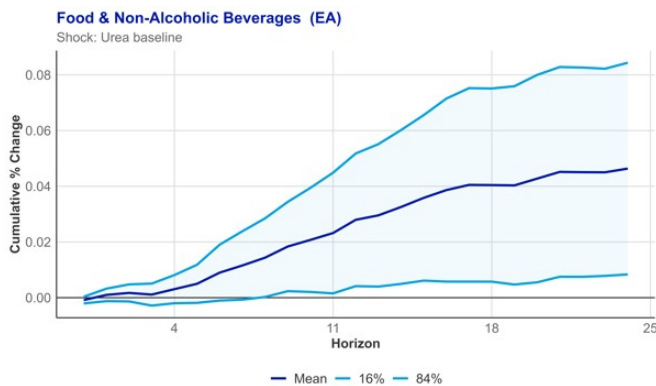
Source : Deutsche Bank

Figure 5:...and just over 10bps in the UK (CPI)



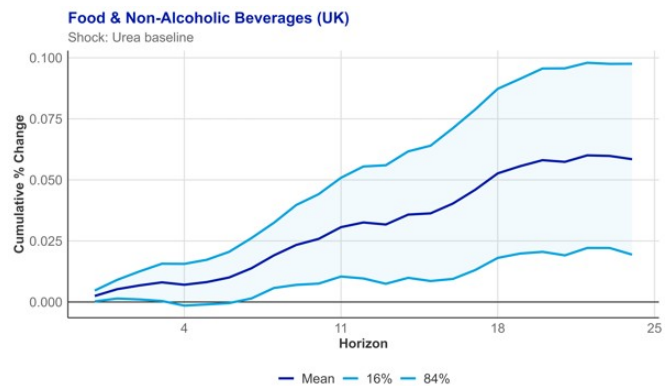
Source : Deutsche Bank

Figure 6: A 1pp shock to urea prices results in just over 4bps to EA HICP (food & non-alcoholic beverages)...



Source : Deutsche Bank

Figure 7:...and close to 6bps for UK CPI (food and non-alcoholic beverages)



Source : Deutsche Bank

In further detail, [figures 4-7](#) present the impact of a 1 percentage point increase in monthly oil and urea price growth on the cumulative food & non-alcoholic beverages price level. Looking first at fertiliser price shocks, we find that the peak impact on UK food prices occurs after around 20 months with a 0.07% cumulative impact on the food price level ([figure 7](#)). In the Euro area, the peak impact is slightly lower at around 0.05% and also occurs after roughly 20 months ([figure 6](#)). A broadly similar pattern is visible for oil price shocks. The UK experiences a larger cumulative increase of 0.1% in food prices while the Euro area sees an impact of about 0.07%.

There are a few possible explanations for the slightly larger estimated effects in the UK relative to the Euro area. One possible explanation that may be driving the difference in pass-through between the Euro area and the UK is the higher imported share in the UK food basket. This would make importers and producers more exposed to externally driven input cost pressures.

A second explanation may come from sterling weakness during global energy shocks.

A third explanation may be in fact that many UK farming sectors operate on narrow margins. When energy, fertiliser or feed costs rise sharply, farmers have less ability

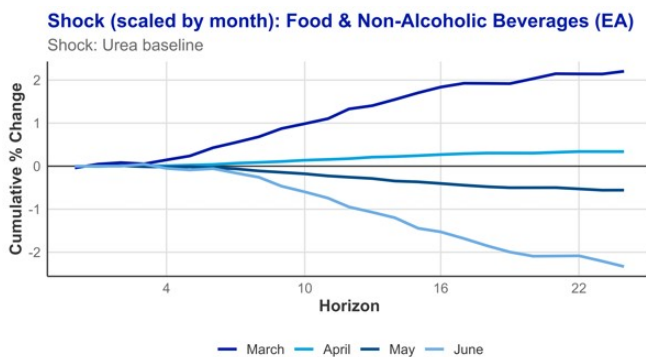


to absorb the shock and are more likely to pass it on through the supply chain. Defra's Farm Business Survey consistently shows that profitability is highly sensitive to movements in input costs. UK agriculture is also relatively intensive, particularly in cereals, oilseeds and grassland systems.

The impact of the Iran war

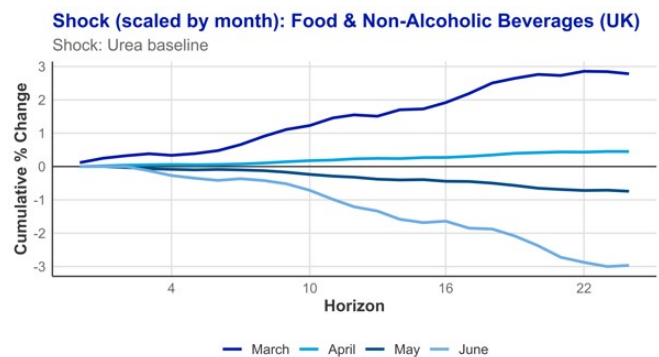
So far, the results have shown a 1 percentage point increase in fertiliser and oil price inflation. However, events such as the Iran war have led to big swings in both urea and oil prices – and in both directions. In order to better gauge these impacts, we scale the shocks in the baseline results to the actual movements observed from March to June. The figures below show the estimated cumulative food price pass-through implied by each monthly commodity price move.

Figure 8: Estimated urea shock pass-through for each month since the conflict began in the EA...



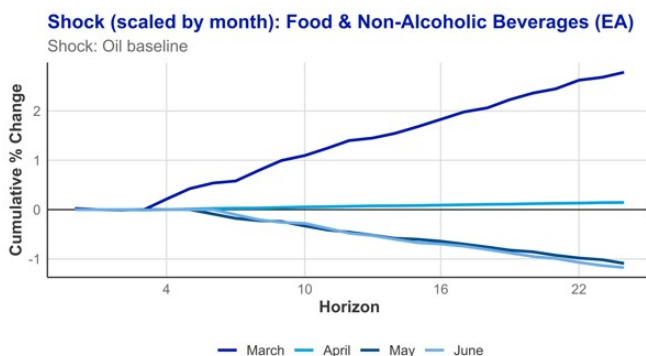
Source : Deutsche Bank

Figure 9:...and the UK



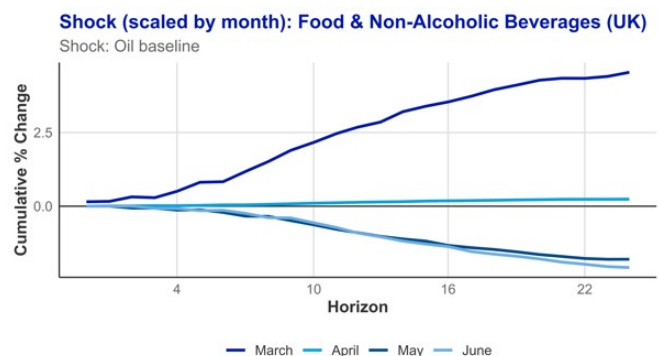
Source : Deutsche Bank

Figure 10: Scaled oil shock between March and June for the EA...



Source : Deutsche Bank

Figure 11: ...and the UK



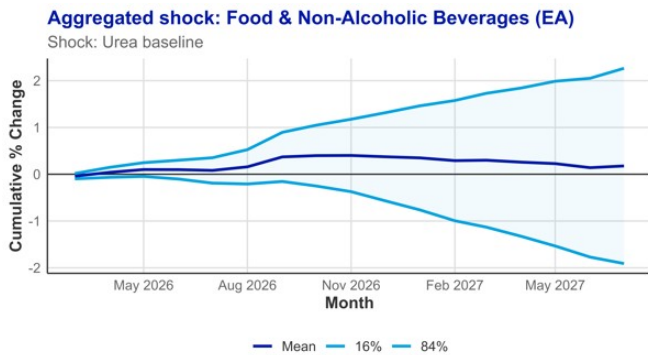
Source : Deutsche Bank

To be sure, the March shock impact alone would have had a big impact on food prices in the Euro area and the UK. For both oil and urea, it was an upside surprise of over 40 percentage points (over 3 standard deviations). However, over the course of the past few months, there have been significant reversals in both oil and urea prices. We attempt to capture these offsetting shocks and obtain the overall impact since March. How do we do this? We do this by aggregating the impacts of the



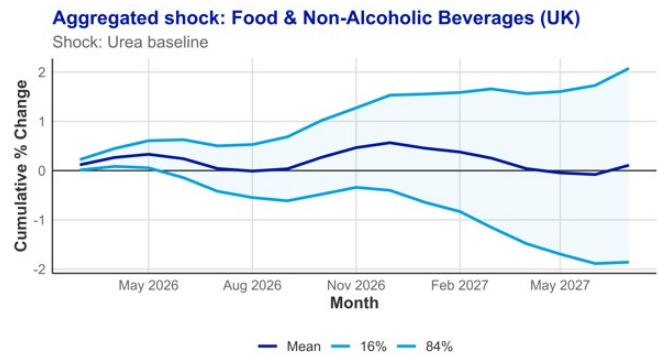
monthly urea and oil price shocks – i.e. we aggregate the shocks from oil and urea via superposition of the estimated local projections.

Figure 12: Estimated aggregate urea shock on food HICP in the EA...



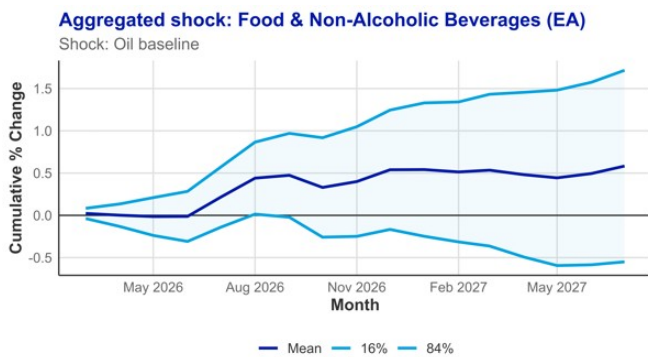
Source : Deutsche Bank

Figure 13:...and on UK food CPI



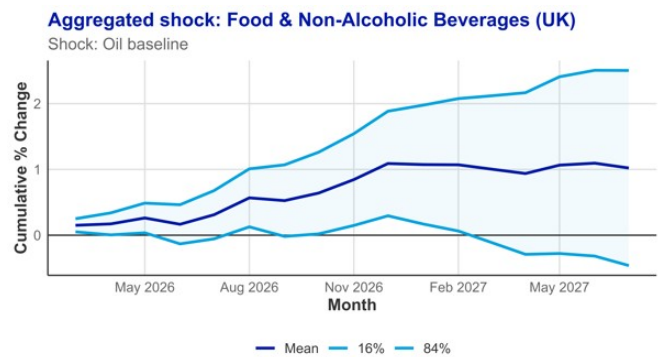
Source : Deutsche Bank

Figure 14: Estimated aggregated oil shock on food HICP in the EA...



Source : Deutsche Bank

Figure 15:...and on UK food CPI



Source : Deutsche Bank

We trace out these shocks and find some interesting implications for the overall impact. For the UK, we see a cumulative impact of around 1% on food prices from the oil shocks a year from March and 0.3% from a urea shock peaking also in a year's time from March, i.e. totalling nearly 130bps (though the effect starts to unwind four quarters out). For the Euro area, the oil shocks lead to about a 0.5% cumulative increase in a year from March and about a 0.3% impact from urea shocks in a year from March. Overall, we observe similar patterns where the UK experiences a slightly stronger impact from the recent shocks compared to the Euro area.

Stress testing our results: persistent vs transitory shocks

A simple scaling of the shocks is unlikely to fully reflect firms' pricing behaviour. In practice, firms may respond differently to the size and duration of the shock. A large but temporary increase in oil or fertiliser prices may generate less pass-through than a smaller but more sustained rise.

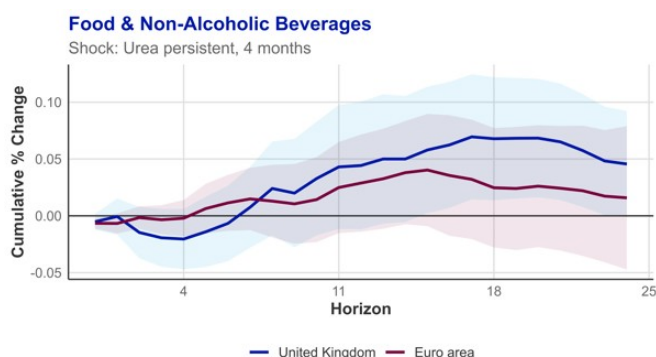
To capture this distinction, we split our oil and fertiliser shocks between persistent



and transitory. We classify a shock as persistent if the price remains at least half as large after four months. If the price falls by more than half over that period, we treat it as transitory. We only apply this classification to sizeable shocks, defined as moves larger than one standard deviation in the relevant shock series. This gives us a simple way to separate temporary price spikes from more sustained input-cost pressures.

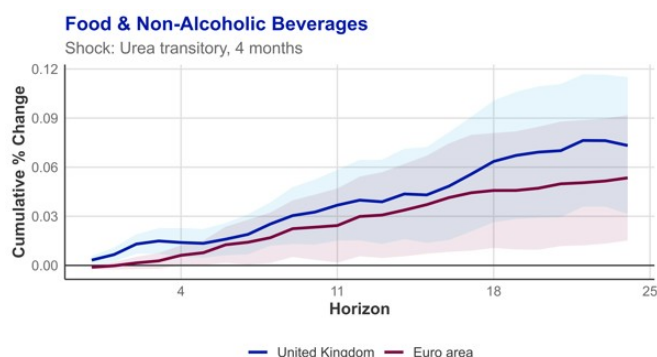
We then run local projections again on the persistent and short-lived shocks and we obtain the following results:

Figure 16: Persistent urea price shock...



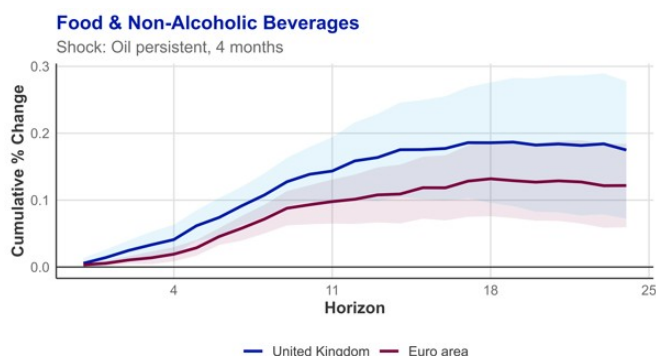
Source : Deutsche Bank

Figure 17: ...vs transitory urea price shock



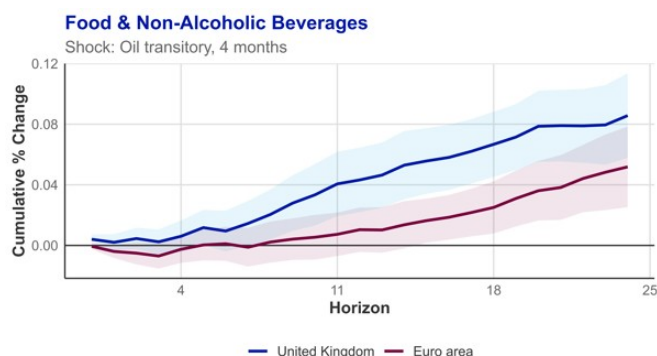
Source : Deutsche Bank

Figure 18: Persistent oil price shock...



Source : Deutsche Bank

Figure 19: ...vs transitory oil price shock



Source : Deutsche Bank

The results suggest that persistence matters, particularly for oil shocks ([figures 18 and 19](#)). When the shock is persistent, we find a stronger peak impact for both the Euro area and the United Kingdom for a given 1 percentage point surprise compared to a short-lived shock. For urea, the pattern is clearer in the UK, where persistent shocks are associated with a larger food price pass-through. In the Euro area, however, the difference between persistent and short-lived shocks is less clear. Overall, the evidence points to a stronger role for shock persistence in the UK, while the Euro area results are more mixed.

While it is too early to definitely classify the latest energy price shocks, the above results highlight an important distinction. If recent recent rises in oil and fertiliser prices prove short-lived and unwind over the coming months, then pass-through to



food prices may be more limited. However, if prices remain elevated, history suggests we could expect a larger and more sustained impact on food inflation.

Based on the shocks observed so far, our classification leans towards the recent developments being transitory in nature. That said, the classification is far from certain and will depend on how oil and fertiliser prices evolve through July and the months ahead.

Conclusions

What to make from the above analyses? We highlight several key findings.

First, the impact of a fertiliser and oil price shock and its pass-through to food price levels tends to be stronger in the United Kingdom than in the Euro area.

Second, after scaling and aggregating the fertiliser and oil price shocks from March up to June, we find that in a year's time from March, the cumulative impact on the UK food price level could be about 1.3% while for the Euro area it would be around 0.8% instead – implying a 0.1-0.15pp increase in headline HICP/CPI.

Third, we also recognise the limitations of simply scaling these shocks. The size and duration of the shock can result in very different implications for food price pass-through. The persistence of a shock can have a much larger impact, especially in the UK. It's too early to say where the current energy shock will fall into. Recent price action suggests a further wave of energy pressures on the back of escalating Middle East tensions. We will be monitoring this closely. The longer commodity prices remain elevated, the more they feed through into food prices.

Lastly, the confidence bands around our estimates suggest wide variation in pass-through. Given the swings in price rises across oil and urea, the staggered pass-through of cost rises into the food supply chain may be more varied. While our estimates provide a measure of the average conditional pass-through, other factors will also impact pass-through. For example, warmer global temperatures, less rainfall, and changing labour costs alongside ongoing geopolitical tensions may also shape pricing strategies.

To be sure, despite the pull back in oil and urea prices from their peak, the net effects still point to a modest jump in food (and non-alcoholic beverages) prices. Furthermore, the recent jump back in price momentum will do little to assuage concerns of indirect effects brewing within inflation. We will reflect some of our findings over the coming week(s) as we update our inflation projections.



Appendix 1

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